Low Temperature Processing for Heterogeneous Integration of Ge-on-Si Materials Systems

By

Eveline Postelnicu

B.S., Materials Science and Engineering
Massachusetts Institute of Technology, Cambridge, 2017

Submitted to the Department of Materials Science and Engineering
in partial fulfillment of the requirements for the degree of

DOCTOR OF PHILOSOPHY IN MATERIALS SCIENCE AND ENGINEERING
at the
MASSACHUSETTS INSTITUTE OF TECHNOLOGY

September 2022

Signature of Author:____
Department of Materials Science and Engineering
September 2, 2022

Certified by:____
Lionel C. Kimerling
Thomas Lord Professor of Materials Science and Engineering
Thesis Supervisor

Certified by:____
Anuradha M. Agarwal
Principal Research Scientist, Materials Research Laboratory
Thesis Supervisor

Accepted by:____
Frances M. Ross
Professor of Materials Science and Engineering
Chair, Departmental Committee on Graduate Studies

This page intentionally left blank

Low Temperature Processing for Heterogeneous Integration of Ge-on-Si Materials Systems

By

Eveline Postelnicu

Submitted to the Department of Materials Science and Engineering on September 2, 2022

in Partial Fulfillment of the Requirements for the Degree of

Doctor of Philosophy in Materials Science and Engineering

Abstract

Data communication relies on interconnects between circuit elements on a chip as well as between chips. As feature sizes decrease for silicon (Si) complementary metal oxide semiconductor (CMOS) transistors, electrical interconnects face increasing delay and increasing power consumption. Photonic interconnects can ameliorate issues with power consumption by removing heat dissipation and allowing for higher bandwidths and higher degrees of multiplexing. Germanium (Ge)'s absorption in the Telecom wavelength range (1.3-1.55 μm), its transparency in the mid-longwave infrared (mid-LWIR) range, and its ease of integration on a Si backbone make it a perfect candidate for photodetectors (PDs) for Telecom optical interconnects and waveguides for mid-LWIR optical sensors. However, optical interconnects are still larger than electrical interconnects, motivating the move to vertical integration of photonic devices. In order to achieve vertical integration, interconnects must be integrated in the back-end-of-line (BEOL) of a CMOS process flow. BEOL integration introduces strict temperature requirements of T<450°C to prevent metal diffusion and oxidation of devices already on-chip.

First, this thesis tackles BEOL-compatible active devices such as Ge-on-Si PDs for the near-IR wavelength range. We developed a BEOL-compatible process flow for Ge on Si epitaxy. Investigating strain in our low-temperature Ge led to a discovery of a residual compressive strain at extended temperature regimes compared to prior work. Based on this residual compressive strain, a model is proposed for misfit dislocation nucleation in Ge-on-Si epitaxy. Point defects and dislocation kinetics prevalent in low temperature growth of Ge are characterized and analyzed via post-processing annealing. The impact of these defects and processing conditions on Ge-on-Si photodetector device performance is also explored. We found that acceptor-like point defects identified via Hall effect are responsible for conductivity type conversion via low temperature post-processing. These point defects impact Ge-on-Si PDs' internal quantum efficiency, while dislocations appear to dominate device leakage current. We also found that effective sidewall passivation, a remote heterojunction structure, and post-processing annealing of Ge-on-Si PDs result in the lowest reported dark current density for Ge-on-Si PDs, 160 nA/cm^2. Lastly, BEOL-compatible passive Ge devices for mid-IR-LWIR sensing applications are investigated. Room temperature-evaporated amorphous Ge on FZ-Si waveguides demonstrate transmission loss of 15 dB/cm at 9.79µm wavelength, which is a promising start for room temperature-evaporated Ge waveguides.

Thesis Supervisors:

Lionel C. Kimerling

Title: Thomas Lord Professor of Materials Science and Engineering

Anuradha M. Agarwal

Title: Principal Research Scientist, Materials Research Laboratory

Acknowledgments

There are so many people (and cats) involved in the making of this PhD without whom I am certain I would have not made it past this finish line (in classic Evi style, coming together in the late hours of the night, beyond Taylor Swift's favorite time – midnight). First and foremost, it has been my honor and privilege to work with my advisor and mentor, Professor Lionel C. Kimerling. I have known Kim since I was a freshman at MIT in his 3.003/4 class, Principles of Engineering Practice. This was the class that introduced me to solar cells, the study of light-matter interactions, and first instructed me in preparing a lab notebook! Four years went past in the blink of an eye and suddenly I was starting my PhD and trying to pick an advisor for the next five years. I wanted an advisor that would be positive and encouraging, and that I would always walk away from having learned something new. I got that and so much more in Kim. Every time I brought a problem I was stuck with to Kim, he pushed me to figure out deeper, more insightful questions in a way that was so exciting, you wouldn't realize he hadn't answered your question until the meeting was done! Every time catastrophe struck in lab and I thought this was it, there was no way out, Kim would always be incredibly calm and find an alternative solution. Kim somehow always knows your research better than you do and manages to learn something out of every little data point. I have learned so much more from Kim than just Ge-on-Si photonics – he has guided me throughout my PhD, gently steering me in the directions he predicted way ahead of me, while still allowing me the space to learn through my own process. I am eternally grateful to have had an advisor who was so encouraging and positive, and always motivated me and supported me even when it seemed like nothing would work.

It has also been an incredible privilege to work with my co-advisor and mentor Dr. Anuradha Agarwal, who led the Ge waveguide and sensing work. As soon as I joined EMAT, I knew I wanted to work with Anu. Not only was Anu leading work I felt was incredibly impactful and exciting, her energy was contagious! Anu's warmth and impact on the EMAT/PMAT community keep our entire lab group running and set the tone for leadership, mentorship, and collaboration top-down. Anu has constantly inspired me even when research was going anything but well. She has taught me to always remember the big picture in my research and to be efficient in the directions I take. Without Anu, I don't know how much would get done in EMAT! I also love that Anu is a fellow night owl like me, though I think she actually has boundless energy because unlike me, Anu has 8 AM meetings! Thank you Anu, I am so grateful to you for all the time you invested in advising me, and all you have taught me about how to conduct research.

I would also like to thank my committee – Professor Polina Anikeeva and Professor Juejun Hu. Thank you to Polina and JJ for always encouraging me to ask better, more precise questions in my committee meetings and giving me invaluable feedback on experimental methodology and analysis. Thank you for making sure I was on track in my PhD and pushing me to finally make definitive conclusions! I would like to also thank my other supervisors Dr. Jurgen Michel and Professor Kazumi Wada. Jurgen and Kazumi took on the roles of advisors along with Kim and Anu, and generously shared their expertise with me. Jurgen taught me everything about UHVCVD and SEL maintenance. Without him, the UHVCVD would not have run at all. Thank you to Jurgen for always stepping in to help with maintenance and going above and beyond to keep the lab running. Thank you to Kazumi for always providing the most interesting theoretical perspectives and encouraging me to think of the "why" behind my results. I learned so much from Kazumi in all of our discussions, and I appreciated every time he helped me get started on answering the hardest research questions.

Next, I would like to thank all the collaborators of this work, starting with Professor Ruitao Wen, Dr. Danhao Ma, Dr. Boaming Wang, and Stephanie Marzen for the Ge-on-Si photodetector work. Ruitao and Danhao helped start the LT Ge project. They contributed so much to this work with their UHVCVD, SEM, and XRD expertise. Thank you to Ruitao and Danhao for teaching me about UHVCVD and device processing and always answering my UHVCVD/threading dislocation questions even after they left MIT! Thank you to Dr. Baoming Wang for his invaluable TEM contributions, and thank you to Stephanie who has been my labmate on the Ge-on-Si photodetector project for the last two years! Stephanie has contributed so much to the device process development, the device design, and the measurement set-up. We couldn't have had her EE expertise come in at a better time! Thank you to Stephanie for working beside me in lab, swapping out processing shifts when the Sigma deadlines were looming, and for always having great and insightful discussions that helped define out project's next steps and always helped me learn more as well. I am also very grateful for all of Stephanie's help in SEL – my favorite SEL story is still decoding the German wiring diagram (before we had Jurgen to help!). Thank you also to Mark Beals, the director of SEL, for all of his help and instruction in SEL maintenance. Thank you to Tarky, James, and David Bono for all of the help with hoisting the roughing pump! Without Dr. Hyunseok Kim and Kuangye Lu who run the MOCVD in SEL, SEL would not run. Thank you so much to Hyunseok and Kuangye for taking charge of all the difficult SEL gas maintenance projects, and for teaching me gas tank maintenance! Thank you for always stepping in to help out and being so generous with your time, I appreciate it so much!

I would like to thank my collaborators for the amorphous Ge waveguide project – Michelle Clark who worked extensively on the measurement set-up and analysis and led the project, and Samarth Aggarwal and Melvin Nuñez – two undergraduates who I had the pleasure to work with

and mentor on this project. Thank you Samarth for all of the many valuable contributions to this work – processing, characterization, and simulation! I am also grateful to how much Melvin contributed on the a-Ge simulation front even though his research program was so short. Next, I would like to thank my collaborators for the Ge-on-SiN work. Thank you to Professor Samuel Serna for all the simulation and experimental expertise he contributed to this work, and for teaching me Lumerical! Thank you to Samuel for also always being a kind and inspiring person to work with – I learned a lot about mentoring by watching Samuel, thank you! Thank you to Katie Stoll who contributed so much in the MZI simulation and design of the Ge-on-SiN work. Katie is such a wonderful person to work with and I am so lucky to have had the kindest collaborator and friend in her. A huge thank you also to Vincent Reboud for all the excellent points and insightful conversations we have had about this project, and for all of the simulation and experimental guidance he provided! Thank you to Vincent and Olivier also for spending every day with me for a week in Grenoble trying to measure transmission loss of our waveguides! I loved visiting and working with these amazing and generous researchers.

Next I would like to thank my entire lab group – EMAT, and our sister lab group PMAT. Thank you for always being a kind and collaborative lab group! Special thank you to Peter, Derek, and Jérome for being my first mentors in the group. Thank you to Derek and Jérome for being the main reason I joined the group since you guys were so fun, and for helping me out so much with processing advice and grad school advice in general (GRT Derek was the best!). Thank you to Peter, the guru of EMAT, for helping me with any and all problems. I appreciate how much time Peter put in to mentoring, and still does! Thank you especially to Xueying for being my first friend in the group and for all the help with processing, UHVCVD, and understanding dislocations! Xueying is one of my best friends and working with her made everything better always. Thank you to Drew for all of the fun conversations we've had in the office, commiserating about processing and about waking up early. Another fellow night owl and an awesome person – thank you to Drew for always making my day a little brighter and for being a great friend! Thank you to Maarten, the newest EMAT member, for all of your UHVCVD help and social planning! You are the only EMAT/PMAT student I know of who has been able to successfully initiate regularly scheduled group lunches – if you can do that, you can do anything! Thank you to Tushar, Gillian, Misha, and Carlos from PMAT for their collaboration and friendship, and thank you to Cosmin from PMAT for being my Romanian buddy! I never thought I'd get to share secret Romanian messages with someone at work, and I am so grateful I met you. Thank you to the visiting students in EMAT as well – Yiding, Ruihan, and Wat for being great presences in our group. Thanks especially to Wat for all of your excitement with experiments and helping out with simulations! I hope we get boba together soon!

Without the MTL and MIT.Nano staff, I wouldn't have been able to complete a lot of this research. Thank you to Vicky, Jorg, and Whitney for everything they taught me on PTC about cleanroom processing and operations! Thank you also to all of the MTL and MIT.Nano staff who go above and beyond to train users, especially Dennis Ward, Dave Terry, Kurt Broderick, Gary Riggott, and Eric Lim. You guys taught me how to process and know more about these tools and materials than anyone. Thank you for your generosity and for the many times you have saved my research in the most desperate of times! I would especially like to thank Dennis, Dave, and Kurt, for always stopping to chat with me in the cleanroom and check in on me. Your support and kindness meant so much to me and helped me get through the most frustrating times. Thank you! I wouldn't be able to have a paragraph thanking cleanroom folks without mentioning by buddy John Niroula. Thank you to my dear friend John for being my 24-hour buddy. When no one else

was available to come in after hours with me on evenings and weekends, John was there. Thank you for all the fun lunch breaks and the desperate deadline times you have saved me, not just in the cleanroom, but in psetting in 8.223 in undergrad and in our optoelectronics grad class too.

Thank you to everyone in OGE Programs for Women with whom I've had the pleasure and honor to work. Thank you to Dean Blanche Staton for leading Programs for Women and always checking in on us and supporting us as graduate community fellows, researchers, and people. Your kindness and generosity are boundless and I can only aspire to have the kind of impact you have had on your community. I am so grateful for all of your support and for how you have led by example. Thank you to Patty and Elaine for taking care of so many behind-the-scenes details, with a special thank you to Elaine for your advice and friendship. Thank you to my fellow GCFs – Tahoura, Bianca, Maddie and Anastasia, who I have had an awesome time working with on Path of Professorship and other OGE events and who I am lucky to also call my friends. Through the GCF program and WXOMS in DMSE, I found a community and network that have empowered and uplifted me throughout my PhD.

Thank you to all my best friends from undergrad for continuing to support me during this crazy PhD journey. To the fantastic five girlies – Erika, Anna, Sabrine, and Wajeeha, thank you for commiserating about grad school together, always remembering birthdays, and all the fun trips/adventures we've been on together! To Juju – thanks for always being down to adventure, I can't wait for the future ones! To the solar car crew – Michelle, Fabiola, Rose, Julia, Kelly, and Kathleen, thank you all for the amazing memories and adventures together. You are the coolest, kindest, and wisest bunch of people and I am extremely grateful for all of your support, especially when I needed to be reminded that there is life outside of MIT, and it's fantastic! Thank you to

Rebecca, the Geetha to my Tresca, for being an incredible friend and always supporting me. Every time we talk, it's as if no time has passed. Love you my soul sister!

Thank you to my first real housemates: Ali, George, Parker, Saurabh, and Christina, for being wonderful people to live with and wonderful friends. Thank you to Georgie for all the (very last-minute) times you shared pset answers with me, and all the great times we've had throughout grad school. Thanks for always being there for me George and always brightening my day with your humor! Thank you to my dear bestie Christina for all of your support and encouragement! Team Juice has had the best adventures and I can't wait to have more together. Thank you for being the best auntie to Catsby and for telling me I could finish my thesis when I outlined all I had to do three days before the deadline (I believed you and I got it done because you believed in me!). Thank you to the Spoopy Crew/Taylor Swift Fan Club/Fall Fan Club – Kate and Ty. You are the most incredible and supportive friends and without your Taylor Swift lyrics, fall puns, and productive vent sessions, I would not have made it through. Thank you for always being there for me and for always lifting me up. Thank you to the Materials Girls who Run the World – Kat, Cécile, and Jackie, for your friendship, support, and all the running and workout inspo you've provided throughout the years! Love you all and can't wait for when we run that marathon together! Thank you to Eesha and Elad for the great dinner outings and the friendship and support! Thank you to my pottery chums Kate and Jackie for doing my favorite activity with me, and thank you to Marc, our incredible instructor who always listens to my research frustrations and encourages me every time. Thank you for your support, and for providing me the best possible outlet and the time I look forward to most in the week!

Finally, thank you to my family who I love more than words can express. Thank you to the Leissrings for taking me in and for all the amazing adventures we've been on! Thank you for introducing me to camping, thank you to Luke's dad, Mr. Leissring, for all the foodie recommendations we've tried together, and thank you to Luke's mom, Mrs. Leissring, for sharing crafting with me. Thank you to Faith for all the fun times we've spent together skiing, climbing, hiking, and just hanging out! Thank you to Bunica and Bunicu for all the love and support – I miss them so much and I know they would want me to live to the fullest. Thank you to Maia for always believing in me and encouraging me. Maia always believes the impossible will be possible and she inspires me to do the same. Thank you to my sister, Iris, for taking care of me since I was young and always being my inspiration and role model. Thank you for teaching me to stand up for myself and to go after the things I want, and that no one can define my place but me. Thank you to my mom and dad for teaching me to dream big since I was little. Thank you to my dad for always spinning the craziest tales and jokes and feeding my imagination when I was a kid. Without my dad doing science experiments and fun math problems with me, I probably wouldn't have pursued engineering. Thank you for running half marathons with me and being there for me every step of the way through my whole life, and for always joking and keeping the energy up no matter what. Thank you to my mom who taught me how to work hard and never give up. Thank you for inspiring me to pursue a PhD and for encouraging me every day in our daily Snapchat! You have been there for me throughout everything, including working with me on Zoom for hours during my hardest times in my PhD (like writing my thesis!). You aren't just my work buddy, you are my role model and inspiration. I dedicate my PhD thesis to you, the first Dr. Postelnicu, because you worked just as hard for this as I did! Te iubesc Mami!

Thank you to my first cats Freddie, Neo, and Katie for inspiring my love of cats and my baby Katie for being the best friend and companion growing up. I miss you so much. Thank you to my cat Catsby for all the times you blocked my notebook/computer screen while I was trying to

work, and for all the Zoom calls you've crashed. It was always a welcome distraction! Finally, thank you to my partner, Luke, without whom I wouldn't have been able to get through this whole PhD. Thank you for always supporting me and taking care of me when I was too busy to take care of myself. Thank you for driving me back and forth from work, sitting with me in lab late at night, helping with the UHVCVD loading arm, helping with the DMSE socials, and being so involved with MIT life that you deserve an honorary degree! More importantly, thank you for all the wonderful times we've had these past four years – for all the camping, hiking, climbing, skiing, and traveling adventures we've had, and for all the normal moments that are so wonderful because we're together. I can't wait for the incredible future we have ahead and for all of the adventures ahead of us! I love you to infinity and beyond!

“To boldly go where no one has gone before.”

– Captain Jean-Luc Picard, Star Trek: The Next Generation

Table of Contents

List of Figures

List of Tables

Chapter 1: Introduction

Data communication relies on interconnects between circuit elements on a chip as well as between chips to relay data. In order to keep up with Moore's Law, devices must become smaller in order to keep up with increasing integration demands for higher densities of devices on microprocessors.[1] As feature sizes decrease for silicon (Si) complementary metal oxide semiconductor (CMOS) transistors, electrical interconnects face increasing delay and increasing power consumption.[2–4] Photonic interconnects can ameliorate issues with power consumption by removing heat dissipation. Along with lower power consumption, data transmitted via optical communication has higher bandwidths and higher degrees of multiplexing. Optical interconnects are made up of a light source, a modulator, a waveguide, and a detector. Germanium (Ge)'s absorption in the Telecom wavelength range (1.3-1.55 μm) along with its ease of integration on a Si backbone makes it a perfect candidate for photodetectors for optical interconnects. However, optical interconnects are still larger than electrical interconnects, motivating the move to vertical integration of photonic devices. This would allow optical interconnects to be fabricated on layers of other devices, thus alleviating the device density issues with larger feature sizes. In order to achieve vertical integration, interconnects must be integrated in the back-end-of-line (BEOL) of a CMOS process flow. BEOL integration introduces strict temperature requirements of T<450°C to prevent metal diffusion and oxidation of devices already on-chip.[5]

State-of-the-art Ge is grown on Si via a two-step process followed by post-growth cyclic annealing to reduce threading dislocation density to 10^6 cm^{-2} for Ge that is selectively grown and 10^7 cm^{-2} for blanket films. Typical growth temperatures are around 650°C-730°C with post-growth anneals at 900°C.[6] Additionally, an 800°C pre-growth anneal in hydrogen (H_2) is conducted in order to remove oxide that may have formed on the Si surface prior to loading the Si substrate into the growth reactor.[7] Figure 1-1 details the thermal budget for current state-of-the-art Ge growth.

This thesis will explore the limitations and opportunities of low temperature BEOL-compatible growth and post-processing of Ge-on-Si. The first application discussed will be BEOL-compatible growth and post-processing for active Ge devices, more specifically Ge-on-Si photodetectors for the Telecom wavelength range. To begin, Chapter 2 describes the growth mode and surface morphology for fully BEOL-compatible Ge growth. Next, Chapter 3 dives into the defect microstructure prevalent for low temperature-grown Ge, and the impact of annealing on defects. Energetic considerations for misfit dislocation nucleation are modeled and evaluated at low temperatures. Point defect-dislocation interactions and their impact on strain are also explored. Chapter 4 expands upon the defect knowledge gained in Chapter 3 and looks at the impact of annealing and processing conditions on Ge-on-Si photodetector performance. Finally, Chapter 5 pushes the limits of low temperature as far as they can go in Ge passive device applications. Amorphous Ge-on-Si waveguides are designed, fabricated, and characterized for mid-longwave infrared (mid-LWIR) chemical sensing.

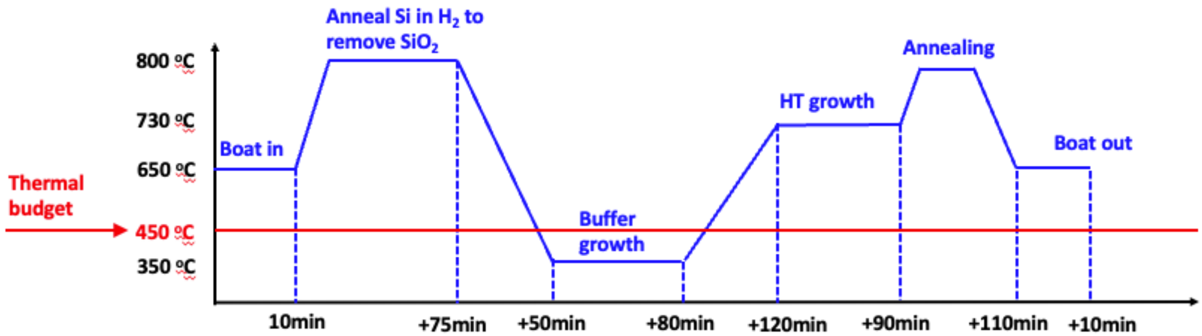


Figure 1-1. State-of-the-art Ge on Si growth thermal budget.

Chapter 2: Low temperature Ge growth for integration of Ge photodetectors

2.1 Background

We utilize Ultra High Vacuum Chemical Vapor Deposition (UHVCVD) to explore epitaxial growth of Ge on Si at temperatures lower than 450°C for CMOS compatibility. Epitaxy is the growth of a crystal on top of another crystal. Heteroepitaxy refers to the growth of material that differs from the substrate and faces issues with regards to lattice matching. Ge faces a 4.2% lattice mismatch to Si. The growing crystal adopts the orientation of the substrate initially. As it grows, the film relaxes in the vertical direction but remains biaxially strained if it maintains the substrate's lattice without any deviation. One way to relieve the build-up of strain is to introduce a dislocation (a missing plane of atoms), as in Figure 2-2. These dislocations are called misfit dislocations because they relieve misfit strain in epitaxial growth. Dislocations cannot terminate within a crystal unless they extend across the entire length of the interface or intersect another surface. Thus, dislocations extend to the film surface in a half-loop, creating a threading dislocation.[8] These dislocations are undesirable because they serve as recombination centers that reduce the signal and thus the efficiency of the detector.

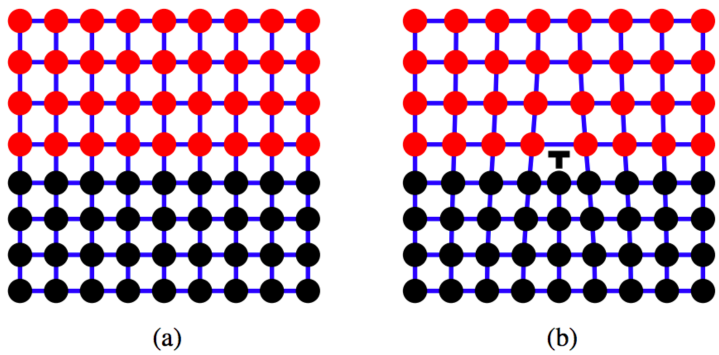


Figure 2-1. Cross section of a heteroepitaxially grown film (a) before and (b) after introduction of an edge dislocation. Image retreived from Albert.[9]

In 1984, Luryi et al were the first to utilize SiGe buffer layers in epitaxial Ge on Si growth to reduce threading dislocation densities.[10] By gradually increasing the amount of Ge and thus the lattice constant, less strain is present in each layer and so there are fewer dislocations. Fitzgerald et al then introduced optimized SiGe layers and mechanically polished off surface roughness ex situ to reduce threading dislocations.[11] Current state of the art growth techniques utilize a two-step growth process sans SiGe buffer layers. The first step involves a thin Ge buffer layer, 30-60 nm, directly grown on Si at low temperatures (320-360°C). Low temperatures suppress Stransky-Krastanov growth, or islanding, of Ge. The second growth step traditionally increases the temperature >600°C to achieve higher growth rates and better crystal quality. If the buffer layer is thicker than 30 nm, the second growth step results in planar, homoepitaxial Ge. Typical dislocation densities for what we will now refer to as "high temperature" growth are 10^8-10^9 cm^{-2}.[6] Post-growth annealing at temperatures >750°C can reduce dislocation densities by two orders of magnitude.[12] Another option other than annealing is to use H_2 annealing at 825°C to reduce surface roughness and threading dislocation density because of enhanced atomic mobility.[6]

Dislocation motion is either conservative (reversible) or non-conservative (non-reversible). Conservative movement is known as glide, movement of dislocations within the slip plane, which can be reversed if the opposite stress if applied. Climb, on the other hand, is non-conservative motion normal to the slip plane that is not reversible. Climb is highly dependent on temperature as it depends on point defect creation or annihilation. Instead of moving as a whole and breaking every bond between the lattice and the dislocation, thus engaging in a high energy process resulting in many dangling bonds, dislocations move via kinks or jogs – where one bond is broken at a time.[13] By utilizing TEM to image dislocations in deformed Si and DLTS to analyze the defect spectra, Kimerling et al observed that after an initial deformation at 750°C and a secondary anneal at 450°C the defect spectra were relatively unchanged but the dislocation lines were straightened out compared to the tangled dislocations after the first deformation. This study effectively demonstrates that point defect clusters produced by deforming Si at high temperatures are the result of non-conservative jog motion since the dislocation core defect energy remains unchanged but the arrangement of dislocations indicates dislocation motion.[14]

Another study of point defects related to dislocations in a high temperature environments comes from Samavedam et al. They demonstrate that in relaxed Ge/SiGe/Si structures, vacancy diffusion increases as growth temperature approaches the melting point and dislocation climb mechanisms which are primarily vacancy diffusion based can be activated.[15] Previous studies of dislocation energy levels in deformed Si conducted using DLTS have shown that dislocations introduce a specific energy band at E_v + 0.34 eV for 60 degree dislocations which are threading dislocations. However, upon applying high temperature deformation, further defect peaks were observed. Utilizing Electron Beam Induced Current (EBIC), dislocation density was tracked along with defect levels in DLTS for a secondary deformation. The dislocation defect peak remained consistent with the density determined from EBIC while the other defect levels were not. Thus, there are point defect energy levels separate from dislocation energy levels present in the material.[16]

2.2 Heteroepitaxy via Ultra High Vacuum Chemical Vapor Deposition

Ultra High Vacuum Chemical Vapor Deposition (UHVCVD) is used to grow Ge heteroepitaxially on Si. The UHVCVD we perform uses germane gas (GeH_4) to grow Ge. The gas flows through the reactor's quartz tube (see Figure 2-2) and arrives at the Si wafer. If one Si wafer side is preferred for growth, that side is oriented facing towards the load-lock so that the germane gas is in turbulent flow at that surface which results in a more uniform deposition. Silicon dioxide (SiO_2) coated wafers are placed before and after the main growth wafers to also aid in obtaining turbulent flow at the growth surface. The growth reaction steps first determined by Cunningham et al[17] are:

$$2GeH_4(\mathrm{g}) + 4\,\_ \rightarrow 2\underline{H} + 2\underline{Ge}H_3 \tag{2-1}$$

$$2\underline{Ge}H_3(\mathrm{g}) + 2\,\_ \rightarrow 2\underline{H} + 2\underline{Ge}H_2 \tag{2-2}$$

$$2\underline{Ge}H_2(\mathrm{g}) \rightarrow H_2(\mathrm{g}) + 2\underline{Ge}\mathrm{H} \tag{2-3}$$

$$2\underline{Ge}H(\mathrm{g}) \rightarrow H_2(\mathrm{g}) + 2\,\_ + \mathrm{film} \tag{2-4}$$

$$4\underline{H} \rightarrow 2H_2(\mathrm{g}) + 4\,\_ \tag{2-5}$$

where the symbol _ is used to indicate a surface site. The first step is adsorption (2-1), followed by surface decomposition and diffusion of GeH_3 (2-2) and GeH_2 (2-3), lattice incorporation (2-4) and finally, hydrogen desorption (2-5) to open up more surface sites for incorporation. The hydrogen desorption step controls the growth rate of the reaction.[17] Once the Ge adatom adsorbs on the surface, it has to have sufficient energy to diffuse to a Si atom and react. If the surface diffusion length is too high, we will enter Stransky-Krastanov growth mode, resulting in islanding of Ge to relieve the compressive strain from the lattice mismatch. However, if surface diffusion is suppressed at low temperatures in the initial growth step, the film will form misfit dislocations to relieve the strain and proceed with planar growth.[6]

A schematic of the UHVCVD reactor is shown in Figure 2-2. The wafers are loaded into a quartz boat which is then loaded into the load-lock. A loading arm automatically loads the boat into the quartz reaction tube while H_2 flows. The reaction begins when germane gas starts flowing through the reactor and proceeds until the germane gas is shut off. The boat is then unloaded back to the load-lock and the wafers are unloaded after the boat cools down. While the growth rate is dominated by surface reaction kinetics described above, mass transport through the reactor will not affect the growth rate. If the reaction is mass transport-limited which occurs at higher temperatures, the growth rate can be controlled through the gas flow and reaction pressure.

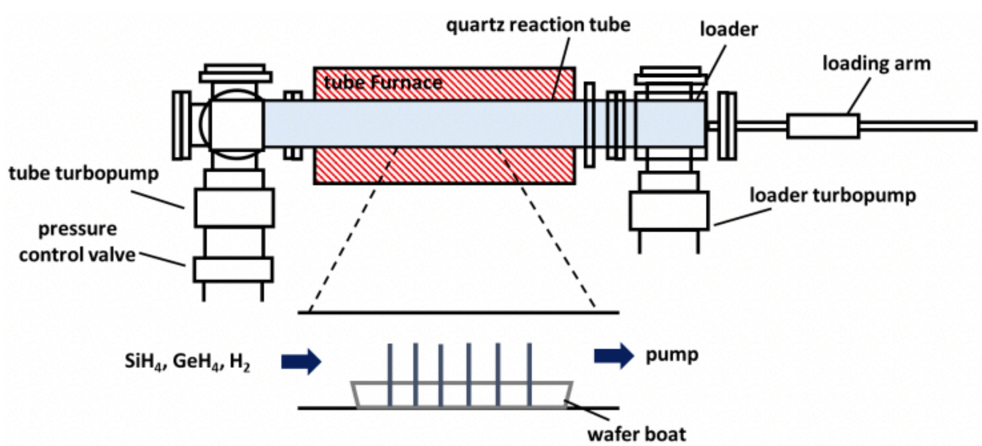


Figure 2-2. UHVCVD growth reactor schematic, obtained from Danhao Ma.[18]

2.3 Room temperature pre-growth Si clean

Currently, standard Ge-on-Si growth for integration in the FEOL includes a pre-growth anneal of the Si wafer in H_2 ambient at 800°C to remove any silicon oxide (SiO_2)) from the surface.

Ge won't grow on SiO_2, so high quality Ge growth requires an oxide-free surface. The RCA clean, a typical pre-growth clean, leaves a thin passivating layer on the surface in order to prevent contamination of Si's bare surface before loading into the growth chamber. Instead of the high temperature (HT) removal step of this passivation layer, we utilize a room temperature. (RT) methanol-iodine clean developed in our group by Tracey Burr[19] after the standard RCA clean. The methanol-iodine clean removes the oxide on the surface using hydrofluoric acid (HF) followed by a methanol-iodine dip that passivates the surface and prevents oxidation for two hours. Ge grown on a Si surface at 400°C with methanol-iodine pre-treatment is specular, comparable to Ge grown utilizing the pre-growth HT step, and demonstrates a much smoother surface than Ge grown on a Si surface utilizing only the HF oxide removal step prior to growth (Figure 2-3).

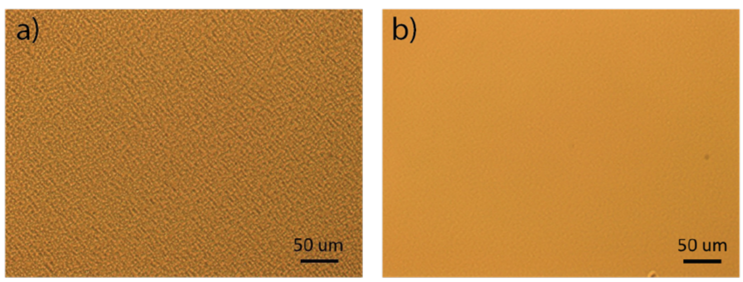


Figure 2-3. Inspection of the Germanium surface after epitaxy on Si for wafers from the same growth run shows (a) visible haze and high surface roughness for wafers without iodine/methanol passivation pre-growth and shows a (b) specular surface when the iodine/methanol passivation is used pre-growth. Optical microscope images courtesy of Dr. Danhao Ma.

2.4 Surface morphology characterization and optimization of growth temperature

We explored various growth temperatures that were BEOL-compatible – 400°C, 415°C, and 430°C. We analyzed surface roughness and morphology via Atomic Force Microscopy (AFM) and Scanning Electron Microscopy (SEM). Figure 2-4 demonstrates how the surface morphology changes with growth temperature. By analyzing the SEM and AFM surface scans, we observe that small pits form on the surface at 400°C, which grow larger at 415 and 430°C with angular sidewalls. By comparison, the standard Ge growth which includes a 350°C buffer followed by a 730°C growth step and post-growth cyclic annealing, whose surface morphology is shown in Figure 2-5c, is very smooth with no evidence of surface pits.

Figure 2-5 includes the surface roughness profiles of these four different growth conditions, as well as the measured root mean square (rms) roughness via AFM. The rms roughness increases from 7nm to 44nm between 400°C and 430°C growth temperatures. The 730°C growth temperature rms roughness is around 1nm, which is comparable to previous high-temperature epitaxy reports.[20] While it has been previously reported that the onset of islanding for UHVCVD growth is 375°C [17], we did not observe a difference in the surface morphology of 400°C-grown Ge on Si both with and without a 350°C buffer before the 400°C growth step (Figure 2-6). The average roughness measured via AFM for the 400°C single-step growth was 9nm, while the average roughness for the two-step growth was 8.5nm. As discussed previously in section 2.1, the 350°C buffer is utilized to suppress surface diffusion during growth so that a planar film can be achieved. The comparable average roughness values as well as the very

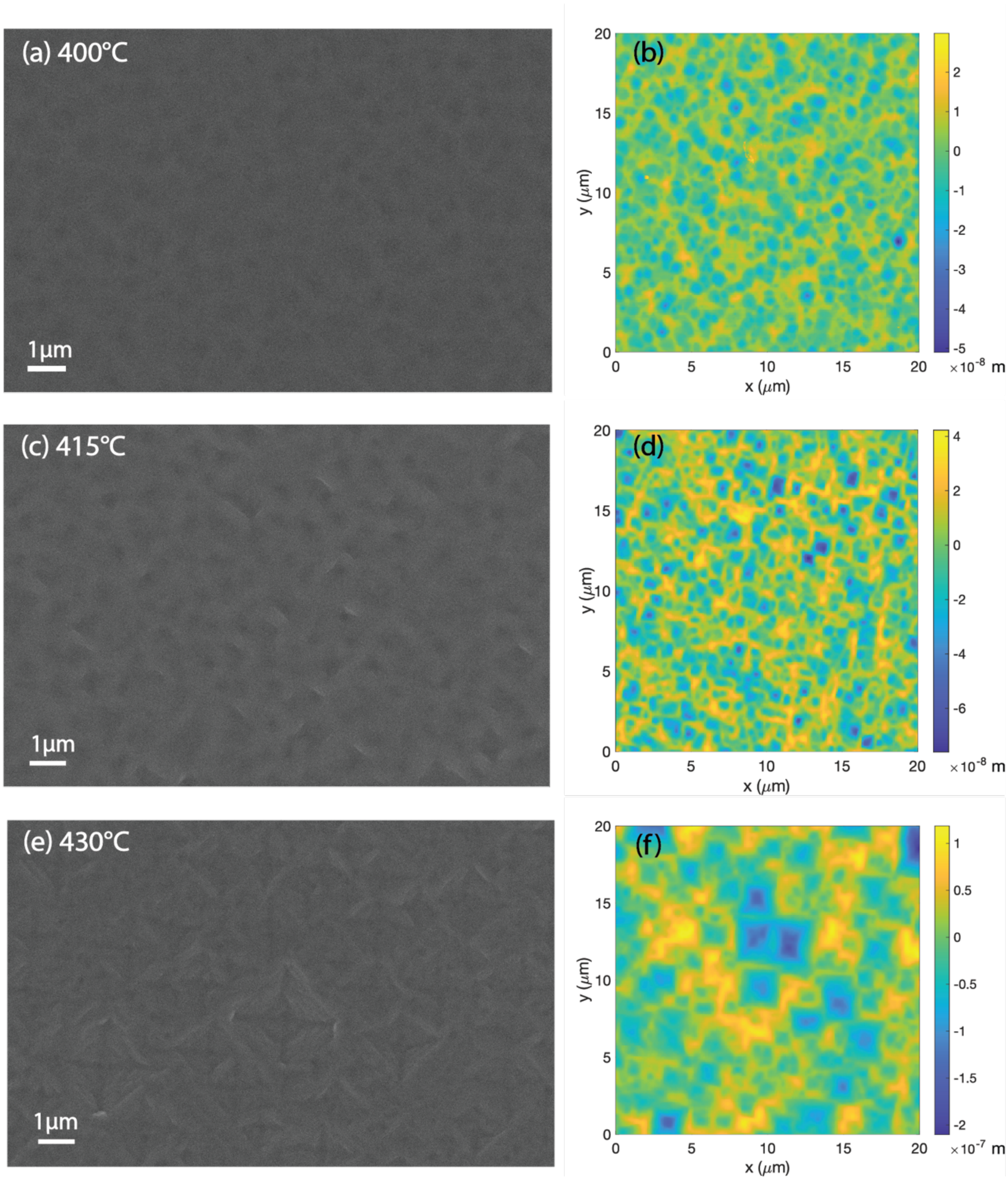


Figure 2-4. Top-view SEM images of Ge surface grown on Si via UHVCVD at growth temperature of (a) 400°C, (c) 415°C, and (e) (1) 350°C + (2) 430°C,, followed by AFM surface roughness characterization for (b) 400°C, (d) 415°C, and (f) (1) 350°C + (2) 430°C, and growth temperatures. Pits becomes more pronounced at 415°C and clearly defined crystallographically-

oriented pits are visible at 430°C, despite a 60 nm buffer layer employed as the first growth step. SEM images and growth courtesy of Dr. Ruitao Wen.

similar surface morphology observed via AFM in Figure 2-6 are evidence that islands are not forming, but that the surface is roughening by other kinetic means.

The average period and amplitude of the pits we observe at growth temperatures between 400-430°C are described in Table 2-1. These pits may correspond to coherent strain relaxation via surface roughening as described by Srolovitz.[21] Srolovitz determined a critical wavelength for surface roughening via undulations of critical wavelength:

$$\lambda_c = \frac{\pi * \gamma\ * M}{\sigma^2} = \frac{\pi * \gamma}{M * \epsilon^2} \tag{2-6}$$

where M is the Young's modulus 102.7 GPa, σ is the stress in the film, ϵ is the misfit strain, and γ is the surface energy of Ge on a Si(100) surface, 1835 ergs/cm^2 obtained from Jaccodine.[22] The critical wavelength for surface roughening for a lattice misfit of 4.2% is thus 33nm. We would expect the surface periodicity to exhibit this critical wavelength for surface undulation/surface roughness if the diffusion length is lower than this critical wavelength.[23] The diffusion length at our growth temperatures was calculated via:

$$l = \sqrt{D_s t_m} \tag{2-7}$$

where D_s is diffusivity of Ge on a Si(100) surface, and t_m is the length of time it take to grow one monolayer of Ge.[23] We assume one monolayer is 5.66Å, the lattice constant of Ge, and we extrapolate the diffusivity of Ge on a Si(100) surface from Dolbak et al[24] via:

$$D_{Ge\ on\ Si(100)} = 3.3 * 10^{-2} \exp\left(\frac{-1.2\ eV}{k_B T}\right) cm^2/s \tag{2-8}$$

where k_B is the Boltzmann constant, and T is the growth temperature. From the growth rates we measured for our 400, 415, and 430°C growths (7, 10, and 13 nm/min, respectively), we calculate the surface diffusion lengths at 400, 415, and 430°C are 128, 135, and 146 nm, respectively. The surface diffusion is higher than the critical wavelength for surface roughening in all cases, so we would expect the periodicity of the roughening to be larger than the critical wavelength. From Table 2-1, we can see that the periodicity of roughening is much larger than the diffusion lengths for all growth temperatures ranging from 400-430°C. The amplitude and period of the pitting increase with growth temperature as we would expect.

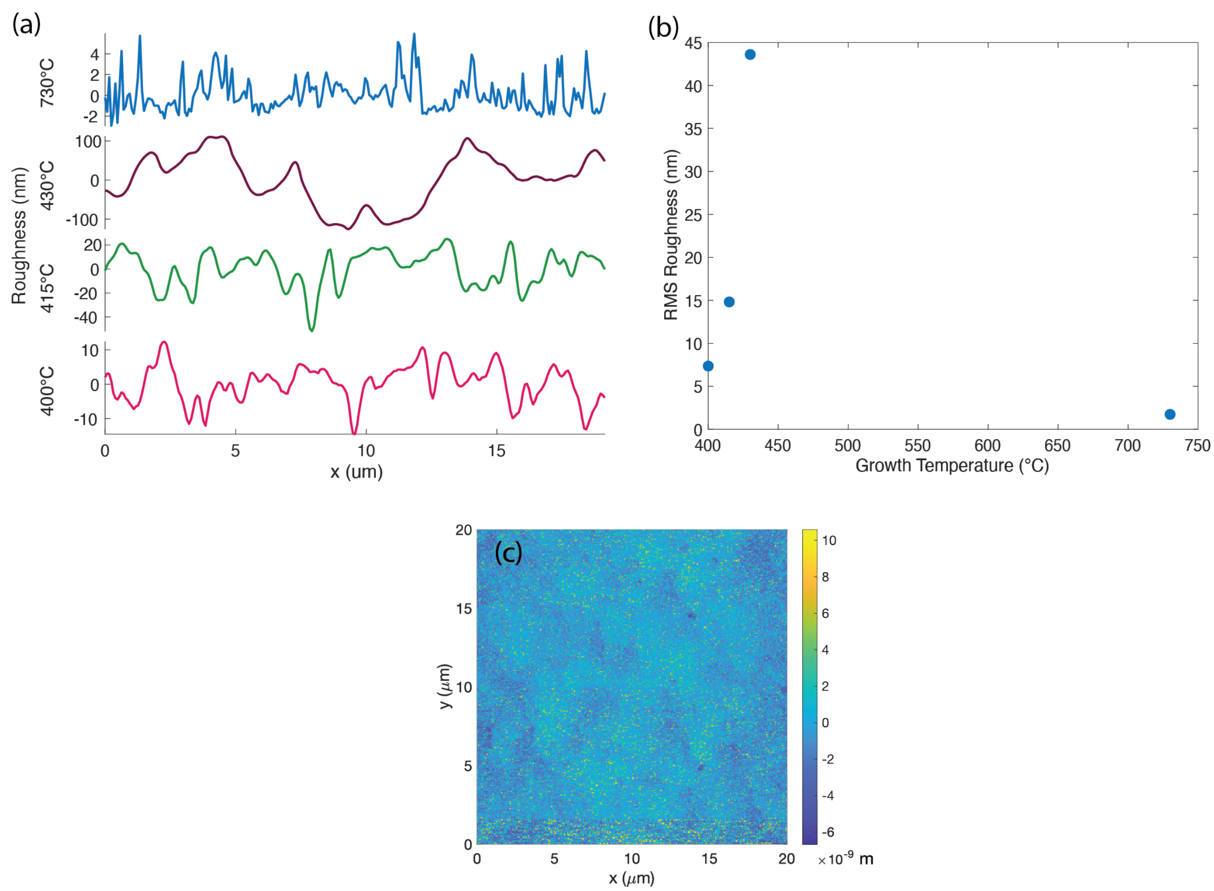


Figure 2-5. (a) Surface roughness profiles and (b) rms roughness measured by AFM for Ge growth temperatures between 400°C and 730°C. 430°C and 730°C growths included a 350°C 60 nm buffer layer prior to the higher temperature growth step. (c) AFM 2D-scan of 730°C growth from surface profile in (a).

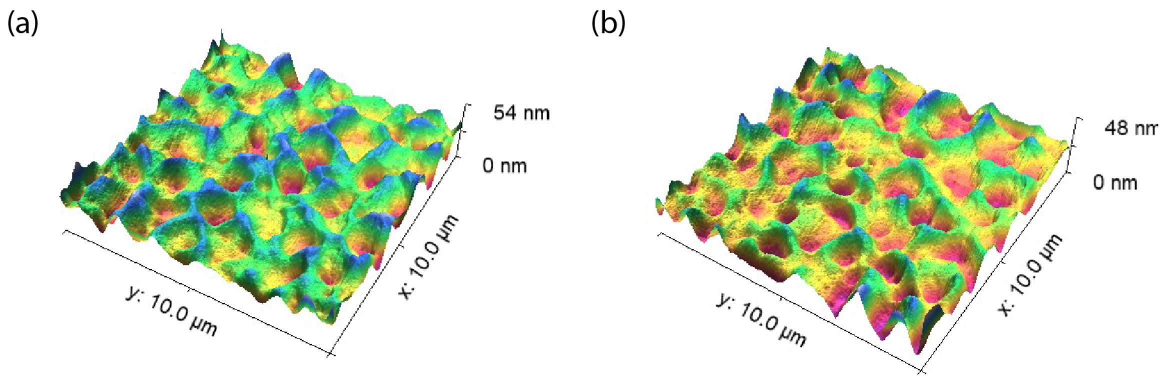


Figure 2-6. AFM 3D plot characterization of (a) Ge grown on Si via a two-step process with a 60 nm buffer growth at 350°C first followed by the 400°C growth step, average roughness 8.5 nm and (b) Ge grown on Si at 400°C in one-step process with average roughness of 9 nm. AFM courtesy of Dr. Danhao Ma.

Growth temperature (°C)	**Buffer**	**Period (nm)**	**Amplitude (nm)**
400	350°C, 60nm	1200	15
415	No	1400	30
430	350°C, 60nm	2500	50

Table 2-1. Average surface pitting period and amplitude for different growth temperatures.

We identified 400°C as the optimal low temperature (LT) growth temperature for BEOL-compatible Ge-on-Si surface roughness due to its lower surface roughness compared to other growth temperatures we explored. Though 400°C is in the germane decomposition limited growth regime (Figure 2-7), the growth rate is much faster than 350°C which is the typical growth temperature for buffer layers and its surface roughness is acceptable for Ge-on-Si photodetector fabrication.

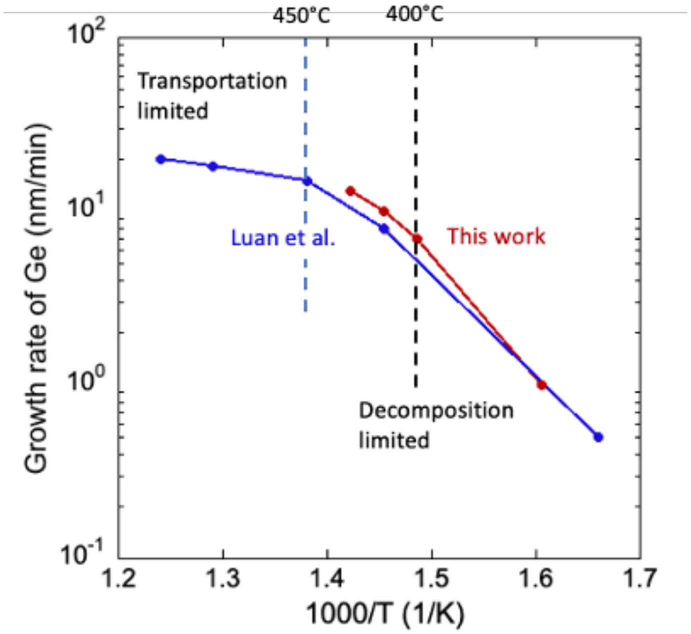


Figure 2-7. Ge-on-Si epitaxial growth regimes where we are limited by the decomposition of germane on the surface up to 450°C and above which we are mass-transport limited. Red data points are from This Work, courtesy of Dr. Ruitao Wen and Dr. Danhao Ma. Blue data points are courtesy of Dr. Hsin-Chiao Luan from prior UHVCVD growths. Figure courtesy of Professor Kazumi Wada.

2.5 Summary and Conclusion

We explored low temperature Ge-on-Si epitaxy and found from XRD that we have single-crystalline Ge at 400°C both with and without a 350°C buffer layer. We determined the optimal growth temperature is 400°C for BEOL-compatible growth due to the lower average surface roughness of 9nm for this growth temperature compared to the much higher roughness of 415°C and 430°C growths. We developed a pre-growth room temperature methanol-iodine clean that results in a specular surface observed via optical microscopy. Together, these two efforts result in a fully BEOL-compatible low thermal budget growth process for Ge-on-Si epitaxy.

We also identified surface morphology resembling pitting that occurs periodically in our Ge-on-Si growths in the decomposition-limited growth regime. The surface morphology was present even in growths that included an initial 60 nm 350°C buffer layer, indicating that the surface roughening was occurring as a kinetics rather than growth-mediated islanding process. It is likely that the surface roughening is indicative of coherent strain relaxation. The periodicity of the surface roughening was higher than we would expect – 10 times higher than the surface diffusion length, ~130 nm for this growth temperature range. We also determined that the 400-430°C growth temperature range is in the decomposition-limited, or surface reaction-limited, rather than the mass transport-limited growth regime. The growth rate is 7nm/min at our ideal growth temperature of 400°C. Due to the lack of impact of a 350°C buffer layer on the surface roughness of the 400°C growth, we decided to proceed with single-step 400°C growths to optimize growth time. While the rms roughness of 400°C-grown Ge-on-Si is 7nm-9nm which is higher than the standard HT Ge rms roughness of 1nm, the surface is still considered sufficiently planar for device processing. The next chapter delves further into the origin of the characteristic LT Ge surface morphology via characterization of LT Ge defects and their interactions upon annealing.

Chapter 3: Effects of annealing on LT Ge materials properties

3.1 Background

Ge's lattice constant is 4.2% larger than that of Si, which leads to misfit dislocation formation to relieve compressive strain during growth. The misfit dislocations generate threading dislocations that extend to the Ge surface along the (111) plane. The lattice misfit, along with Ge's lower surface energy, can also lead to islanding or the Stranski-Krastanov growth mode. Ge grows layer-by-layer until it reaches a critical thickness where island formation occurs. In order to obtain a planar surface, the surface migration length is reduced by growing a lower temperature buffer layer first before the secondary high temperature growth step.[25]

Epitaxial growth of high-quality Ge grown on Si has been studied throughout the last 60 years, with a focus on achieving a planar surface and reducing threading dislocation density (TDD) due to ease of further device integration via a planar surface and the reported negative impact of threading dislocations in the epi layer on device performance. Threading dislocation density reduction to $<10^6$ cm^{-2} has been achieved via growth of a graded Si_xGe_{1-x} buffer, low temperature buffer layer, and/or post-growth annealing.[26] Low temperature (where T<450°C) growth of Ge on

Si has been explored for thin (<50nm) buffer layers at 350°C for a two-step growth process where the second high temperature growth step is at 600-750°C.[27,12] It is well-established that misfit dislocations form at the Ge/Si interface to relieve strain when the buffer layer reaches the critical thickness. These misfit dislocations then have to terminate at a surface, via threading dislocations.[28] Halbwax et al investigated Ge grown at 330°C in-situ via reflection high-energy electron diffraction (RHEED) and found that the in-plane lattice parameter does not fully reach that of bulk Ge. They concluded that the misfit dislocation density they measured via TEM was only partially responsible for the incomplete strain relaxation in the epitaxial Ge layer.[29] Marzegalli et al investigated the prominence of vertical dislocations in selectively-grown Ge on Si at 490°C via low-energy plasma-enhanced chemical vapor deposition (LEPECVD) and found that vertical dislocations are of both screw and edge quality, but mainly of screw quality. They determined that the screw dislocations were formed via interaction of two 60° dislocations, while the edge dislocations were most likely formed from the two residual pairs of threading armers of a Lomer misfit segment at the interface. They also conducted post-growth cyclic annealing between 600°C-835°C and found the 60° and 90° threading dislocation densities were reduced, for 90° dislocations of both screw and edge quality. They conclude that the pure edge vertical dislocations may be annihilated via interaction with other 60° threading dislocations.[30]

Point defect-dislocation interactions have also been studied for conductivity type conversion of plastically deformed bulk Si[31,32,33] and bulk Ge.[34] Plastic relaxation also caused conductivity type conversion in SiGe/Si heterostructures epitaxially grown via UHVCVD and characterized by Grillot et al. They postulate that increasing the compositional grading rate of the epitaxial growth leads to increased residual strain within each graded region which then promotes increased dislocation reactions which generate acceptor-like point defects that are responsible for the type conversion observed after annealing.[35] In this paper, we have developed a single-step low temperature (400°C) BEOL-compatible Ge growth process and characterized dislocations and point defects present in the low-temperature Ge (LT Ge). We have also observed that vertical dislocations play a crucial role in point defect-dislocation interactions responsible for type conversion upon annealing. We have also attempted to understand the nature of residual compressive strain in epitaxially grown Ge on Si via a dislocation interaction energy model. Understanding these defect interactions and correlated strain properties are essential in developing BEOL-compatible interconnects using Ge-on-Si photodetectors.

3.2 Experimental Methods

Ge was grown via UHVCVD) on various Si substrates – p-type doped Si, n-type doped Si, as well as silicon on insulator (SOI). The base pressure of the UHVCVD chamber is 10^{-9} -10^{-8} Torr. Germane (GeH_4) gas at 10 sccm flow is utilized as the Ge source.

In order to analyze materials quality and conductivity type, Hall effect samples were fabricated by cleaving the Ge on SOI wafers into 1.2 x 1.2 cm pieces and patterning metal contacts on the corners via E-beam evaporation. The contact structure consisted of 20 nm of titanium (Ti), followed by 20 nm of titanium nitride (TiN), followed by 300 nm of aluminum (Al). The TiN acts as a diffusion barrier between Al and Ge and the Ti is used as an adhesion layer. Hall effect measurements were conducted using 4-point probes and the van der Pauw technique to measure resistivity and current. A magnet with a 4200 Gauss field was utilized to measure the samples' Hall voltage and thus determine conductivity type.

Transmission Electron Microscopy (TEM), Atomic Force Microscopy (AFM), and X-ray diffraction (XRD) were also utilized to characterize LT Ge's material properties. Samples were prepared for Transmission Electron Microscopy (TEM) by using focused ion beam along the [110] direction to make a sample cross section. Cross-sectional TEM measurements were conducted along the g vector for imaging from the Kikuchi line [220]. X-ray diffraction (XRD) is performed using 1.54 Å Cu X-rays to obtain relevant material peaks and thus information about the material's lattice constant and subsequent strain.

3.3 Results and Discussion

3.3.1 Origin of strain relaxation

Due to the 4.2% lattice mismatch between Ge on Si, Ge is compressively strained during growth. The volume strain energy is relieved either via dislocations when the film thickness is above the critical thickness, or via islands if the temperature is high enough for Ge atoms to diffuse across the surface faster than the adatom arrival rate. Ge and Si also have different thermal expansion coefficients which leads to a thermal shear stress experienced by Ge upon cooling, resulting in a build-up of tensile strain in the Ge. From XRD, we can calculate the perpendicular strain in the film:

$$\epsilon_{\perp} = \frac{d_{400}^{film} - d_{400}^{bulk}}{d_{400}^{bulk}} \tag{3-1}$$

where d_{400}^{film} is the spacing of the Ge(400) planes obtained from the $K\alpha_1$ and d_{400}^{bulk} is the spacing between bulk Ge(400) planes, 0.14 nm. The film is under biaxial stress so the perpendicular strain is related to the in-plane strain via:

$$\epsilon_{\parallel} = \epsilon_{\perp} * -\frac{C_{11}}{2C_{12}} \tag{3-2}$$

where $C_{11} = 126.5\ GPa$ and $C_{12} = 48.3\ GPa$ are Ge's elastic constants.[36] The theoretical thermal shear strain that builds up upon cooling is given by:

$$\epsilon_{\parallel} = \int_{T_0}^{T_1} [\alpha_{Ge}(T) - \alpha_{Si}(T)] dT \qquad (3\text{-}3)$$

where T_1 is the initial growth/annealing temperature and T_0 is the final temperature after cooling (room temperature), and $\alpha_{Ge}(T)$ and $\alpha_{Si}(T)$ are the thermal expansion coefficients of Ge and Si at a temperature T $(°C)$ and are given by:

$$\alpha_{Ge}(T) = 6.050 * 10^{-6} + 3.60 * 10^{-9}T - 0.35 * 10^{-12}T^2 \ (°C^{-1}) \qquad (3\text{-}4)$$

$$\alpha_{Si}(T) = \begin{pmatrix} 3.725 * \{1 - \exp[-5.88 * 10^{-3}(T + 149.15)]\} + \\ 5.548 * 10^{-4}T \end{pmatrix} * 10^{-6} \ (°C^{-1}) \qquad (3\text{-}5)$$

in the 25-900°C range.

The theoretical thermal shear strain from cooling from various growth/annealing temperatures in the 400-800°C range is compared to the experimentally measured strain from XRD in Figure 3-1a. We found that Ge grown at low temperatures of 400-430°C (LT Ge) exhibited very low strain, 0.05%. We compared these low temperature Ge-on-Si growths to high temperature (600°C+) growths we conducted in This work as well as high temperatures growths conducted previously in our group by Cannon et al[37] and Liu et al[7] via UHVCVD. The strain expected from growth at in this temperature range is shown by the dashed line in Figure 3-1a, while the dotted line shows the strain expected from annealing in the various temperature regimes. The solid line shows the expected tensile strain from thermal expansion alone. The deviation from theoretical tensile train is about 0.086% for the entire range of growth temperatures from 400-750°C. The blue shaded region I is the region of LT Growth where strain remains a consistent residual compressive strain away from the expected tensile strain. This deviation is also consistent for annealed films at annealing temperatures above ~650°C, in the Dislocation Recovery regime, or region II, the orange shaded region. However, in region II (the Point Defect Recovery regime, or purple shaded region), the Ge film's strain increases at a much lower rate with increasing temperature due to lack of dislocation motion in this range, or lack of plastic relaxation. Due to the elastic nature of the film's strain in this regime, the tensile strain in the annealed films stays mostly the same and doesn't reach the equilibrium value would during growth conditions. The slight increase in tensile strain, or slight reduction in residual compressive strain, could be due to slight dislocation recovery via the point defect-mediated climb process we describe in section 3.3.2. Point defect-dislocation interactions.

The residual compressive strain that remains in the material leads to a constant increase in tensile strain that matches the slope of the expected tensile train from thermal expansion, until plateauing at around 800°C. Previously, this deviation from the expected tensile strain, or additional compressive strain, was attributed to relaxation from dislocation reduction. Dislocation reduction occurs because 60°C threading dislocations that exist along the (111) plane glide under the thermal shear stress that is also along the [111] direction. These dislocations either glide to a surface to annihilate or participate in dislocation interactions that result in annihilation. The dislocation glide velocity is given by:

$$v_{dislocation} = v_0 * \sigma_{exc} * \exp\left(\frac{-E_v}{kT}\right) \tag{3-6}$$

where $v_0 = 3 * 10^{-3} \frac{m^2 s}{kg}$, σ_{exc} is the excess stress that drives dislocation motion, $E_v = 1.6\ eV$ is the energy barrier to dislocation glide[38], k is the Boltzmann constant, and T is the temperature in Kelvin. In our case, the excess stress is the thermal shear stress minus the dislocation line tension stress. However, since the dislocation line stress is much smaller compared to the thermal stress, we can approximate the excess stress as just the thermal shear stress given by:

$$\sigma_{exc} = \int_{T_0}^{T_1} (\alpha_{Ge} - \alpha_{Si}) dT * \frac{2 * (1 + \upsilon)}{(1 - \upsilon)} * \mu \tag{3-7}$$

where $\upsilon = 0.25$ is the Poisson ratio in Ge epitaxial films[39] and $\mu = 102.7\ GPa$ is the Young's modulus.[40]

In Figure 3-1b we plot the dislocation glide velocity calculated via (3-6) and (3-7) that occurs during cooling from a growth/annealing temperature T_1. We take 60 minutes as the typical duration of the glide process during cooling to compute the estimated distance a 60° dislocation could travel based on its glide velocity. We find that below 630°C, dislocations do not have sufficient energy to glide the distances that would allow them to participate in dislocation interactions resulting in annihilation, ~13 nm (dislocation separation and dislocation density are covered in more detail in section 3.3.2. Point defect-dislocation interactions). Thus, the deviation from the expected tensile strain must be due to residual compressive strain during growth for our low-temperature grown Ge. Halbwax et al grew Ge at similarly low temperatures, 330°C-350°C, via UHVCVD. The in-plane lattice parameter for their Ge was lower than the lattice parameter of bulk Ge at 330°C, indicating a residual compressive strain in their material as well. Halbwax et al also observe a larger dislocation separation via TEM in the <110> direction than the dislocation separation necessary for complete relaxation. This led them to conclude that the relaxation of the Ge film was only partially mitigated by misfit dislocation nucleation.[29] We observe that the dislocation separation via TEM for our as-grown LT Ge is ~13nm in the <110> direction, much larger than the 9.5nm we calculate is necessary for complete relaxation. The misfit dislocation density for complete relaxation would be about 10^{12} cm^{-2} while the threading dislocation densities we measure for LT Ge (see Table 3-1) are on the order of 10^{11} cm^{-2}. Since we have about two threads for every one misfit dislocation, we can approximate our misfit dislocation density as 0.5-1*10^{11} cm^{-2}. As we cannot explain the observed lower dislocation density dislocation by relaxation mechanisms and dislocation annihilation that occur at higher temperatures, we plot the stress field of the misfit dislocations to see if dislocation repulsions are relevant at these misfit dislocation separation distances.

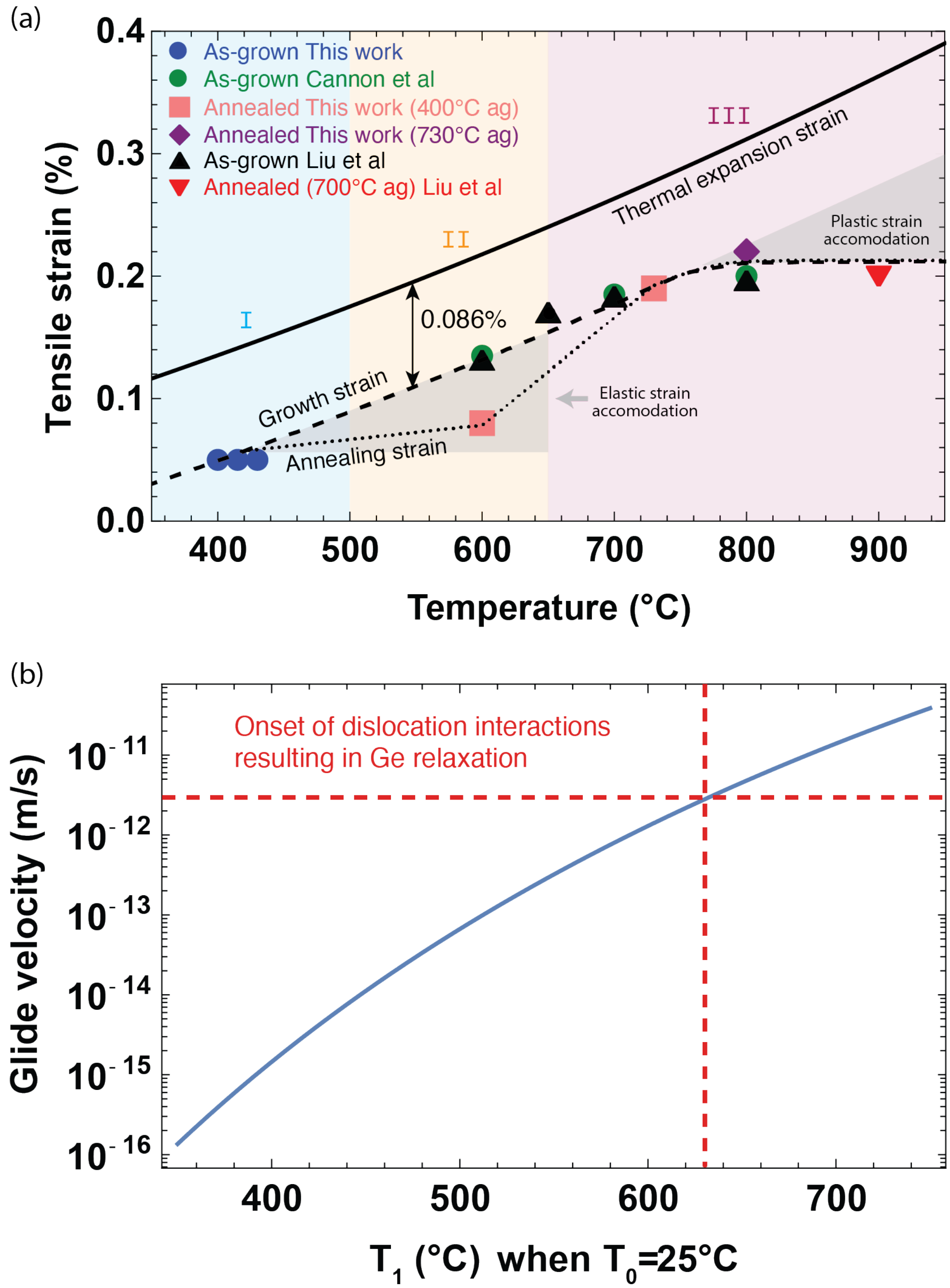


Figure 3-1. (a) Strain measured from X-ray diffraction for Ge on Si grown at temperatures from 400-800°C from This work as well as Cannon et al[37] and Liu et al[7] and for LT and HT Ge annealed at temperatures ranging from 600-800°C. The solid line shows the calculated tensile strain due to

thermal expansion. The blue shaded region I is LT Growth regime, the orange shaded region II is the Point Defect Recovery regime, and region III is the Dislocation Recovery regime. The dashed line follows strain vs growth temperature and the dotted line follows strain vs annealing temperature. The grey regions exhibit elastic strain recovery in the low temperature region and plastic strain build-up in the high temperature region. XRD courtesy of Dr. Ruitao Wen. (b) Dislocation glide velocity vs growth/annealing temperature in Ge on Si resulting from the thermal shear stress associated with cooling to room temperature. The dashed red line shows the dislocation glide velocity necessary to result for dislocation interactions to result in relaxation.

Misfit dislocations in Ge-on-Si have been reported to be pure edge dislocations.[41] Taking two edge dislocations that are parallel two each other at the Ge-on-Si interface, their burger's vectors are both:

$$\vec{b}_1 = \vec{b}_2 = \frac{a}{2}[110]$$

and line vectors are both:

$$\vec{\xi}_1 = \vec{\xi}_2 = [001]$$

We can then compute the radial stress field exerted from one dislocation at (r, θ) on another at (0,0) via:

$$\sigma_{rr} = \frac{Gb\sin\theta}{2\pi(1-\nu)r} \tag{3-8}$$

where G is the shear modulus in Ge, $4.1*10^{10}$ Pa[40], and b is 4 Å, the magnitude of the burger's vector in Ge.[13] From Figure 3-2a we can see that the stress at the dislocation separation observed via TEM (indicated by the red dashed line) is ~70% of the stress at the dislocation separation for total relaxation, 9.5 nm, indicated by the green dashed line.

In order to explore our residual compressive strain, we model the added kinetic barrier to misfit dislocation nucleation that dislocation interactions present. From the Matthews' model for homogeneous half-loop nucleation[8], the total energy balance of the system consists of the half-loop formation energy $E_{half-loop}$ which costs the system energy, and the strain energy E_{ϵ} and energy of the step E_{step} which is formed/removed when the half-loop is nucleated at the surface which reduce the system's energy:

$$E = E_{half-loop} - \left(E_{\epsilon} + E_{step}\right) \tag{3-9}$$

To see if our theory of dislocation repulsion results in lower MD density is valid, we add the dislocation loop interaction energy to this balance:

$$\begin{aligned} E = \left[\frac{GbR}{8(1-\upsilon)}\right] * [b * (2-\upsilon) * \ln\left(8 * \eta * \frac{R}{e^2 b}\right) \\ -8\pi R\epsilon(1+\upsilon)\cos\lambda\cos\phi - 2b(1-\upsilon)\sin\alpha\,] + E_{loop\ interaction} \end{aligned} \tag{3-10}$$

where R is the half-loop radius, $\eta = 4$ for most diamond structures, ϵ is the misfit strain, α is the angle between the Burger's vector and dislocation line vector, which we take as 90° for complete edge dislocations, and $\cos\lambda\cos\phi$ (or the Shmid factor) is a geometric factor where $\cos\lambda = 1/2$ and $\cos\phi = \sqrt{2/3}$.[8] We follow Hirth and Lothe's model[13] for dislocation loop interaction energy:

$$For\ z{\sim}a{:}\ E_{loop\ interaction} = \frac{Gb^2 rk}{1-\nu}(K-E) \tag{3-11}$$

where E and K are the elliptical integrals:

$$\mathrm{E} = \int_0^{\pi/2} \sqrt{(1-k^2\sin\eta^2)}\,d\eta \tag{3-12}$$

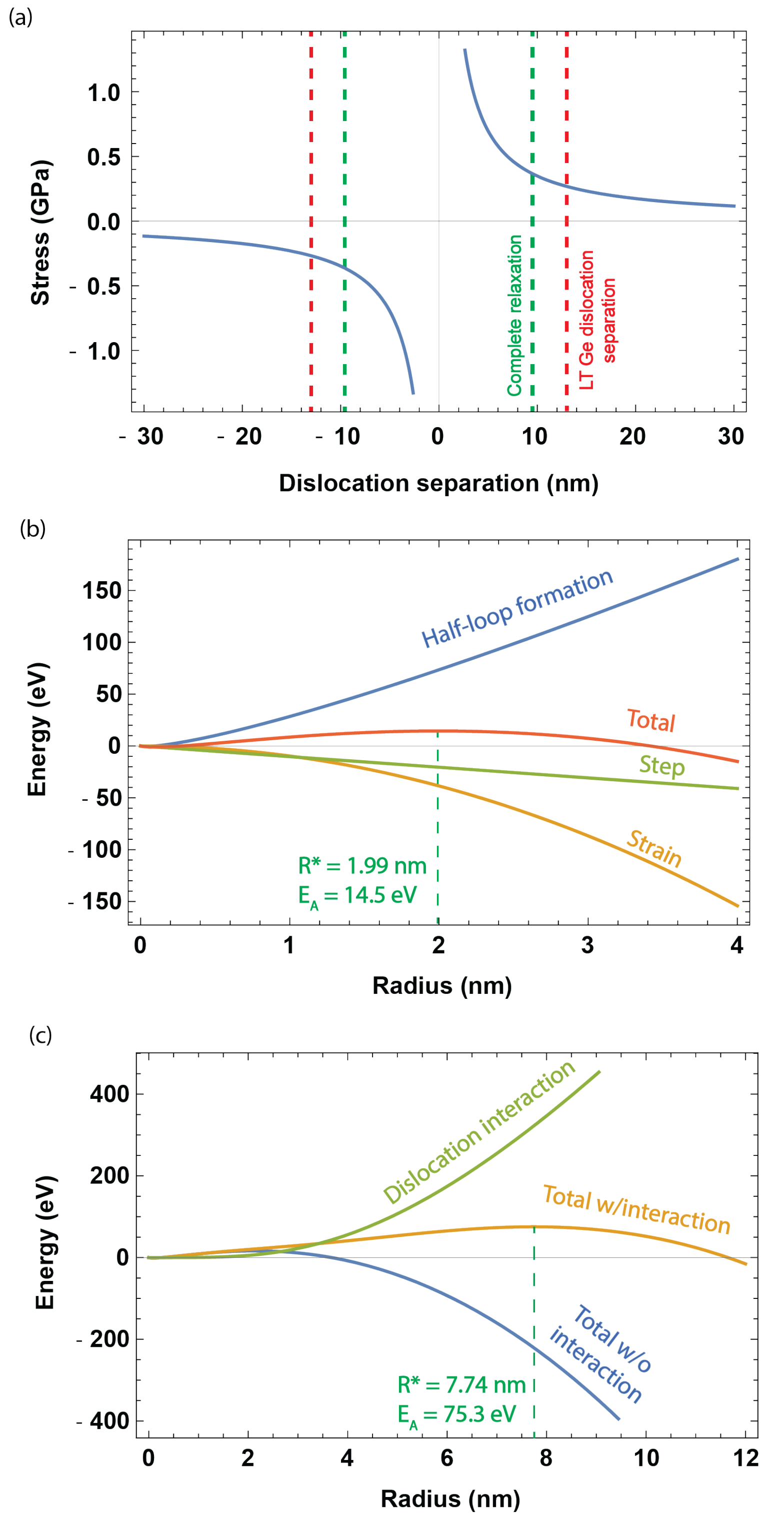


Figure 3-2. Radial component of misfit dislocation stress field based on dislocation separation. The green dashed line indicates the stress field at the dislocation separation necessary for

complete relaxation of the lattice mismatch strain in Ge on Si, and the red dashed line indicates the existing dislocation separation measured via TEM in as-grown LT Ge on Si. (b) Misfit dislocation nucleation energy vs dislocation half-loop radius with relevant energy contributions (c) Misfit dislocation formation energy including kinetics correction of dislocation loop interactions compared to formation energy without interaction energy.

$$\mathrm{K} = \int_0^{\pi/2} \frac{1}{\sqrt{(1 - k^2 \sin \eta^2)}} d\eta \tag{3-13}$$

and k is given by the relation:

$$k^2 = \frac{4a^2}{z^2 + 4a^2} \tag{3-14}$$

where z is the distance between the two half-loops, which we calculate as 9.48 nm (the distance between dislocations in the [110] direction for total relaxation at 4.4% lattice strain which is present at 400°C), and a is the half-loop radius. In order to find the activation energy for half-loop nucleation E^* and the critical loop radius for surface nucleation, R^*, we have to maximize E with respect to R:

$$E^* = E(R^*) \tag{3-15}$$

We obtain that the critical radius for half-loop nucleation is 1.99 nm without dislocation loop interaction energy and the misfit dislocation formation energy is 14.5 eV (Figure 3-2b)). This is comparable to previous reports of modeled nucleation energy.[42] When we add the dislocation interaction energy, the critical radius for nucleation is pushed out further to 7.74 nm and the activation energy for misfit dislocation nucleation is much higher, 75.3 eV (Figure 3-2c). This is evidence that dislocation interactions at the distances relevant for complete relaxation in Ge-on-Si do affect misfit dislocation nucleation. However, both 14.5eV and 75.3 eV are activation energies far above kT – or the thermal energy that is available to the system, 0.058 eV. So where is all of this energy coming from to even be able to nucleate misfit dislocations homogeneously at all? The energy contributions to the nucleation energy balance are all on the right order of magnitude to contribute (for example – strain energy alone easily reaches 100s of eVs as dislocation radius increases, see figure 3-2b). Dong et al conducted molecular dynamics simulations that demonstrate misfit dislocations form via surface roughening that allows very high strain concentrations to arise and thus nucleate dislocations in those regions.[43] Tersoff and LeGoues similarly show that for high misfit epitaxy, nucleation of surface pits lowers the nucleation energy barrier for misfit dislocation nucleation.[44] This implies that strain concentrations at the surface can lower these barriers to misfit dislocation nucleation, thus allowing us to overcome the large energy contributions required for homogeneous misfit dislocation nucleation.

We can quantify these strain fluctuations by evaluating the strain necessary to overcome the lower activation energy barrier to the level of the thermal energy available to us during growth. We take the strain contribution to the total energy balance and set it equal to the strain energy at the critical radius minus the additional activation energy it needs to overcome:

$$\begin{aligned} E_\epsilon &= \left[\frac{GbR}{8(1-\upsilon)}\right] * \ 8\pi R\epsilon(1+\upsilon)\cos\lambda\cos\phi \\ &= E_\epsilon(7.74\text{nm}) - E_A(7.74nm) \\ &= -550 - 75.3\text{ eV} \end{aligned} \tag{3-16}$$

We can then solve for the strain value, ϵ that satisfies this equation, 4.77%. Thus, strain fluctuations of 0.37% higher than the lattice mismatch at 400°C, 4.4%, are required to nucleate misfit dislocations. Failure to reach such high strain energy fluctuations as dislocations are nucleated and strain energy is relieved is likely we see a residual 0.086% compressive strain in Ge-on-Si films throughout the 400-800°C growth temperature range. The periodic pitting detailed in Chapter 2 that we observed for our LT Ge-on-Si epitaxy is further evidence of misfit dislocation nucleation mediated via surface roughening-induced strain fluctuations.

At growth/annealing temperatures over 630°C, the strain deviation is due to either residual compressive strain during growth or relaxation. The constant deviation in the entirety of regime I and II implies that either the deviation is from the same source – residual compressive strain during growth, or that there is a maximum compressive strain offset that is energetically favorable for Ge-on-Si films. We observe a plateau in regime III at around 800°C of 0.2% strain, where dislocation relaxation mechanisms are highly prevalent, resulting in plastic deformation that brings down the tensile strain as more dislocations annihilate.

3.3.2 Point defect-dislocation interactions at low temperatures

We utilized TEM to explore the dislocation microstructure in 1um-thick LT Ge. We find that as-grown (ag) LT Ge has a total threading dislocation density of ~10^{11} cm^{-2}. At low temperatures of 400°C, we do not expect any dislocation reduction via glide so the fully relaxed layer should maintain a threading dislocation density close to that necessary to fully relieve the lattice mismatch strain between Ge and Si, ~10^{12} cm^{-2} misfit dislocation density (MDD) thus, twice as large threading dislocation density (TDD), ~10^{12} cm^{-2}. Figure 3-3 shows cross-sectional TEM images of LT Ge ag (a), and LT Ge annealed at 500°C (b), 600°C (c), and 730°C (d), all for 60 minutes in vacuum. Table 3-1 provides TDDs calculated from the cross-sectional TEM images and rounded to the nearest order of magnitude. Any dislocation that occurs in the bulk of the material is considered a threading dislocation here, not solely the dislocations that thread all the way to the surface. This is so that we can approximate the misfit dislocation density at the interface as closely as possible. As-grown LT Ge has 60° threading dislocations making up the majority of the dislocations and 90° threading dislocations in ~10^{10} cm^{-2} density. Annealing at 500°C for 60 minutes results in similar microstructure with both 60° and 90° dislocations present in similar densities to the ag sample. Annealing at 600°C shows a drastically different microsctructure, with mainly 60° threading dislocations present in the same density as the ag sample and a big reduction in 90° threading dislocations to ~10^{8} cm^{-2}. Annealing at 730°C results in 0 visible 90° dislocations, meaning 90° dislocation density is lower than the TEM detection limit (10^{6}-10^{7} cm^{-2}). Total threading dislocation density for the 730°C annealed sample was ~10^{8} cm^{-2}, an order of magnitude higher than the sample grown via the standard two-step high temperature (HT) process at 730°C followed by post-growth annealing at 800°C and 650°C.

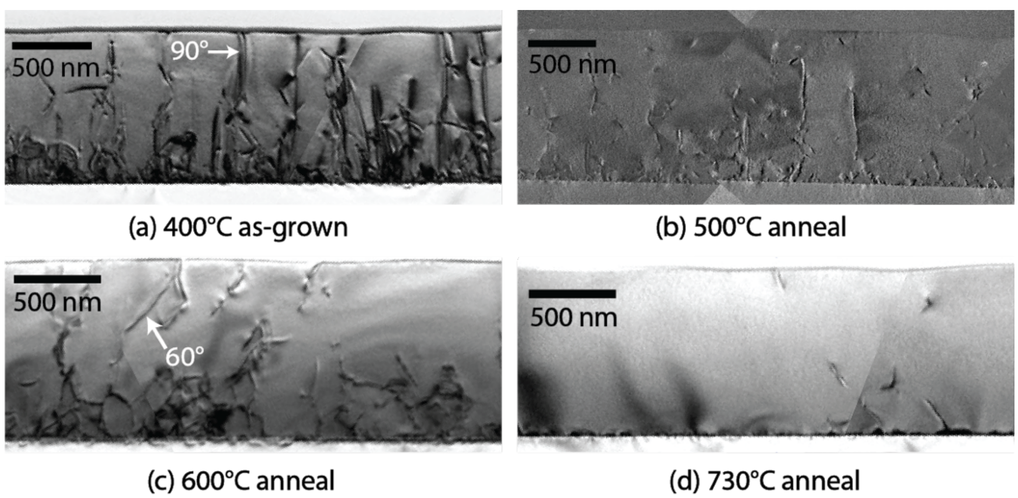


Figure 3-3. Characteristic dislocation distribution in LT Ge as-grown (a) and after post-growth 60 minute anneals at (b) 500°C, (c) 600°C, and (d) 730°C. Vertical dislocations, or 90° dislocations as well as 60° threading dislocations are highlighted in yellow in (a) and (c). TEM courtesy of Dr. Baoming Wang.

From g•b analysis (see Figure 3-4) of the as-grown LT Ge sample, we see that some of the 90° dislocations appear only in cross-sections imaged along the [220] direction, but don't appear in the same cross-section imaged along the [004] direction. This means that some of the 90° dislocations are pure edge dislocations with Burgers vector in the <110> direction.[30] These pure edge dislocations are sessile dislocations that cannot glide under the thermal shear stress that occurs upon cooling.[45]

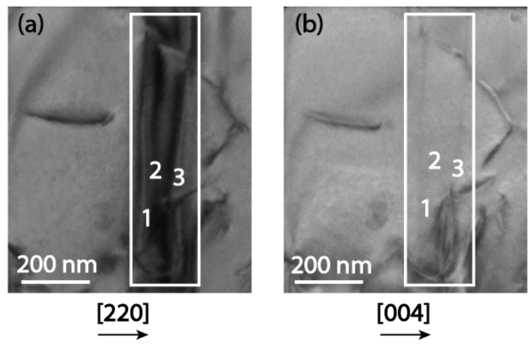


Figure 3-4. LT Ge as-grown cross-section imaged along the (a) [220] and (b) [004] direction for g•b analysis. Vertical dislocations 1, 2, and 3 appear in the [220] cross-section but 1 and 3 only appear very faintly in the [004] cross-section and 2 is completely absent, indicating 1, 2, and 3 are pure edge dislocations with Burgers vectors parallel to the <110> direction. TEM courtesy of Dr. Baoming Wang.

Sample growth T (°C) + annealing T (°C) (mins)	**60 ° TDD (cm^{-2})**	**90 ° TDD (cm^{-2})**	**Total TDD (cm^{-2})**
400ag	10^{11}	10^{10}	10^{11}
400ag + 500 (60)	10^{11}	10^{10}	10^{11}
400ag + 600 (60)	10^{11}	10^{8}	10^{11}
400ag + 730 (60)	10^{8}	$<10^{6}$	10^{8}
730ag + 800 (60) + 650 (60)	10^{7}	10^{7}	10^{7}

Table 3-1. 60°, 90°, and total threading dislocation density in as-grown and annealed 1 um thick Ge on Si samples measured via cross-sectional TEM.

A possible mechanism for the removal of sessile edge dislocations is that a 60° glissile dislocation will glide to a sessile dislocation and produce another glissile dislocation via:

$$\frac{a}{2}[1\bar{1}0] + \frac{a}{2}[01\bar{1}] = \frac{a}{2}[10\bar{1}] \tag{3-17}$$

and then the glissile dislocation that is formed can glide to a surface or another dislocation to annihilate.[45] However, we see a significant reduction in dislocation density at 600°C and lower, where dislocation glide is insufficient to result in dislocation interactions that would result in annihilation as shown in Figure 3-1b. From TEM, we see that the typical dislocation separation for as-grown LT Ge is ~30nm. We need to anneal to at least 630°C to be able to achieve 10nm of dislocation glide in 60 minutes, which is a conservative estimate for our 30nm dislocation separation. We assume dislocation interactions as our main method of dislocation annihilation because gliding to a surface would require even longer times or large glide velocities in our 500nm – 1um thick films. Since these sessile vertical edge dislocations cannot glide and other 60° dislocations cannot glide to them, we must be relying on climb in order to remove these vertical dislocations.

We conducted both 60-minute isochronal anneals and isothermal anneals at 450°C to characterize LT Ge's material properties and explore the bounds of its performance potential. Hall effect measurements were conducted to obtain carrier density and mobility. As-grown LT Ge exhibited p-type conductivity with carrier density of $3.0*10^{16}$ cm^{-3} ± $0.7*10^{16}$ cm^{-3}, despite being undoped. Annealing at 500°C for 60 minutes consistently results in n-type conversion for 500nm-thick films. This type conversion and the temperature range at which it occurs implies we either 1) start off with acceptor-like point defects resulting in p-type compensation that are then annihilated upon annealing, or 2) we have a donor-like point defect forming upon annealing. It is important to note that both case 1 and 2 will result in the same analysis of point defect energetics because we are either tracking removed p-type carriers or added n-type carriers. Moving forward, we assume case 1 and that any reduction in p-type carrier density or increase in n-type carrier density is the density of removed acceptors. By making this assumption in our analysis, any p-type carrier density measured is interpreted as p-type carrier compensation of the amount measured added to the underlying n-type conductivity. Thus, the carrier density we track in our analysis of annealing energetics is:

1) If the annealed sample exhibits p-type conductivity, the removed acceptor density is the sample's p-type density substracted from the initial p-type density :

$$N_{removed}^{p} = N_0^{p} - N_{sample}^{p} \tag{3-18}$$

or 2) If the annealed sample exhibits n-type conductivity, the removed acceptor density is the sample's n-type carrier density added to the initial p-type density :

$$N_{removed}^{p} = N_0^{p} + N_{sample}^{n} \tag{3-19}$$

The change in acceptor density follows the Arrhenius behavior:

$$dN = N_0 * dt * \nu * \exp\left(\frac{-E_a}{k_B T}\right) \tag{3-20}$$

where N_0 is the underlying donor density in our assumed case of acceptor removal, ν is the exponential pre-factor, E_a is the point defect reaction's activation energy, k_B is the Boltzmann constant and T is the temperature in Kelvin. The exponential pre-fractor ν can be expressed as the jump attempt frequency divided by the number of jumps it takes a point defect to reach a sink to annihilate[46]:

$$\nu = \frac{\nu_{attempt}}{N} \tag{3-21}$$

The annealing rate is thus:

$$\frac{\left(\frac{dN}{N_0}\right)}{dt} = \nu * \exp\left(\frac{-E_a}{k_B T}\right) \tag{3-22}$$

By plotting the annealing rate vs 1/T in the Arrhenius plot in Figure 3-5a, we utilize Matlab to fit an activation energy of 1.7 eV for the point defect reaction we observe. We also find that the line of best fit yields a pre-factor of a $2.3*10^{7}$ s^{-1}. The Debye frequency in Ge at room temperature is $7.5*10^{12}$ s^{-1} [47] and the expected number of jumps through the 500nm – 1000nm thickness of the material is ~1000. This would result in an expected pre-factor of ~10^{9} s^{-1} if the point defect reaction was a single jump event. The prefactor we fit is two orders of magnitude lower, which is consistent with a diffusion-limited process.[46] We also conducted isothermal anneals at 450°C and plotted the acceptor fraction as it changed with time (see Figure 3-5b), assuming the same acceptor removal process we assumed for isothermal annealing analysis. Utilizing Matlab, we fit the isothermal anneals to a time constant of 0.0033 min^{-1} with remarkably good fit at 450°C, which is further indication that the acceptor removal process follows first order kinetics.

When Ge is grown in far-from-equilibrium conditions, such as LT growth of 400°C, supersaturations of point defects are readily present in Ge. A supersaturation of vacancies may be the acceptor-like point defect we tracked via Hall effect. Further evidence for acceptor-like point defects being present in our LT Ge is the removal of vertical dislocations via climb because climb is a point-defect mediated process.[48] We also see a slight increase in tensile strain (or increase in residual compressive strain) upon annealing in the point defect recovery regime, regime II, in Figure 3-1a. We estimate that this is the maximum strain recovery possible from point defect/dislocation interactions prevalent in the point defect recovery regime. The difference in compressive strain is about 0.03%. The amount of acceptors per Ge atom is around ~10^{-6} cm^{-2}. This means that each acceptor would have to relieve ~$5*10^{-10}$ % of strain to account for the extra

0.03% compressive strain recovery. Otherwise, the annealing strain stays largely the same in the point defect recovery regime due to the lack of dislocation motion in this temperature range. Since there is no dislocation motion induced by the thermal shear stress, the material snaps back to its original equilibrated strain state as dictated by the original dislocation microstructure.

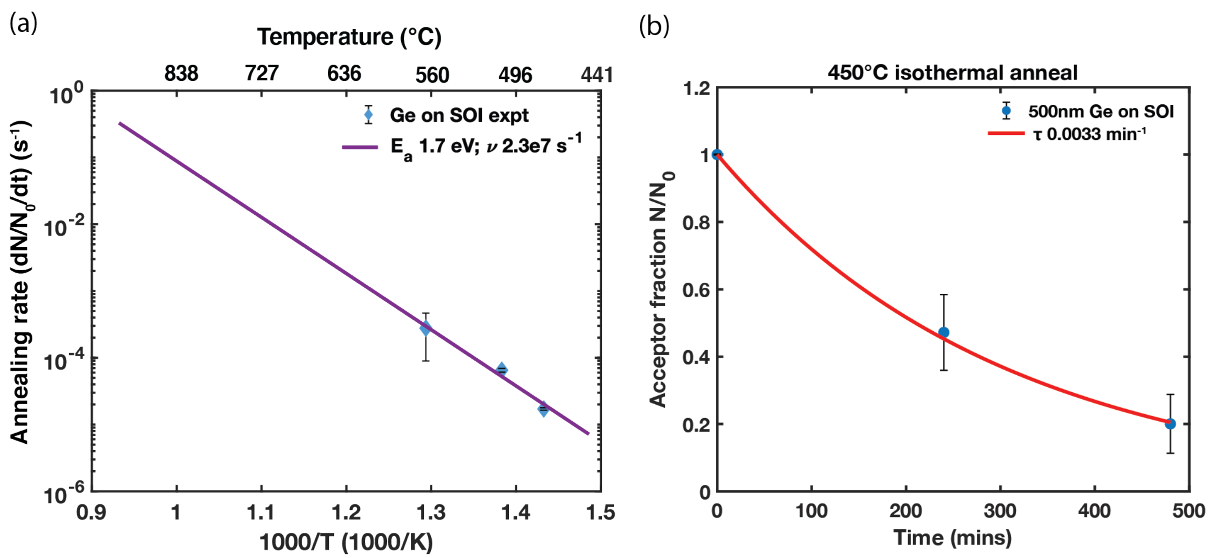


Figure 3-5. (a) Effect of 60 minute isochronal anneals on acceptor annealing rate in LT Ge. (b) Effect of 450°C isothermal anneals on acceptor fraction in LT Ge.

3.4 Summary and Conclusion

We have developed a BEOL-compatible Ge-on-Si epitaixal growth process, incorporating a pre-growth Si clean that can be conducted at room temperature to achieve specular, planar Ge films. We have found that 0.086% residual compressive strain remains in epitiaxially grown Ge which is consistent in our LT Ge growth range. We propose a new model for misfit dislocation formation that includes dislocation interaction energy. We find that inlcuding dislocation repulsion interactions hinders misfit dislocation nucleation and increases the critical half-loop radii and activation energy from 1.99 nm to 7.74 nm and 14.5 eV to 75.3 eV, respectively. We then utilized our new model for misfit dislocation formation to determine that strain fluctuations of 4.77% are required to overcome the barrier to MD nucleation with the thermal energy available at the growth temperature. These strain fluctations are correlated to the surface pitting morphology we observe for LT Ge both with and without a buffer layer. As Dong et al[43] show, strain concentrates in the bottom of pits in order to nucleat misfit dislocations. The observed residual compressive strain, incomplete relaxation in Ge observed via TEM, and the misfit dislocation energy hindered by dislocation repulsion are all evidence that the pitting surface morphology of LT Ge is necessary for misfit dislocations formed via strain concentrations at the bottom of the pit.

By comparing our LT Ge with Ge also grown via UVHCVD and annealed at higher temperatures, we have identified three annealing regimes: I) the LT Growth regime, II) the Point Defect Recovery regime, and III) the Dislocation Recovery regime. The growth strain follows the thermal expansion strain with residual compressive strain offset until reaching the plastic strain accomodation regime above 800°C where tensile strain plateaus at ~0.2% due to rapid dislocation annihiltion at these temperatures. We identifed the elastic strain accomdation regime for annealing temperatures lower than 650°C. Tensile strain is frozen in at the growth temperature in this regime due to limited dislocation mobiltiy and virtually no dislocation interactions to accommodate the extra tensile strain.

We utilized Hall effect to identify undoped Ge intiially exhibits p-type conductivity with carrier density in the 10^{16} cm^{-3} range. We observe conductivity type conversion occurs in our materials upon post-growth anneals between 450-500°C with an activation energy of 1.7 eV. Due to the low base pressure of UHVCVD, it is unlikely we have interstitial-like point defects. It is far more likely that we have acceptor-like point defects, or vacancies, that are favorable at low temperatures as they relieve the residual compressive strain. This has revealed that LT Ge has an n-type background rather than a p-type background as was previously believed. Upon annealing, the lower dislocation density of pure edge and mixed character 90° dislocations in the temperature range for point defect removal is evidence that the acceptor-like defects are engaging in climb of 90° dislocation. Understanding these defect interactions and the effects of low temprature growth and post-growth annealing on the strain characteristics of Ge is key to optimizing Ge-on-Si photodetectos for BEOL-compatible optical interconnects. The next chapter focuses on the impact of annealing and processing conditions on Ge-on-Si photodector device performance.

Chapter 4: Effects of annealing and processing conditions on LT Ge photodetector device performance

4.1 Operating principles of photodetectors

Types of photodetectors include pn photodiodes, p-i-n photodetectors, avalanche photodiodes, and metal-semiconductor-metal photodiodes. Our main goal is to probe the electrical properties of LT Ge in the near IR, thus we want to maximize Ge's absorption. We will start by analyzing a pn junction device of p-Ge on n-Si and then move towards a remote Si pn junction with a Ge absorber in order to probe heterostructure properties further.

A space charge region is formed when a p-type and n-type material are brought together because of a "lack" of free carriers left over when electrons and holes diffuse across the junction to maintain equilibrium. This space charge region is referred to as the depletion region, the region over which an electric field is present due to "depleted" free carrier. This electric field is what causes drift current in pn junctions as electrons from the p-side are swept across the junction by drift to the n-side and holes from the n-side are swept up the diffusion gradient to the p-side. A photon with incident energy greater than the band gap energy will be absorbed and an electron is excited from the valence band to the conduction band, leaving behind a hole. This is called the process of generation and is responsible for photocurrent in semiconductors. The generation rate of carriers[49] is given by:

$$\Delta G(x) = (1 - R)\left(\frac{P_0}{Ah\nu}\right)e^{-\alpha x} \tag{4-1}$$

where R is the reflectivity at the surface of the semiconductor material, P_0 is the incident photon power, A is the area the light is incident on, ν is the frequency of the incident light, and α is the absorption coefficient of the material. When a carrier is generated in the depletion region, the electron-hole pair immediately splits and is swept to the edge of the depletion region due to the electric field, and is thus collected and measured at the contacts as drift photocurrent. The drift photocurrent across a junction is given by:

$$J_{drift} = -q\int_0^W \Delta G(x)dx = -q\,(1 - R)\left(\frac{P_0}{Ah\nu}\right)\int_0^W e^{-\alpha x}dx \tag{4-2}$$

where q is the charge of an electron, W is the width of the depletion region for the absorbing material.[50] When a carrier is generated outside of the depletion region in the quasi-neutral region, it must be within a diffusion length of the depletion region in order to be split and felt by the electric field and thus collected by the contacts. The total diffusion photocurrent is given by:

$$J_{diffusion} = q\left[\frac{D_n}{L_n}N_{p_0} + \frac{D_p}{L_p}P_{n_0}\right]\left(e^{\frac{qV_A}{k_BT}} - 1\right) - q\Delta G\left(L_n + L_p\right) \tag{4-3}$$

where $D_{n,p}$ is the diffusivity of electrons/holes in the p-type material, $L_{n,p}$ is the diffusion length of electrons/holes in the p/n-type region, N_{p_0} is the concentration of minority carriers in the p-type region, P_{n_0} is the concentration of minority carriers in the n-type region, V_A is the applied voltage, $x_{p,n}$ is the depletion width on the p/n-type side, ΔG is the generation rate, and $\tau_{n,p}$ is the respective minority carrier lifetime.[50]

A device similar to our remote heterojunction structure is the separate-absorption-multiplication avalanche photodiode (SAM APD). An APD works through the process of impact ionization. A photon is absorbed generating an electron-hole pair. The electron accelerates under the influence of a strong electric field, its energy increasing. Due to random collisions from the lattice the electron eventually reaches a saturation velocity. If the electron has accelerated enough to reach an energy larger than the band gap of the material, a secondary electron-hole pair is generated by impact ionization. The electrons and holes then both continue to accelerate, generating further impacts and contributing to gain. The ionization ratio, κ, is given by the ratio of the ionization coefficient of holes to the ionization coefficient of electrons. When most of the ionization occurs due to the electron, the avalanching proceeds from the p side to the n side of the device. It is undesirable to have an ionization coefficient close to 1 because even though the gain is increased in the device, it is time consuming, contributes to noise, and can be unstable leading to breakdown. Ideally, the ionization coefficient is close to 0 or infinity to maximize device performance.

In a SAM ADP, photons are absorbed in an intrinsic or lightly doped region that is very large. Photoelectrons drift across this region under a moderate electric field and then enter a thin multiplication layer with a strong electric field where avalanching occurs.[51] The modified structure that we will explore includes a Ge absorber with a contact layer on top and a pn Si junction on the bottom to exploit the fact that Si's ionization coefficient is about 0, an ideal case for single carrier multiplication. Depending on the size of the depletion width in the Ge, we will be able to utilize drift to reach the junction, but we will mainly rely on diffusion to reach the junction if the depletion width is small in the Ge. Thus, the generated photocurrent depends on the diffusivity of carriers in the material:

$$D = \frac{\mu k_B T}{e} \tag{4-4}$$

where μ is the mobility and

$$L_{n,p} = \sqrt{D_{n,p}\tau_{n,p}} \tag{4-5}$$

implying high mobilities leads to higher diffusivities which leaders to higher diffusion lengths and thus larger current.

The desired performance metrics of detectors are low dark current/low noise, high quantum efficiency, and high sensitivity/responsivity, which all lead to high signal to noise ratio. In communications applications, high bandwidths are also desired as they are a measure of how fast a detector can respond to an optical signal. The quantum efficiency is given by:

$$\eta = \frac{J/q}{P_0/(A\,h\upsilon)} = \frac{(J_{drift} + J_{diffusion})/q}{P_0/(A\,h\upsilon)} \tag{4-6}$$

From (4-6) and (4-3) we can see that the diffusivity and diffusion lengths of the minority carriers in the absorbing layer directly affect the quantum efficiency, meaning the materials quality is of the utmost importance to carrier collection and detector performance. Beyond quantum efficiency, this work will focus on reducing dark current through defect characterization in order to reduce dark current noise and improve signal to noise ratio.

4.2 Defect contributions to leakage and recombination pathways

As we discussed in Chapter 3, post-growth annealing has become a standard practice to reduce threading dislocation density in order to reduce recombination centers present in the material. More precisely, studies have shown that photodiodes fabricated from annealed films exhibit much higher responsivity than those from unannealed films.[52] From Douglas Cannon's work on near-infrared Ge photodetectors, we can see that low leakage current suggests high film quality and a stable interface with oxide used for passivation. Leakage current can either scale with the area or perimeter of the diode. Cannon found that for diodes over 120 microns wide, area leakage dominates. Bulk leakage is reduced by increasing film quality, while perimeter leakage can be affected by stress, high electric fields, interface states, defects, and other anomalies at the device edge.[53] Giovane showed that the photodiode leakage current per unit of dislocation length follows the relation:

$$J_{TD} = \frac{J_0{}^{bulk}}{(TDD)(W)} = 2.1\ \mu A/cm \tag{4-7}$$

where $J_0{}^{bulk}$ is the bulk leakage current density, TDD is the threading dislocation density, and W is the depletion width of the diode.[54]

While threading dislocations are considered to be the key elements contributing to poor electrical characteristics, some studies indicate that point defects may play a more important role in dark current. Yi et al studied the effect of defects in the low temperature Ge buffer layer used for Ge-on-Si diodes by comparing DFT results with DLTS spectra. They concluded that unintentional point defects arising from the low temperature growth can be electrically active and induce acceptor-like levels within the band gap. The DFT calculations show that vacancies have a significantly lower formation energy and thus pose a bigger risk than interstitials. The vacancy-related charge-state defect energy level at 0.33 eV above the valence band has been detected with DLTS and agrees with DFT calculations for vacancies in Ge. At 0.33 eV above the band gap (mid-gap for Ge), the defect state postulated as vacancies by Yi et al can severely affect the dark current and thus the noise present in Ge-on-Si photodetectors.[55] Thus, when studying dislocations in LT Ge and their interactions with point defects, we must also pay close attention the point defect energy levels as they can be even more inconvenient sites of recombination than dislocations in LT Ge.

Besides recombination centers due to the dislocations caused by the Ge-on-Si lattice mismatch we've discussed and point defects studied above, generation-recombination current is also enhanced at the Ge/Si interface by trap-assisted tunneling. Dark current is a superposition of the diffusion current, generation-recombination current governed by the Shockley-Read-Hall (SRH) process where recombination occurs in the bulk, generation-recombination enhanced by trap-assisted tunneling at the Si/Ge layer interface, band-to-band tunneling current, avalanche current that occurs under high electric fields, and shunt resistance current. However, band-to-band tunneling current and avalanche current are only generated at electric fields greater than $3*10^5$ V/cm and shunt resistance is always present due to testing circuitry. The components of dark current that are thus governed by the materials properties of the device are the diffusion current, and generation-recombination current due to the SRH process and the interface. The diffusion current is affected by threading dislocation density as discussed previously. The interface states and defects present also have been shown to reduce the absorption coefficient around the interface because of compressive stress and high dislocation density.[56]

4.3 State-of-the-art Ge-on-Si photodiodes and prior arts for low temperature Ge growth

Because we expect a much higher dislocation density at lower temperature, we will utilize high temperature Ge photodetectors as a source of comparison. Ahn et al report leakage current densities of 0.41 A/cm^2, high efficiency of 90%, 3dB bandwidth of 7.5 GHz, and responsivity of 1.08 A/W at 1550 nm for waveguide integrated Ge photodetectors grown with UHVCVD and two step growth of 360°C followed by 730°C for the high temperature second growth step with the Ge detectors defined by dry etching.[57] Feng et al utilize selective Ge growth as well as large core Si waveguides for their integrated waveguide detector and achieve slightly better results with dark current density of 0.028 A/cm^2 at -0.5 V, responsivity of 0.7 A/W at 1550 nm, and a 3dB bandwidth of 12 GHz.[58]

Low temperature Ge-on-Si photodetectors grown at less than 450°C like our target CMOS compatible temperature have also been reported. Bandaru et al grew 200 nm of p-Ge on n-Si photodetectors using molecular beam epitaxy (MBE) and E-beam evaporation with a hydrogen desorption treatment at 450°C prior to Ge deposition. They report extremely low dark currents of 0.3 mA/cm, and mobility of 100 cm^2/Vs for intentionally Boron doped p-type Ge with carrier concentration 10^{17} cm^{-3}. The diffusion length is estimated from measured responsivity, with diffusion lengths reaching 25 nm for the low temperature growth. The quantum efficiency, measured at 1310 nm, where the absorption coefficient of Ge is 10^4 cm^{-1}, is 1.37% and thus the responsivity is 1.45 A/W.[59] While the quantum efficiency of this detector is very low as anticipated due to the high defect density from low temperature growth, the low dark currents are very promising and demonstrate Si surface quality plays a big factor in the dark current density of low temperature grown Ge. Masini et al used reduced pressure chemical vapor deposition (RPCVD) at 350°C with no high temperature annealing to grow 200 nm of Ge on Si and fabricate a waveguide-integrated photodetector. They reported large dark currents of 1.6 A/cm^2 and low responsivity of 0.6 A/W at 1550 nm.[60] Plasma enhanced chemical vapor deposition (PECVD) has a lower thermal budget than UHVCVD, so it is of interest in terms of process scalability.[6] Littlejohns et al report a PECVD grown low temperature waveguide-integrated Ge on Si detector with 12.5 GHz bandwidth and a very low responsivity of 0.1 A/W at 1550 nm, but do not report dark current characteristics.[61] While PECVD is attractive due to low thermal budget, MBE and UHVCVD yield better device performance.

4.4 Experimental Methods

4.4.1 Photodetector fabrication

We developed several different process flows for our Ge-on-Si photodetectors. The first process flow (see Figure 4-1) is the standard process flow including patterning and etching to form Ge/Si mesas followed by patterning, metal deposition, and lift-off to form metal contacts. This is the process flow we used for photodetectors that included a 2nm Si cap as well as for direction junction photodetectors and photodetectors that never had a Si cap (detailed designs are presented in the next section). We also developed process flows for Si cap removal both at the beginning of the process (see Figure 4-2) and in the middle of the process (see Figure 4-3) to determine whether the Si cap's network of dislocations or Ge surface exposed to oxidation affected device performance. In the process flows that removed the Si cap, the subsequent processing steps were done as quickly as possible to avoid exposing the Ge to air, thus attempting to prevent oxidation as much as possible. This was harder to do for the process flow where we removed the Si cap at the beginning of the process, though we did spin on resist as soon as possible to protect the Ge surface. The Ge surface was most protected when we left the Si cap on, followed by the mid-process Si cap removal process flow. In Figure 4-3, atomic layer deposition of aluminum oxide (Al_2O_3) was conducted immediately after Si cap removal in order

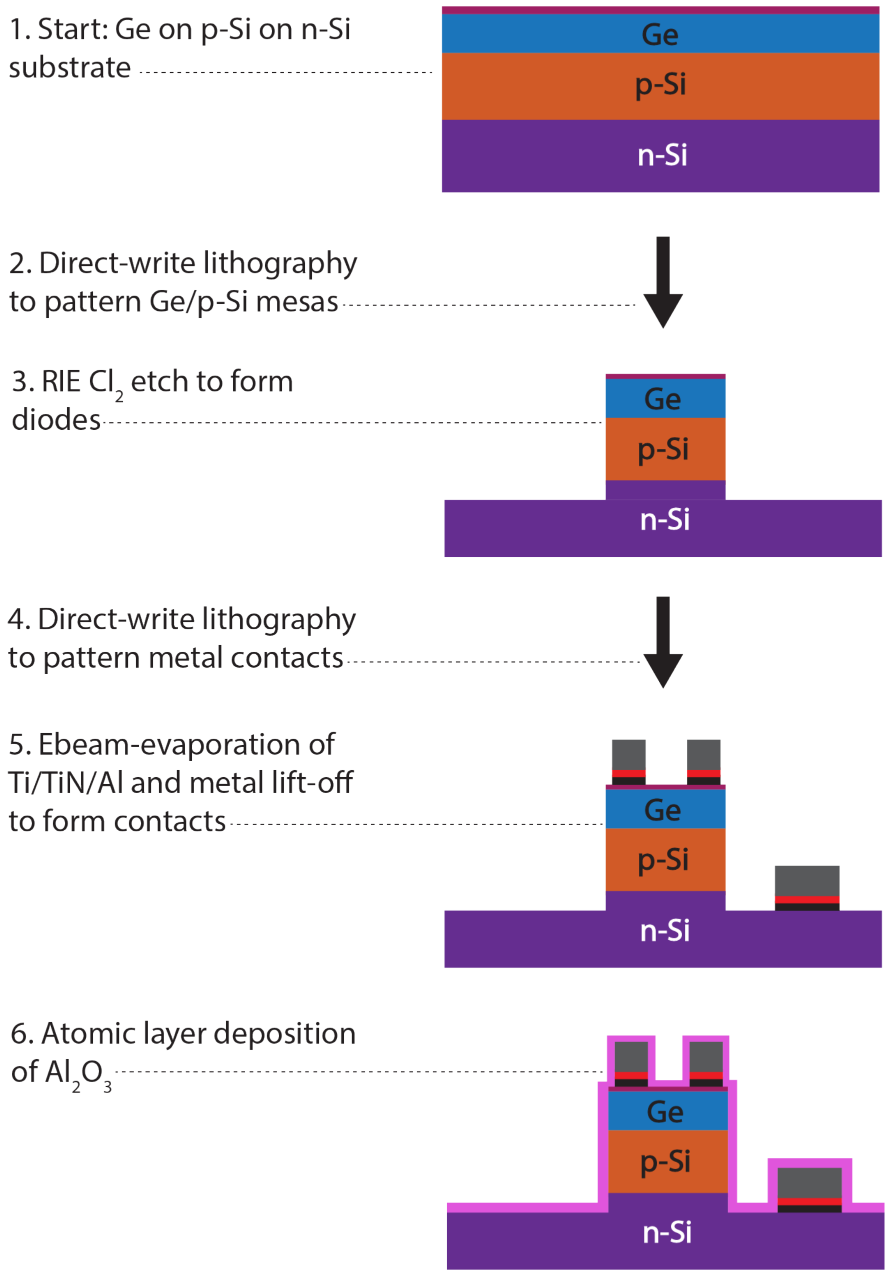


Figure 4-1. Ge-on-pn-Si diode with Si cap process flow. The burgundy layer on top of the Ge layer represents the 2nm Si cap on the Gen2/Gen3 diodes. The black/red/grey metallization represents the 20nm Ti/20nm TiN/300nm Al p-Ge and n-Si (ground) contacts. The bright pink overlayer represents the 20nm thick Al_2O_3 passivation layer.

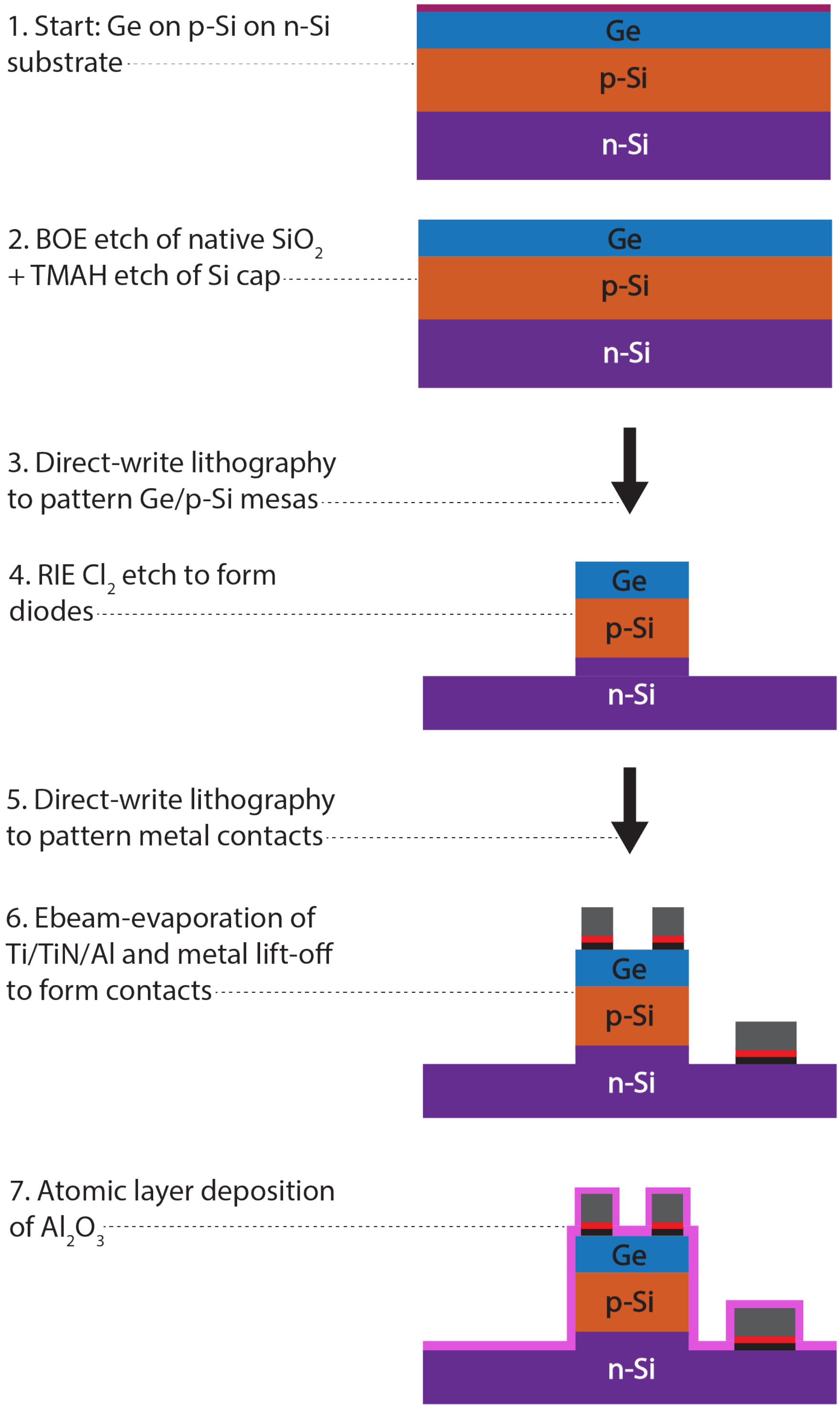


Figure 4-2. Ge-on-pn-Si diode with Si cap process flow where Si cap is removed at the beginning of the process. Ge-on-pn-Si and Ge-on-Si diodes that were not grown with a Si cap were processed starting from Step 3, with the start structure appearing as in Step 2.

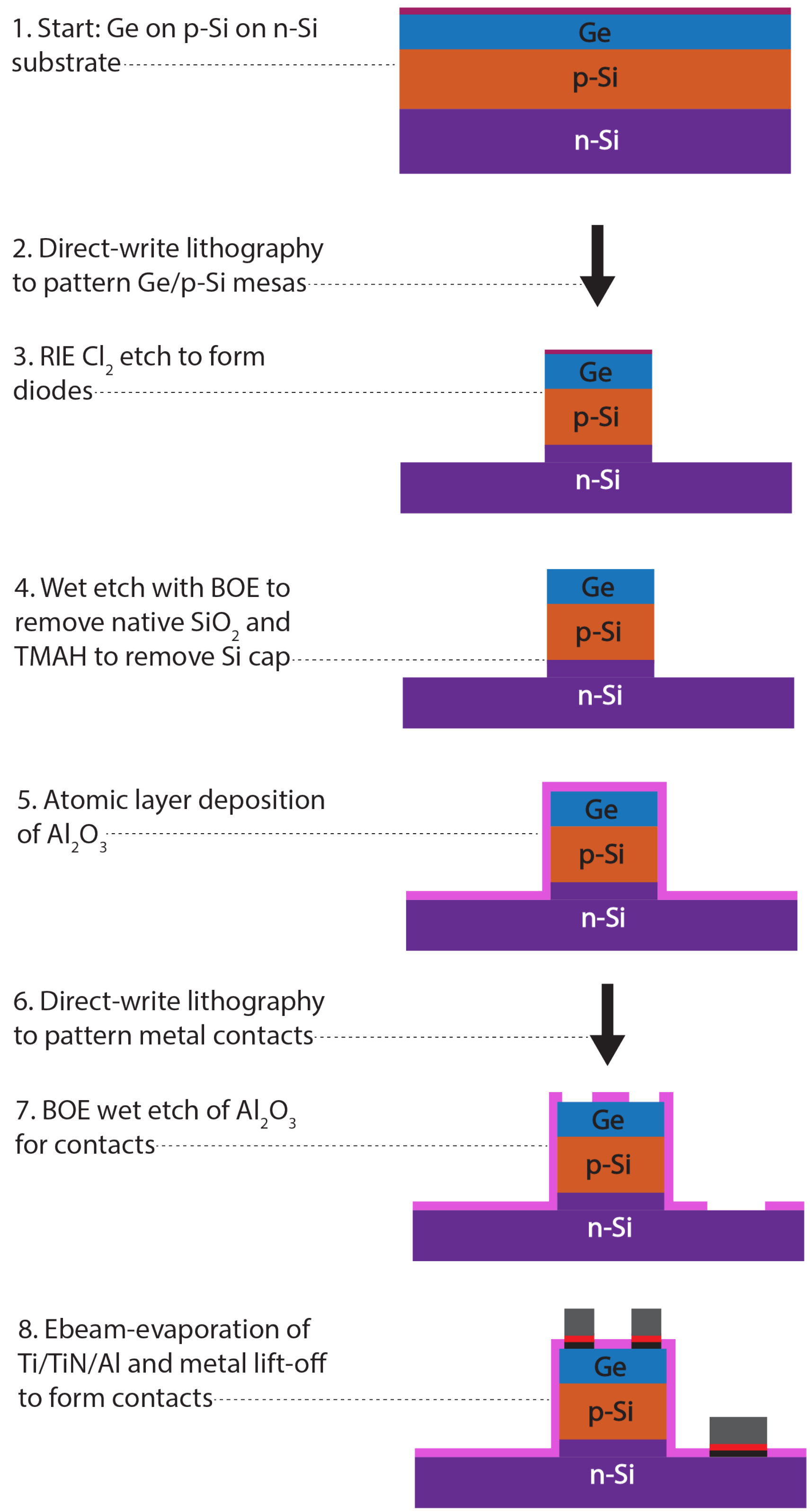


Figure 4-3. Ge-on-pn-Si diode with Si cap process flow where Si cap is removed mid-process.

to prevent oxidation. After the Al_2O_3 was etched away to allow for direct metal contact to the Ge surface, samples were immediately loaded in the Ebeam-evaporation chamber which was then pumped down. A detailed process flow for Si cap removal mid-process is given in Table 4-1.

We also developed a process flow for remote bond pads that had silicon dioxide (SiO_2) or silicon nitride (Si_3N_4) for the insulation layers, in order to be able to wirebond our devices. The process flow for remote bond pads (Figure 4-4) differs from that of direct bond pads due to the need to etch vias through insulation layer and the different metallization needs. Since we have Ge/Si mesas as tall as 3.5-3.7µm in our heterostructures, we cannot utilize Ebeam-evaporation to create a continuous bond pad/metal contact. Instead, we had to switch to sputtering for more conformal deposition. We also had to change the metallization since the wirebonder we used had gold (Au) wire. We moved from our standard titanium (Ti)/titanium nitride (TiN)/aluminum (Al) metallization to Ti/platinum (Pt)/Au metallization. Ti acts as an adhesion layer, and TiN and Pt are diffusion barriers to prevent the main metal spiking into the Ge. Before depositing the insulation layers via plasma enhanced chemical vapor deposition (PECVD), we deposited half of the Ti metallization to act as an etch stop when it came time to etch vias in the insulation layers (SiO_2 and SiN). A detailed process flow for remote bond pads is included in Table 4-2.

Step	Lab	Machine	Action
1	Nano	Spinner-Resist-U12	Spin coat nLOF 2035/2020
2	Nano	Hotplate	Softbake
3	Nano	DirectWrite-MLA150-AirAF	Expose patterning
4	Nano	Hotplate	Post-exposure bake
5	Nano	Develop-U12 or Develop-Brewer-200BX	Develop photoresist

6	Nano	RIE-Mixed-Samco-230iP/RIE-F-Samco-230iP	Dry etching to make Ge on p-Si cylinders using Cl_2
7	Nano	Dektak-XT	Check etch depth
8	Nano	Asher-Barrel-Thierry/Asher-Chuck-ESI	Photoresist strip
9	Nano	Acid-Etch-General-U10	Remove native oxide: BOE 1:7
10	Nano	Develop-U12	Remove 2nm Si cap layer: 2.5% TMAH for 1 minute
11	Nano	ALD-Ozone	Deposit 20 nm Al_2O_3 to passivate sidewalls
12	Nano	Spinner-PGMI-U12	PGMI layer(s)
13	Nano	Spinner-Resist-U12	AZ3312 layer
14	Nano	Hotplate	Softbake
15	Nano	DirectWrite-MLA150-AirAF	Expose patterning
16	Nano	Develop-U12 or Develop-Brewer-200BX	Develop photoresist
17	Nano	Asher-Barrel-Thierry/Asher-Chuck-ESI	Descum
18	Nano	Acid-Etch-General-U10	BOE 1:7, remove ALD for metallization
19	Nano	E-beam AJA/E-beam AU	Deposit 20 nm Ti/20nm TiN/300 nm Al on wafer pieces to form top Si and top Ge contacts
20	Nano	Liftoff-L08/Liftoff-L10	NMP for resist lift-off
21	Nano	Asher-Barrel-Thierry/Asher-Chuck-ESI	Resist removal
22	Nano	ALD-Ozone	Deposit 20 nm Al_2O_3 to passivate

Table 4-1. Detailed process flow for Ge-on Si photodiodes with Si cap removed mid-process. For photodiodes with Si cap removed at the beginning of the process, steps 9 and 10 are conducted before step 1. These steps are skipped for photodiodes that did not have a Si cap or that kept the Si cap throughout the process.

1. Start: Ge on p-Si on n-Si substrate

2. Direct-write lithography to pattern Ge/p-Si mesas

3. RIE Cl_2 etch to form diodes

4. Direct-write lithography to pattern metal contacts

5. Ebeam-evaporation and metal lift-off of Ti etch stop

6. Atomic layer deposition of Al_2O_3

7. PECVD of SiO_2 or Si_3N_4 isolation layer

8. RIE CHF_3 and CF_4 etch to create vias for metallization

9. Direct-write lithography to pattern metal contacts

10. Sputter Ti/Pt/Au and lift-off to form bond pads

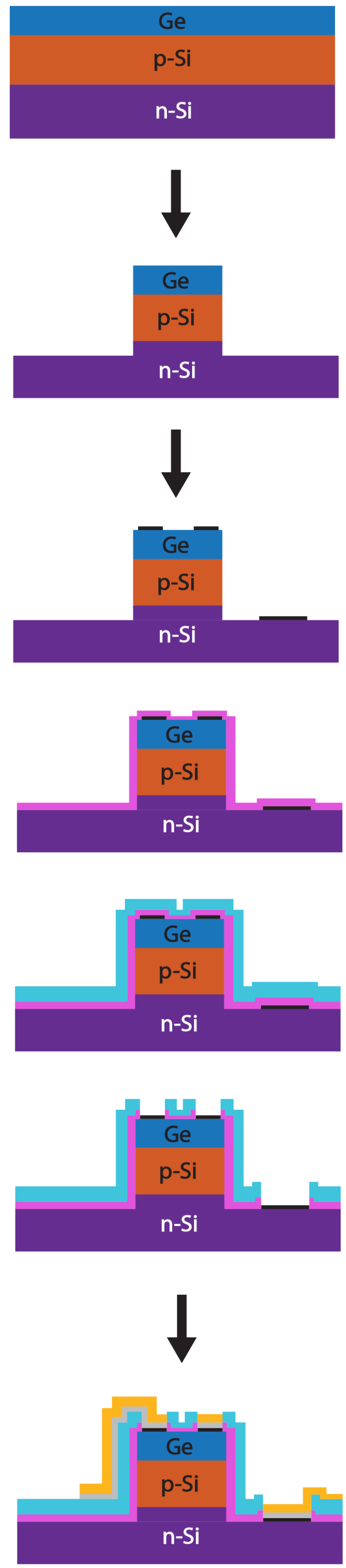


Figure 4-4. Ge-on-pn-Si diode process flow to obtain isolated bond pads for wirebonding where the first part of the process (steps 1-5) is aligned with previous process flows for direct contacts as in Figures 4-1 to 4-3 and the second part (steps 6-10) which includes the dielectric isolation layer (either SiO_2 or Si_3N_4) used to isolate the bond pads, as well as subsequent steps to etch vias and pattern the bond pads.

Step	Lab	Machine	Action
1	Nano	Sonicator	Sonicate 5 minutes (acetone), 5 minutes (methanol), 5 minutes (IPA)
2	Nano	Spinner-all purpose	Spin coat nLOF2035 at 1000 rpm
3	Nano	Hot plate	Bake 110C for 90s
4	Nano	DirectWrite-MLA150-AirAF	Expose patterning
5	Nano	Hotplate	Postbake 110C for 60s
6	Nano	Develop-U12 or Develop-Brewer-200BX	Develop photoresist
7	Nano	RIE-Mixed-Samco-230iP	Dry etching for Ge on p-Si cylinders using Cl_2 → 10 minutes in increments of 3.3 minutes (recipe: Recipe #89), etch depth: 1/2 um in n-Si
8	Nano	Dektak-XT	Check etch depth
9	Nano	Asher-Barrel-Thierry	Strip right after etch – 2 hours superstrip.
10	Nano	Spinner-PGMI-U12	PGMI layer(s)
11	Nano	Spinner-Resist-U12	AZ3312 layer
12	Nano	Hotplate	Softbake
13	Nano	DirectWrite-MLA150-AirAF	Expose contact patterning
14	Nano	Developer	Develop
15	Nano	Sputter-AJA	Deposit 10nm Ti
16	Nano	Solvent hood	Sonicate in NMP or acetone for resist lift-off
17	Nano	Solvent hood	Sonicate in acetone/IPA for contaminant removal

18	Nano	TRL-asher or Asher-Barrel-Thierry	Ash 5 minutes
19	Nano	ALD	Deposit 5-20 nm Al_2O_3 at 250C to passivate sidewalls
20	Nano	STSCVD/PECVD-Samco-PD220	Deposit SiO2 insulating layer (250nm) on diodes or 160 nm Si_3N_4
21	Nano	LAM590/OxideEtch-HF-BOE-L06/RIE-Mixed-Samco-230iP	RIE etch with CHF_3 and CF_4 to create vias in the Al_2O_3 and SiO_2/Si_3N_4layers
10	Nano	Spinner-PGMI-U12	PGMI layer(s)
11	Nano	Spinner-Resist-U12	AZ3312 layer
23	Nano	Hot plate	Softbake
24	Nano	DirectWrite-MLA150-AirAF	Expose patterning for metal contact bond pads
25	Nano	Develop-U12 or Develop-Brewer-200BX	Develop photoresist
26	Nano	Sputter-AJA	Deposit remaining metallization, Ti 10nm/50nm Pt/200nm Au
27	Nano	Solvent hood	Sonicate in NMP for resist lift-off
28	Nano	Diesaw-DAD3240 or Cleave-LatticeAx	Cut wafer into pieces

Table 4-2. Detailed process flow to create remote bond pads for our Ge-on-Si photodiodes.

4.4.2 Photodetector characterization

Photodetector dark current and photocurrent was measured using DC probes and Keithley sourcemeter that applied bias and measured current simultaneously. An optical microscope was used to place the probe tips on the diode contacts. All measurements were conducted with the room lights off as the bias was swept and current was collected. A 1310 nm pigtail fiber-coupled laser was collimated and supplied illumination. The laser was cooled using a thermo-electric cooler and was mounted on a three-axis stage that was moved in while the microscope was moved out of the way for photocurrent measurements. While the sample was connected to the sourcemeter, a constant bias was applied and constant current measurements were displayed on the sourcemeter screen. Rough alignment was conducted with a red (635 nm) laser that was mounted interchangeably on the three-axis stage with the 1310 nm laser. Once the diode was roughly aligned to the red laser (photocurrent maximized on the sourcemeter current read-out), the 1310 nm laser was swapped in and the alignment was refined. Once alignment was conducted and photocurrent was maximized, photocurrent sweeps were conducted at 0.5-2.5 mW driving laser power. Laser power was measured by a Newport power meter and reference photodiode. The reference photodiode aligned to the laser while the laser stayed at the same height in order to obtain an accurate power reading. The power incident at the diode was always about half of the driving power, and was further reduced by the actual exposed area of Ge.

The top contact for the diodes (the p-Ge contact) was designed to have a donut structure, with an open inner 50 μm radius aperture for Ge illumination. The outer ring of exposed Ge was a processing necessity to allow alignment tolerance between the metallization and mesa etch pattern layers. The ring width varied from 5-20 μm for different process lots. The diodes' mesa radii were 250 μm, 500 μm, 750 μm, 1000 μm, and 2000 μm. These diode radii were chosen to have an order of magnitude span to be able to measure peripheral vs areal leakage dependence, and to have more points to fit the leakage dependence curves. We found that the

smallest probable area for the DC probes we used was about 150 μm x 150 μm – thus, the smallest allowable diode radius was around 250 μm to allow for ~200 μm of probable contact area.

We also explored wirebonding our photodetectors to a PCB to reduce noise in our measurements and to improve the accuracy of our alignment and power calibration. We were able to measure laser power directly by splitting off the laser between the photodetector we were measuring and a reference photodiode that was connected to a power meter to measure laser power (see Figure 4-6).

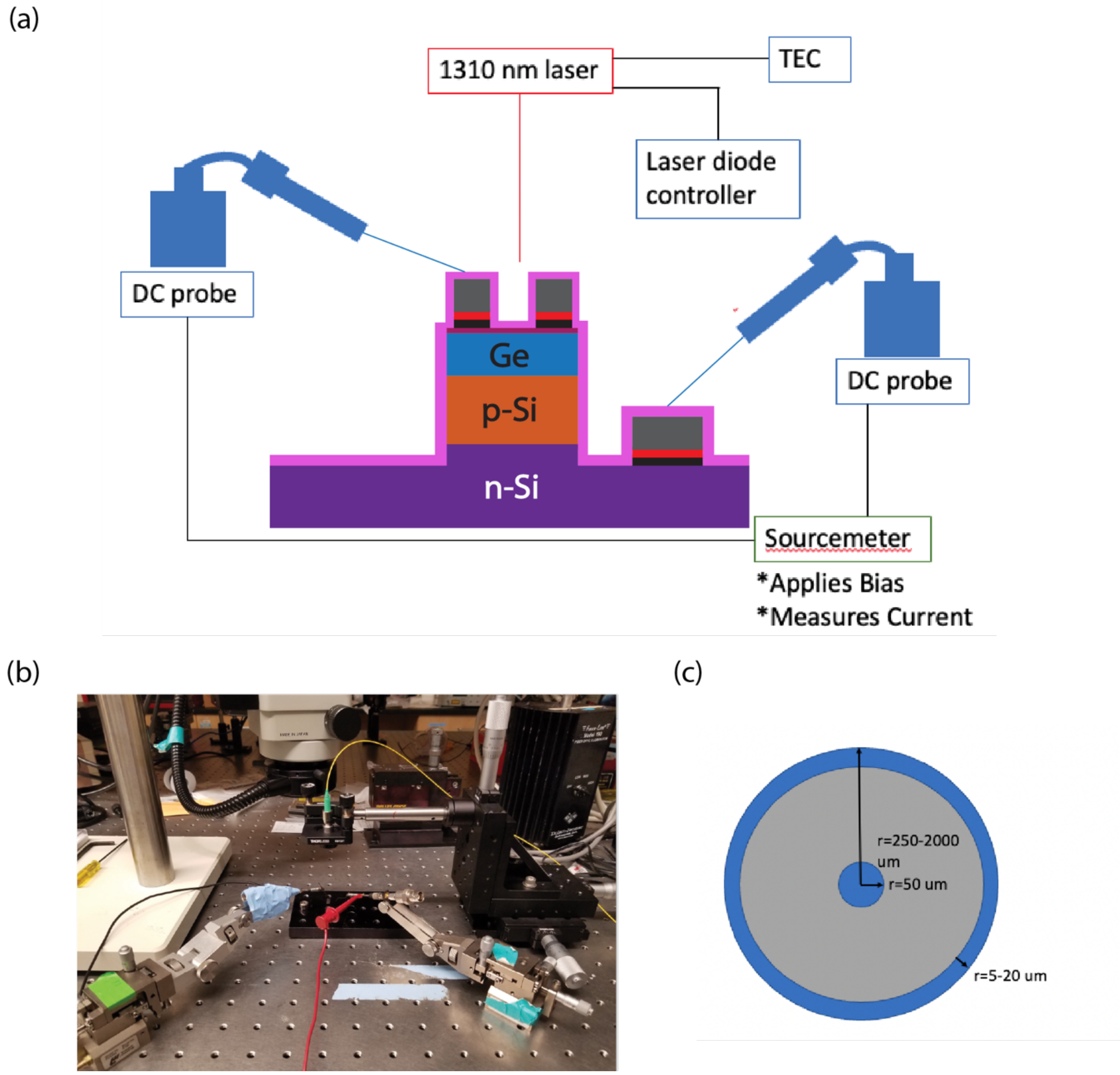


Figure 4-5. Schematic (a) and picture (b) of DC probing set-up to measure dark current and photocurrent from photodiodes, and (c) top-view of metal contact on p-Ge/positive terminal.

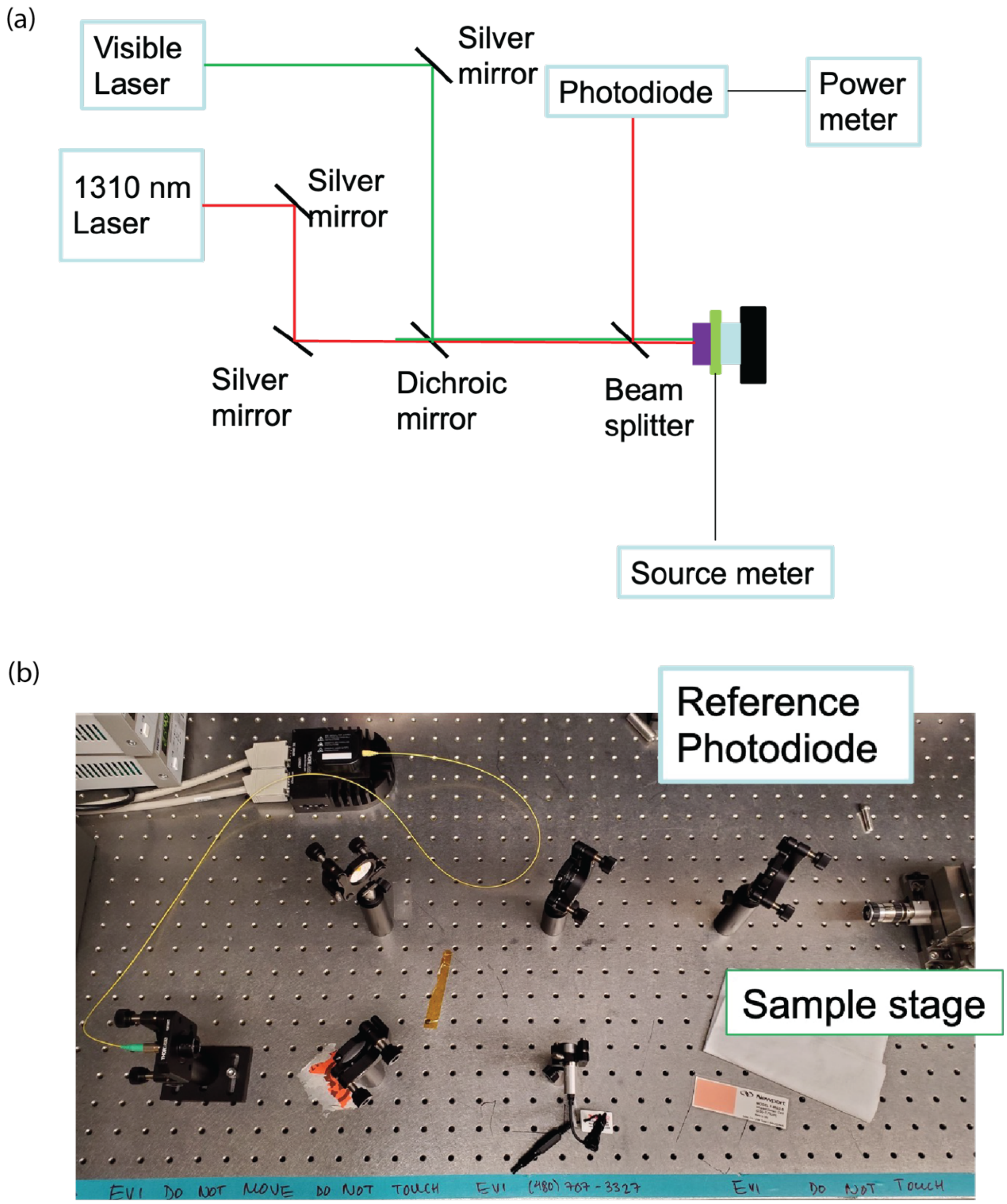


Figure 4-6. Schematic (a) and picture (b) of measurement set-up for dark and photocurrent measurements of wirebonded diodes.

4.4.3 Beam profile calibration

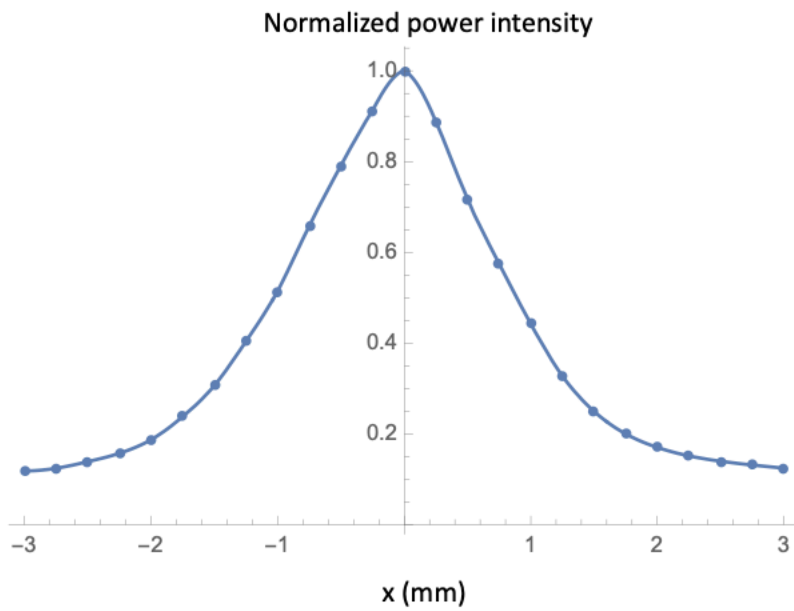


Figure 4-7. Normalized beam profile of 1310nm pigtail fiber-coupled laser. Data and figure courtesy of Stephanie Marzen and Professor Kazumi Wada.

The 1310nm pigtail fiber-coupled laser had a Gaussian beam profile that we roughly measured covering the reference photodiode with a 250 µm diameter aperture and mounting the reference photodiode to a stage capable of making adjustments accurate to 10 µm in the x and y directions. The reference photodiode was then swept in 250 µm increments under the laser and the power was measured and normalized to the beam profile in Figure 4-7. In order to calculate the fraction of power reaching the exposed Ge compared to the measured incident power, the beam profile was interpolated in Mathematica and cylindrically integrated to measure the volume under the curve for both the inner apertures and outer rings of the contacts for the various diode sizes.

For example, for the 250 µm radius diode, which we label D5, with a 5 µm thick outer ring, the power fraction reaching the Ge is given by:

$$Power_{D5\ Ge\ aperture} = \frac{Power_{50um\ r\ inner\ aperture} + Power_{5um\ outer\ edge}}{Power_{3mm\ total\ profile}}$$

$$= \frac{\int_0^{0.05} 2 * \pi * x * InterpolatedProfile(x)\ dx + \int_{0.245}^{0.25} 2 * \pi * x * InterpolatedProfile(x)\ dx}{\int_0^{1.5} 2 * \pi * x * InterpolatedProfile(x)\ dx} \quad (4\text{-}8)$$

The 1.5 mm bound in the denominator is due to the reference photodiode's 3mm diameter. Even though the beam profile extends further out than 3mm in either x direction, the incident power measured by the reference photodiode is only reaching a 1.5 mm radius area, thus the power fraction must compare to this volume under the beam profile. The power fraction calculated by Equation 4-8 is then multiplied by the measured incident power for all subsequent calculations.

4.5 Results and Discussion

4.5.1 Device design and optimization

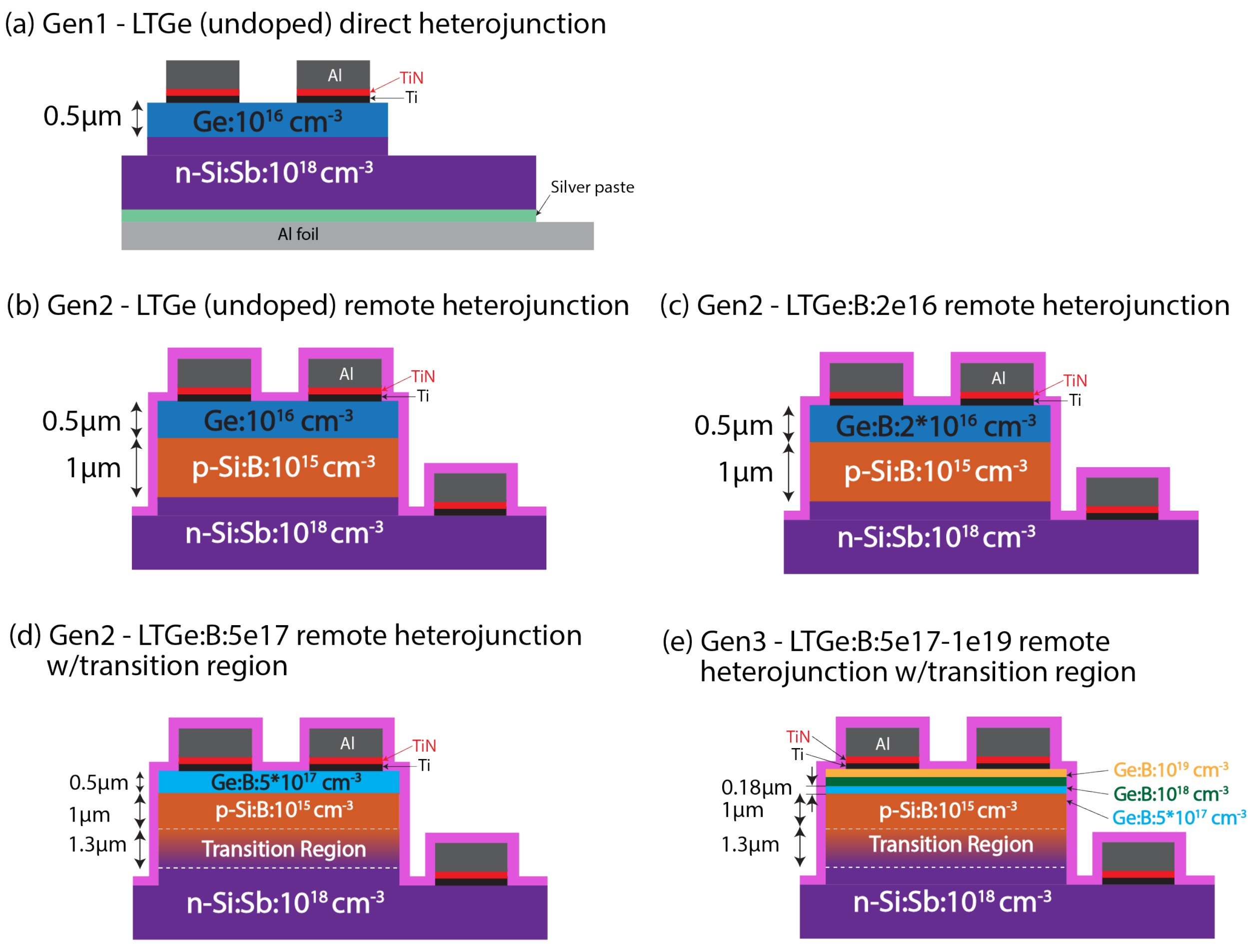


Figure 4-8. The different photodiode structures investigated include (a) Gen1: Ge-on-Si direction heterojunction device where Ge was intentionally undoped for this device, though it exhibited p-type conductivity and 10^{16} cm^{-3} carrier concentration measured via Hall effect (detailed in Chapter 3). Also investigated were (b) Gen2: Ge-on-Si remote heterojunction, where the junction is between the p-Si and n-Si and the p-type intentionally undoped Ge acts as an absorber layer, (c) a Gen2 structure where Ge is intentionally doped with boron (B) at a concentration of $2*10^{16}$ cm^{-3}, (d) a Gen2 structure where Ge is intentionally doped with B at a concentration of $5*10^{17}$ cm^{-3} and a 1.3 μm-thick transition region exists between the p-Si and n-Si regions, and (e) Gen3: a non-

*uniformly doped Ge remote heterojunction, where the Ge region includes three distinct 0.18 µm-thick layers of increasing doping density to the surface (5*10^{17} cm^{-3}, 10^{18} cm^{-3}, 10^{19} cm^{-3}). This Gen3 structure also includes the 1.3 µm-long transition region between p-Si and n-Si that the Gen2:B:5e17 structure in (d) included as well.*

We first begin to explore the effects of LT Ge material properties on Ge-on-Si photodetector performance by examining various photodiode structures (see Figure 4-8). In order to achieve low dark current and high quantum efficiency in our desired Telecom wavelength range, 1.3-1.5 µm, we move from a Ge-on-Si direct heterojunction device where the pn junction is at the Ge/Si interface (see figure 4-8a) to a remote heterojunction device where the Ge layer acts as an absorber layer for our desired wavelengths, while the actual pn junction is at the p-Si/n-Si interface (see Figure 4-8b-e). The impact of device structure on experimental device performance will be examined further in Section 4.3.2. Since we are targeting low dark current densities for our photodetectors, we utilized our Gen1 direct junction devices purely to characterize the limits of our material. We focus on simulating and optimizing Gen2 remote heterojunction devices. This allows us to capitalize on Ge's band gap of 0.66 eV[62] for absorption of our wavelength of interest, while avoiding the high dark current prevalent in Ge-on-Si heterojunction detectors due to thermionic emission from Ge and defects at the Ge/Si interface.[6] It has been shown that threading dislocations act as leakage pathways thus increasing dark current in devices[54] and dislocations at the interface act as recombination centers[52], thus limiting quantum efficiency. We capitalize on the far lower interface defect density of the p-Si/n-Si interface and lower thermionic emission of Si by moving the diode junction to the p-Si/n-Si interface, though the tradeoff is we rely on diffusion of photocarriers across the Ge layer in order to reach the junction and be collected at the contacts.

Thus, understanding the limits of materials quality from Chapter 3, and being able to optimize device design with these materials considerations is paramount to achieving a photodetector with high quality device performance.

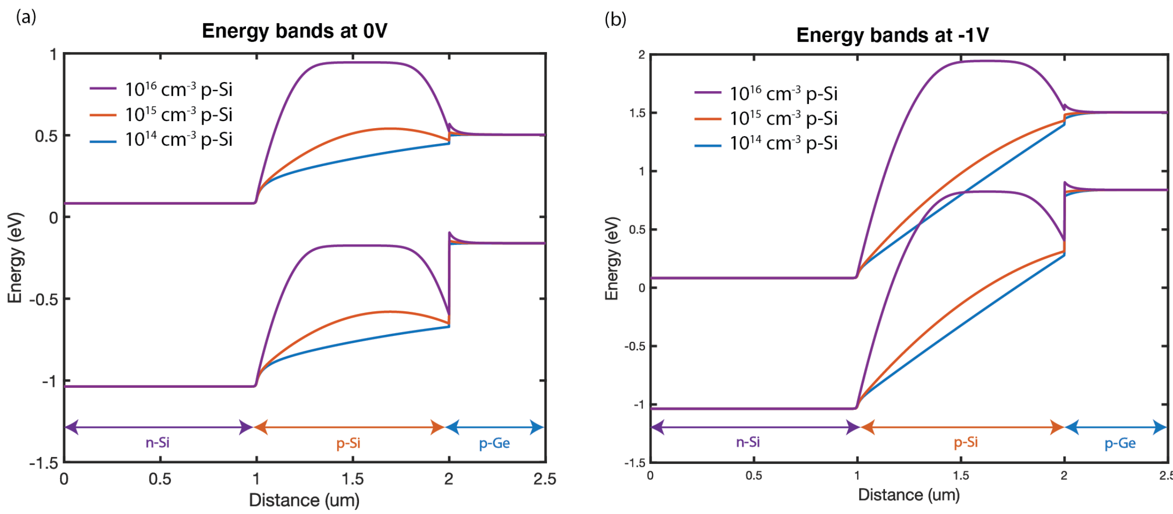


Figure 4-9. (a) Effect of p-Si doping density on band structure at 0V bias. (b) Effect of p-Si doping density on band structure at -1V bias. Modeled using Comsol.

Utilizing Comsol's Semiconductor Module, we modeled the effect of the p-Si doping density on the device's overall energy band structure. We observe an energy barrier exists for electrons in the conduction band between p-Ge and p-Si that is dependent on their doping densities. Keeping Ge's doping density fixed at 10^{16} cm^{-3}, and the n-Si doping density fixed at 10^{18} cm^{-3}, the energy barrier goes from 0.5 eV at 10^{16} cm^{-2} p-Si doping density to about 0.1 eV at 10^{15} cm^{-3} doping density to 0 eV for 10^{14} cm^{-3} doping density (see Figure 4-9). A high energy barrier would aid in keeping dark current low as carriers generated from Ge's thermionic emission or from dislocations at the Ge/Si interface would not be collected across the junction. However, a high energy barrier also prevents photocarriers from diffusing across the Si junction and being collected at the n-type

Si metal contact (which is the ground contact). We see that at a reverse bias of 1V (Figure 4-9b), the energy barrier remains for higher p-Si doping densities but is lowered for 10^{15} cm^{-3} p-Si doping densities and lower. In order to achieve both low dark current and high quantum efficiency, we choose the intermediate condition of 10^{15} cm^{-3} p-Si doping density. Further optimization of p-Si doping density and thickness will allow us to understand the effects of the energy barrier on device performance better.

We also modeled the electric field for Gen2 devices with both an abruption junction (Figure 4-8b and 4-8c) and a large transition region (Figure 4-8d and 4-8e). In order to get the highest photocarrier collection, we want to find the reverse bias at which the Si junction is fully depleted and that extends furthest into Ge. The more we rely on an electric field causing carriers to drift and be swept across the junction, the less we rely on diffusion where carriers can readily recombine and thus be lost before they can be collected at the metal contacts. From the plots of electric field at various reverse biases in Figure 4-10, we can see that in order to fully deplete the Si junction, we have to bias our transition region devices to at least 4V while the abrupt junction devices need only be reverse biased to 1V to be fully depleted. To compare these device structures' IQE fairly, we should compare Gen2 structures with an abrupt junction at -1V bias and transition region devices at -4V bias or lower. Figure 4-10c demonstrates the carrier concentration measured via Secondary Ion Mass Spectroscopy (SIMS) for Ge:B:5e17 Gen devices and the modeled carrier concentration via PC1D for which the electric fields in Figure 4-10b are plotted. While the PC1D model does include a lower carrier density at the cross-over from the transition region to the p-Si region, the bias that it takes to deplete the junction should be similar since we need less and less bias to deplete to lower carrier concentrations.

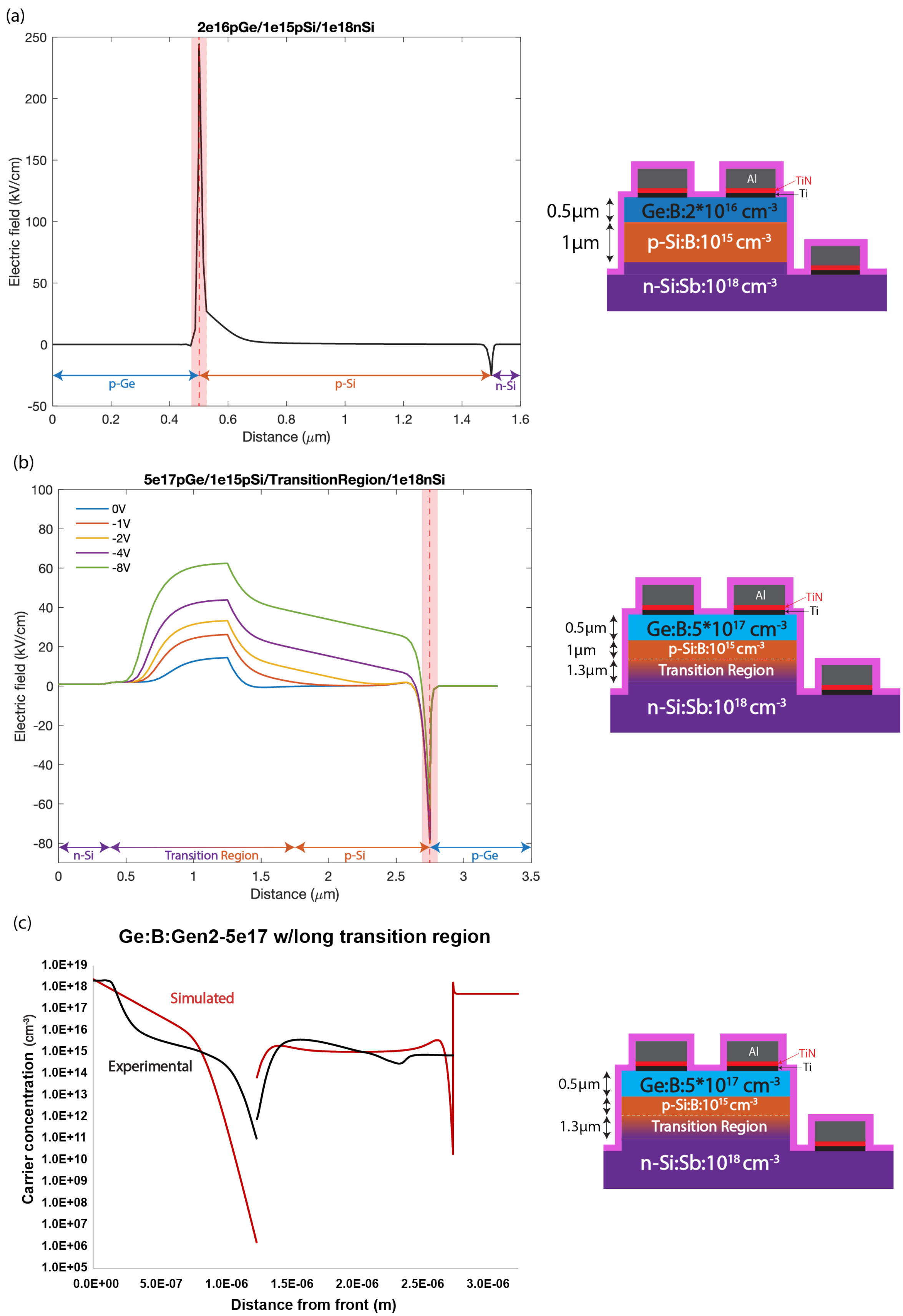

(a)
2e16pGe/1e15pSi/1e18nSi
Electric field (kV/cm)
Distance (μm)
p-Ge
p-Si
n-Si
0.5μm
1μm
Ge:B:2*10^16 cm^-3
p-Si:B:10^15 cm^-3
n-Si:Sb:10^18 cm^-3
Al
TiN
Ti
(b)
5e17pGe/1e15pSi/TransitionRegion/1e18nSi
0V
-1V
-2V
-4V
-8V
Electric field (kV/cm)
Distance (μm)
n-Si
Transition Region
p-Si
p-Ge
0.5μm
1μm
1.3μm
Ge:B:5*10^17 cm^-3
p-Si:B:10^15 cm^-3
Transition Region
n-Si:Sb:10^18 cm^-3
Al
TiN
Ti
(c)
Ge:B:Gen2-5e17 w/long transition region
Simulated
Experimental
Carrier concentration (cm^-3)
Distance from front (m)
0.5μm
1.3μm
Ge:B:5*10^17 cm^-3
p-Si:B:10^15 cm^-3
Transition Region
n-Si:Sb:10^18 cm^-3
Al
TiN
Ti

Figure 4-10. (a) Electric field at 1V reverse bias as modeled by PC1D for LTGe:B:2e16-Gen2 structure. (b) Electric field for 0V to 8V reverse bias as modeled by PC1D for LTGe:B:5e17-Gen2 structure w/transition region between p-Si and n-Si. (c) Carrier concentration measured by SIMS for Gen2/Gen3 devices with the long transition region between p-Si and n-Si (black) compared to the PC1D constructed model's carrier density. PC1D model constructed by Stephanie Marzen and carrier density figure (c) courtesy of Stephanie Marzen.

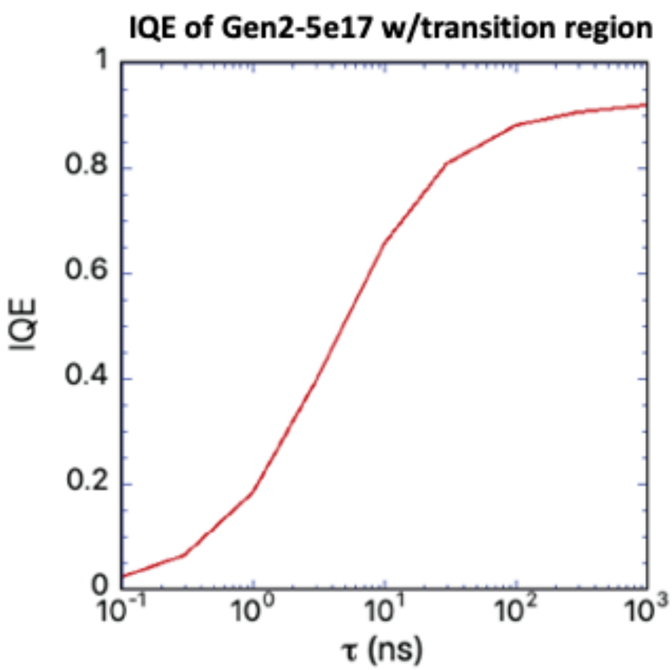


Figure 4-11. IQE based on minority recombination carrier lifetime τ as modeled using PC1D by Professor Kazumi Wada.

From the same Hall effect measurements we detail in Chapter 3, we obtain mobility on the order of 100 cm^2/Vs in our as-grown LT Ge. As-grown Ge exhibits p-type conductivity so holes are the majority carriers. Using equation 4-4, we can calculate diffusivity from the Hall mobility and we obtain as-grown LT Ge's diffusivity is about 2.6 cm^2/s. Since we see a photoresponse in our as-grown 500 nm thick structures, we know that the diffusion lengths of our photocarriers are at least on the order of 500 nm. Using the relationship between diffusion length L, diffusivity, and recombination lifetime τ:

$$L = \sqrt{D\tau} \tag{4-9}$$

we find that our majority carrier recombination lifetime is around 1 ns. If we assume the minority carrier lifetime is around the same or an order of magnitude worse than the majority carrier lifetime, we can make a worst-case assumption that our minority carrier lifetime is around 0.1-1 ns. Figure 4-11 demonstrates the simulated internal quantum efficiency (IQE) versus minority carrier recombination lifetime. The simulations were performed by Professor Kazumi Wada via PC1D for the device structure Gen2:B:5e17 where the p-Si doping density is $8*10^{15}$ cm^{-3} and p-Si thickness is 0.5 μm. The simulation predicts that for our estimated minority carrier lifetime of 0.1-1ns, our as-grown devices will exhibit IQE of 5-20%. Upon increasing recombination lifetime via annealing, the simulation dictates that we would expect to see an increase with IQE upon annealing and reach a plateau when the minority carrier recombination lifetime reaches ~1μs. Our as-grown Gen2:B:5e17 and Gen3:B:5e17-1e19 devices exhibit IQE of around 20-30%, which corresponds to a predicted minority carrier recombination lifetime of 1 ns. Experimental results of device performance (leakage current and IQE) will be discussed in more detail later. It is important to point out that these PC1D simulations were conducted on a device structure that is different from experimental structure – namely the lack of transition region in the simulation between p-Si and n-Si as well as the higher p-Si doping density of $8*10^{15}$ cm^{-3} in the simulation versus the experimental structure whose doping density is $1*10^{15}$ cm^{-3}. The simulated structure would result in a higher depletion width into the Ge for the higher-doped p-Si, and lower depletion width in the p-Si for the higher-doped p-Si. Due to the higher depletion width in the p-Ge, the simulation actually underestimates the effects of recombination lifetime on IQE since IQE would rely even more on diffusion for the lower p-Si doping density. Thus, we can expect experimental results to exhibit either a similar dependence of IQE on minority carrier recombination lifetime or an even faster increase of IQE with an increase in minority carrier recombination lifetime.

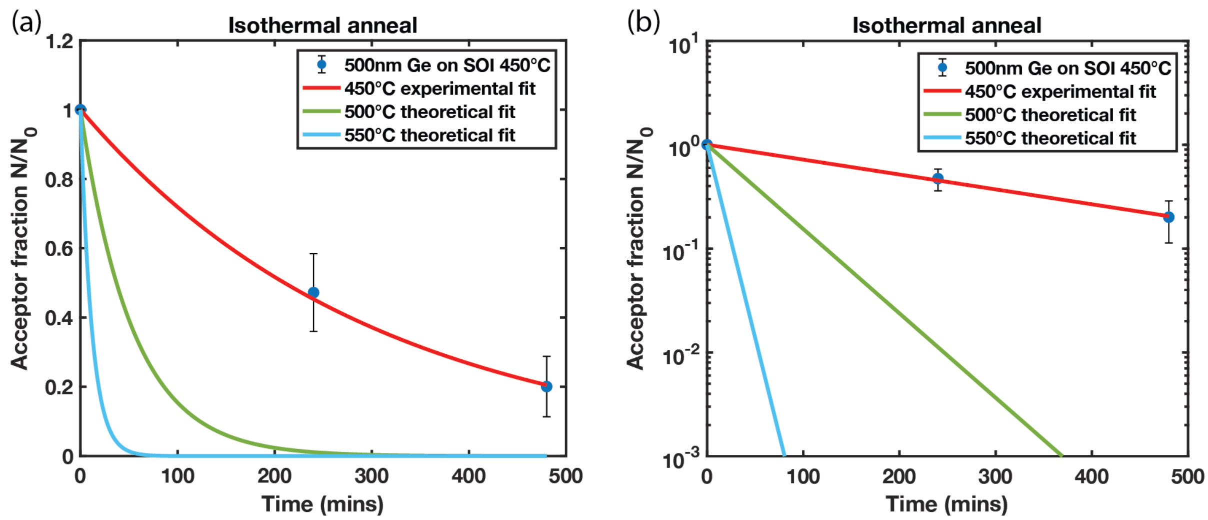


Figure 4-12. (a) Linear plot of 500°C and 550°C modeled acceptor fraction isothermal anneal based on 1.7 eV activation energy and 0.0003 min^{-1} 450°C isothermal annealing rate measured via Hall effect in Chapter 3. (b) Log-linear plot of modeled and experimental isothermal anneal.

Based on the 450°C isothermal anneals we conducted for as-grown LT Ge on SOI detailed in Chapter 3, we can model acceptor point defect removal at 500°C and 550°C to identify the time scales where we would actually see the predicted improvement in IQE from recombination lifetime improvement. The 450°C isothermal anneal demonstrated that the acceptor annealing reaction follows first-order rate kinetics:

$$[A] = [A_0] * \exp(-\mathrm{kt}) \quad (4\text{-}10)$$

Where $[A]$ is the acceptor concentration, $[A_0]$ is initial acceptor concentration, k is the rate constant, and t is time. The rate constant will follow the same Arrhenius behavior that the isochronal anneals identified in Chapter 3:

$$k = [A] * \exp\left(-\frac{E_A}{k_B\,T}\right) \tag{4-11}$$

where E_A is the activation energy, 1.7 eV, k_B is Boltzmann's constant, and T is the temperature. We solve for $[A]$ by inputting $k = 0.0033\ min^{-1}$, identified at $T = 450°C$. We then utilize this model to predict the acceptor annihilation rate at 500°C, 0.0187 min^{-1}, and at 550°C, 0.0857 min^{-1} which are then used to plot the expected acceptor fraction annihilated with annealing time at 500 and 550°C in Figure 4-12. Table 4-3 details time and temperature parameters associated with annihilating different percentages of the acceptor fraction [N_A] which aided in designing our experimental matrix to understand effects of low temperature annealing on device performance.

Annealing temperature (°C)	**Time for [N_A] to 50% (mins)**	**Time for [N_A] to 20% (mins)**	**Time for [N_A] to <1% (mins)**
450	210	490	>490
500	37	86	246
550	8	19	54

Table 4-3. Time to reach various acceptor densities based on annealing temperature.

4.5.2 Effects of annealing and processing conditions on device dark current

We examined the impact of annealing and sidewall passivation of aluminum oxide (Al_2O_3) deposited via Atomic Layer Deposition (ALD) on our Gen2 devices with abrupt junctions with heterostructures in Figure 4-13b and 4-13c. We analyze dark current at 1V reverse bias for diodes with radii of 0.025-0.2 cm (see Figure 4-13a). By plotting the dark current versus the diode radius, we can observe whether the dark current has an areal (r^2) or peripheral (r) dependence. We see that for diodes with sidewalls passivated with Al_2O_3, areal leakage dominates, while for diodes with no sidewall passivation, peripheral leakage dominates. We conclude that 20nm of Al_2O_3 successfully passivates Gen2 Ge-on-Si diode sidewalls and results in stronger areal leakage contributions than peripheral leakage contributions, as well as two orders of magnitude lower dark current overall.

We also anneal Gen2 devices at 500°C and 800°C. As described in Chapter 3, 500°C is in the point defect recovery temperature regime while 800°C is in the dislocation recovery temperature regime. We observe that annealing at 500°C does not have an impact on dark current, but annealing at 800°C does decreases dark current significantly by another 1-2 orders of magnitude. This indicates that for our duration of a 500°C 60 minute anneal, the resulting annihilation of about ~50% of acceptor-like point defects did not impact device dark current, while annealing at 800°C for 2 mins eliminated both point defects and threading dislocations and had a significant impact on dark current. This could indicate that either dislocations do impact dark current more significantly than point defects, or we did not anneal long enough to observe the impact of point defects on dark current.

As-grown intentionally doped Ge photodetectors were fabricated and measured with and without a Si cap on the surface. The devices with a Si cap followed the process flow outlined in Figure 4-1, while the devices without a Si cap had them removed at the beginning of the process as is outlined in Figure 4-2. All of the devices plotted in Figure 4-14 had ALD deposition of Al_2O_3 as the last step, but the devices without a Si cap still exhibit peripheral leakage dependence rather than areal leakage dependence. This is due to a processing issue as the ALD chamber was contaminated during the sidewall passivation step, leading to less effective sidewall passivation.

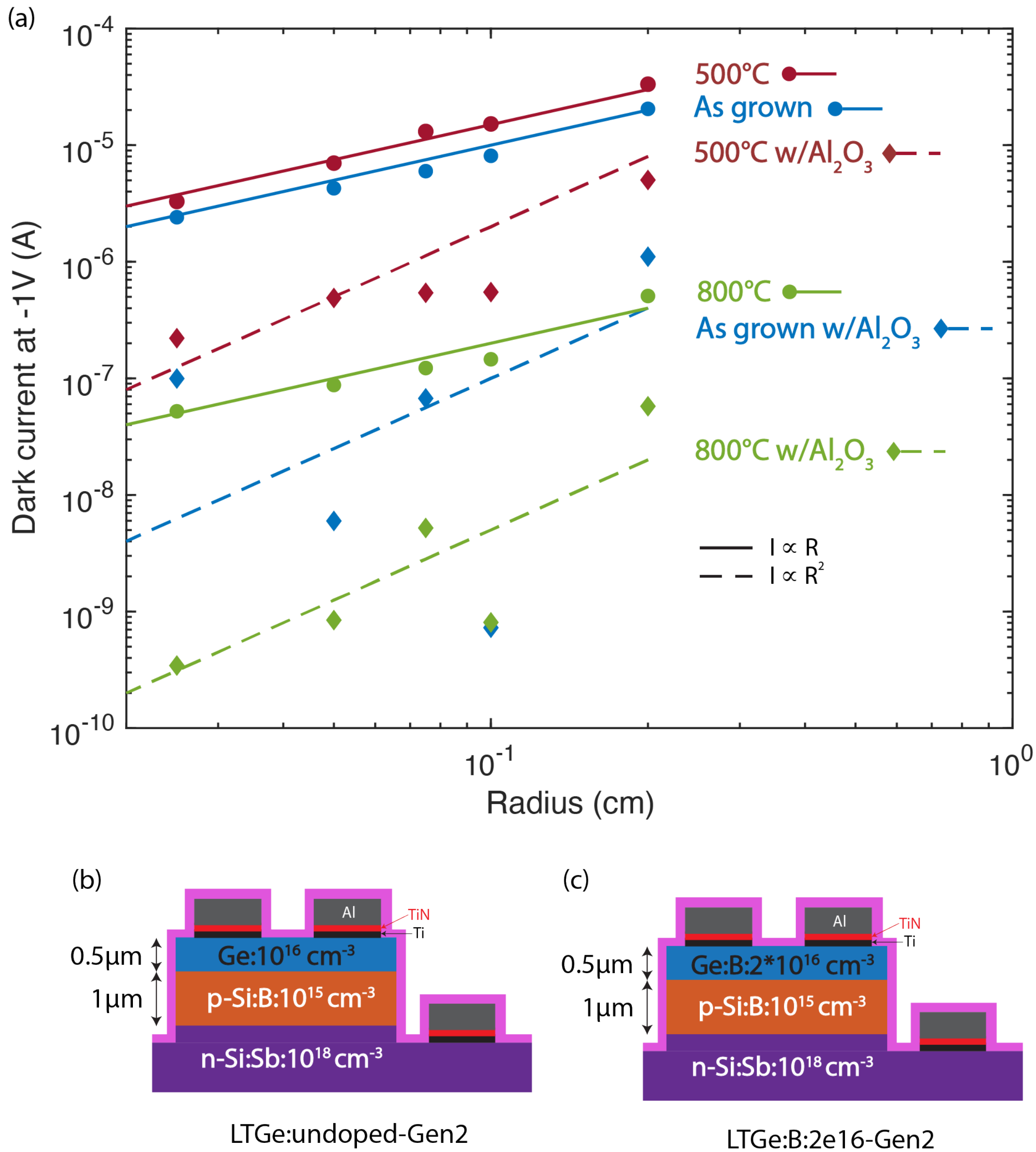


*Figure 4-13. (a) Effect of 500°C and 800°C temperature anneals on dark current of (b) as-grown and 500°C annealed Ge that is undoped during growth and has intrinsically ~1016 cm-3 p-type carrier concentration, as well as (c) 800°C annealed Ge that is intentionally doped with boron at 2*1016 cm-3 doping density.*

=

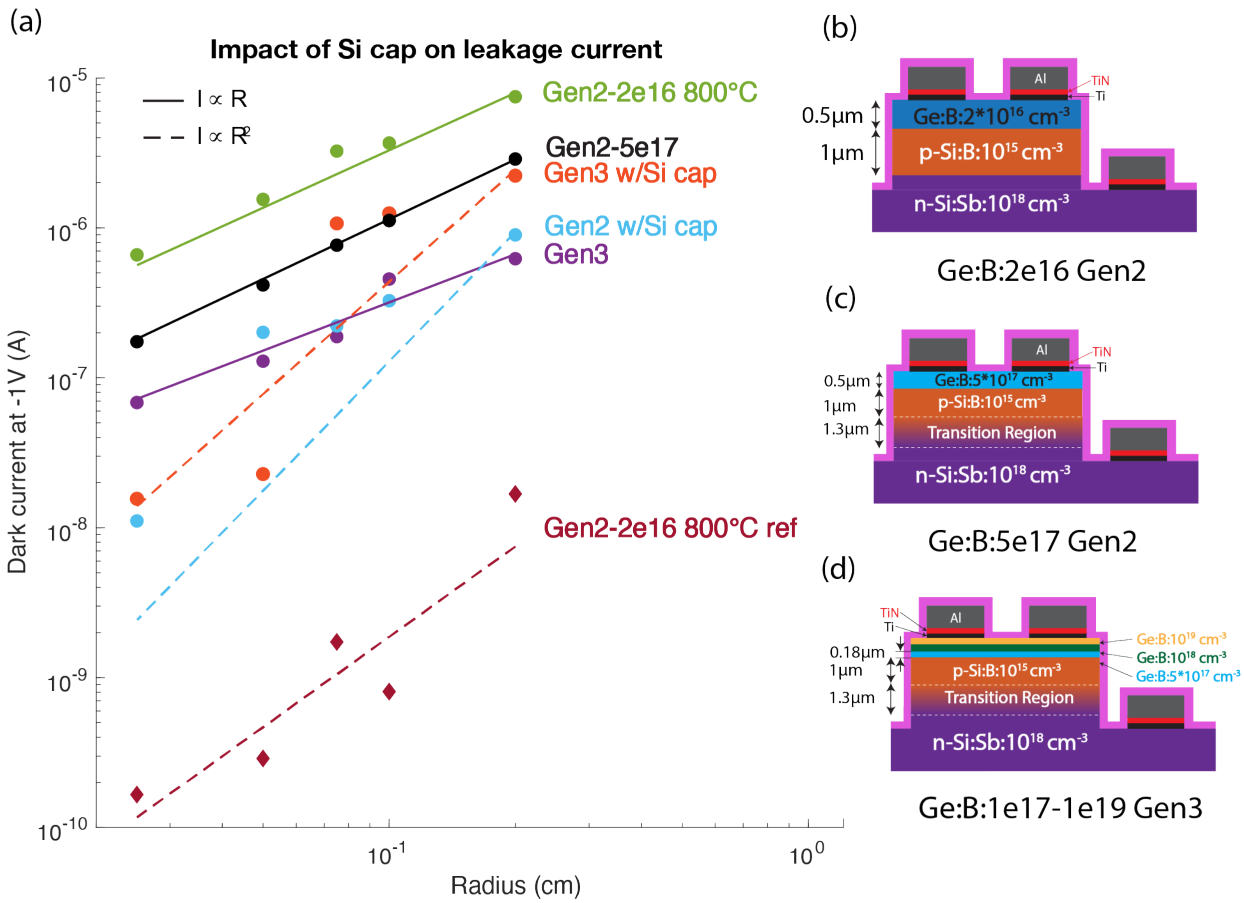


Figure 4-14. (a) Effect of Si cap on device dark current for devices that had ineffective sidewall passivation via Al_2O_3 compared to 800°C-annealed device with effective sidewall passivation. Schematics of the device structures that are plotted here include (b) Ge:B:2e16 Gen2 w/abrupt junction (c) Ge:B:5e17 Gen2 w/transition region and (d) Ge:B:1e17-1e19 Gen3 graded doping device.

We plot the dark current vs radius and use Matlab's power law fit to fit the radial dependence curves. Dark currents are higher due to this ineffective passivation, the lack of annealing, and the added effect from the large transition region between the p-Si and n-Si region in these devices. However, we see that devices with a Si cap tend to have lower dark current and higher areal leakage than devices without a Si cap. This may be due to a network of dislocations that are distributed uniformly along the surface area from the 2nm Si cap on the Ge layer,[63] or due to the Si cap protecting the Ge surface from oxidation. While we are not able to isolate completely the effects of the Si cap on leakage from the leakage pathways due to dangling bonds at the sidewalls, our results show that the Si cap does not adversely affect the dark current. Thus, we chose to proceed with devices that have a Si cap on the Ge.

Due to the aforementioned large transition region between p-Si and n-Si in our higher doped Gen2 and Gen3 devices (Figures 4-8d and 4-8e), the characteristic IV of these devices differs from that of the Gen2 devices with abrupt junctions (Figures 4-8b and 4-8c). These dark and photo IVs under 1310nm driving laser power of 0.5-2.5 mW are plotted in Figure 4-15. Due to the larger reverse biases necessary to fully deplete the large transition region, we do not reach dark current saturation until around 4V reverse bias. In order to reduce noise in our dark current measurements and improve alignment in our photoresponse measurements, we experimented with developing remote bond pads for our photodetectors and wirebonding to those remote bond pads with the process flow outlined in Figure 4-4. We used 250nm SiO_2 and 160nm of SiN as isolation layers. These thicknesses were chosen since they would also act as anti-reflective coatings for Ge at the 1310nm wavelength. As can be seen from Figure 4-16, the dark current increased dramatically after wirebonding for the majority of diodes. Our wirebonding process yield was only about 13%. The low yield of the process as well as the added difficulty and extra steps in fabricating remote bond pads resulted in our decision to continue with probed measurements and direct bond pads. Fabricating remote bond pads was more difficult due to the large height differential that the remote bond pad had to span to connect the p-Ge to the insulator layer. The p-Ge to n-Si etch depth was between 3.5-3.7µm, so the sputtered metal had to be continuous for this height. This along with the low thickness of the insulation layers may have

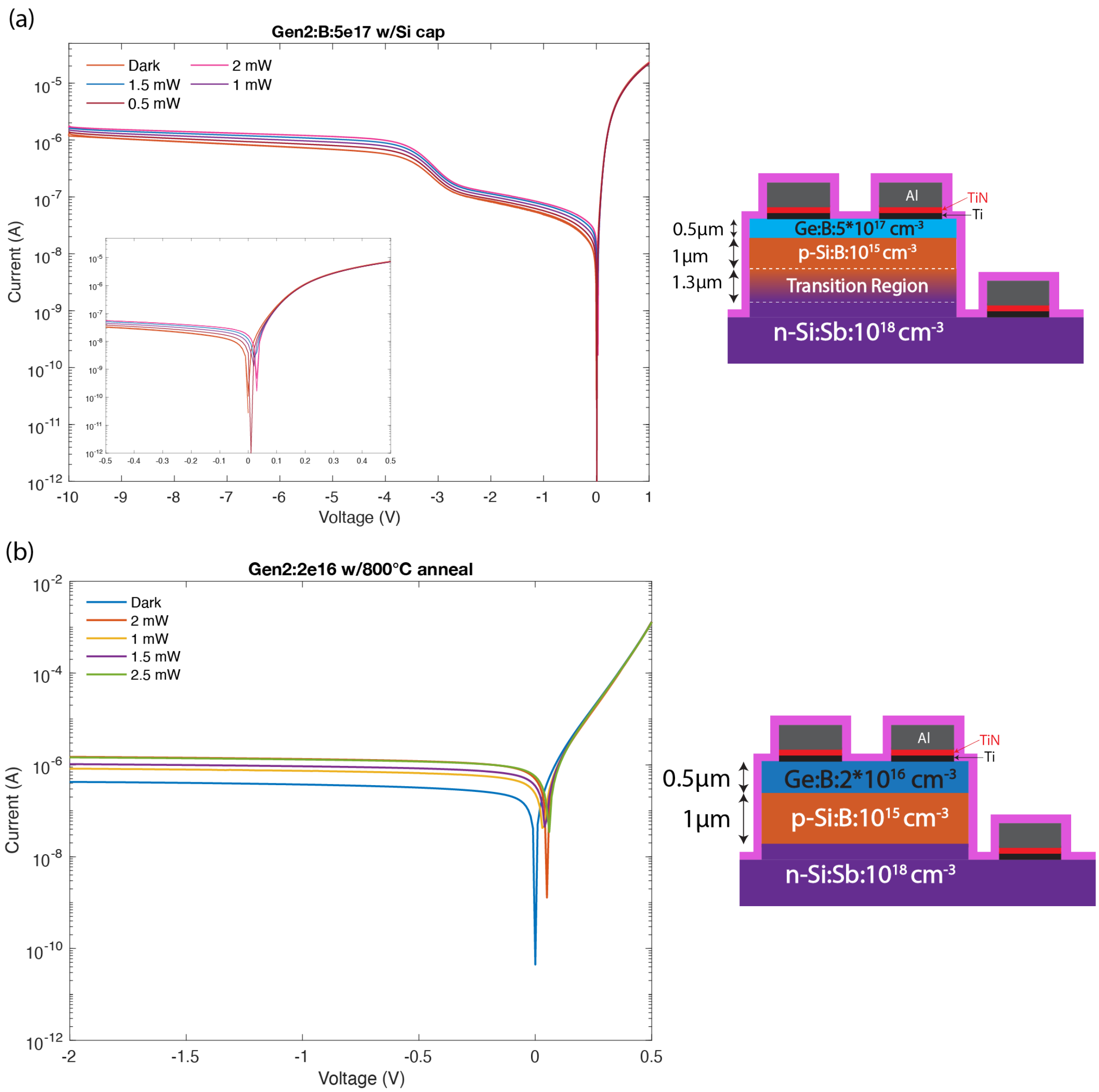


Figure 4-15. Typical dark and photo IV curves of (a) Gen2 and Gen3 devices with long transition region between p-Si and n-Si and (b) Gen2 devices with abrupt junction between p-Si and n-Si. Curves labeled 0.5 mW, 1 mW, 1.5 mW, 2mW, 2.5 mW are photo IV curves from 1310nm laser driving power of 0.5 mW, 1 mW, 1.5 mW, 2 mW, and 2.5 mW, respectively. To the right of the IV plots are representative diode cross-sections.

contributed to the low yield of our devices, as typical industry standard is thicker for higher yield.[64] Since we rely on fabricating five different diode sizes to analyze geometric leakage dependence, the yield/fabrication time product is too low to make wirebonding a process worth pursuing further.

Sidewall passivation via Al_2O_3, high temperature annealing in the dislocation recovery regime, and a remote heterojunction structure lowered dark current significantly in our Ge-on-Si photodiode structures, and allowed us to achieve the lowest reported dark current density (160 nA/cm^2) in literature for Ge-on-Si heterojunction photodetectors (see Table 4-4). Our LT Ge remote heterojunction diodes with sidewall passivation have dark current densities comparable to the lowest reported in literature as well. Our 800°C-annealed diode record provides us a new optimal target to aim for with low-temperature processing. Future work optimizing low temperature processing conditions should be done to evaluate leakage current limits for BEOL-compatible photodetectors.

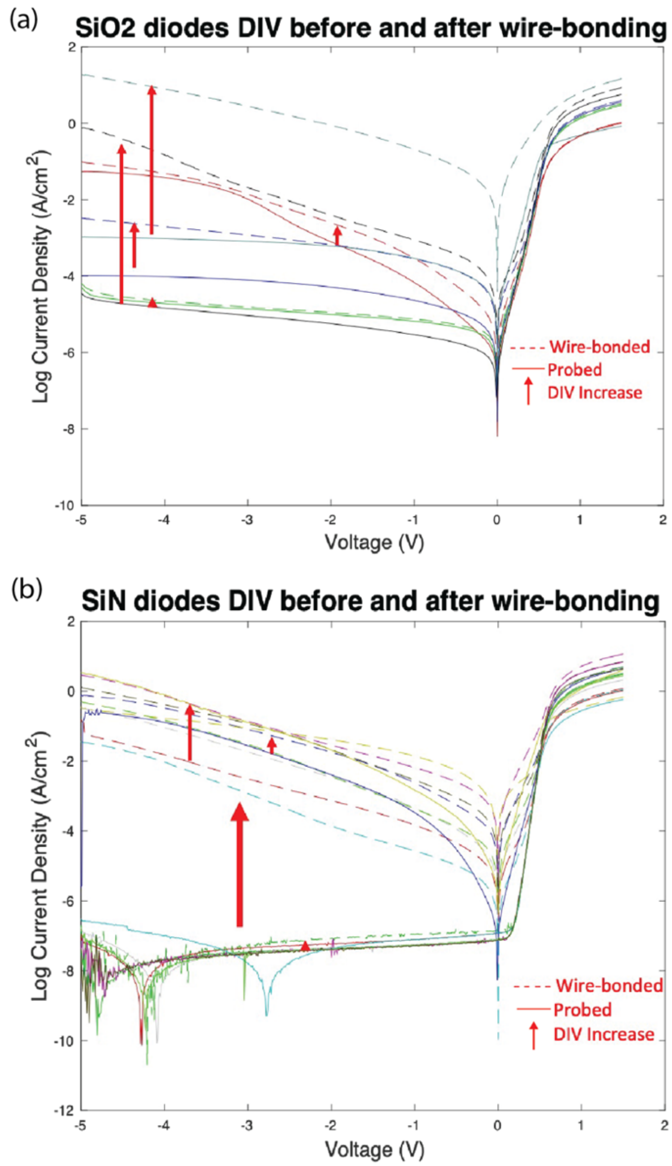


Figure 4-16. Devices were processed via Figure 4-4 with remote bond pads including (a) SiO_2 and (b) SiN as isolation layers. Solid lines indicate probed diode results before wirebonding, dashed lines indicate wirebonded diodes, and the red arrows indicate how much dark current increased with wirebonding. Figure and data courtesy of Stephanie Marzen.

	This work (2021) Ge p/Si p-n	**Bowers (2010) APD Ge p/Si p-i-n**[65]	**Osmond (2009) Ge p-i/n-Si**[66]	**Samavedam (1998) Ge p-n/Si**[67]
J_d @ -1V ($\mu A/cm^2$)	0.16	5.0	41	150
	Lin (2021) Ge p-i-n/Si[68]	**Piels (2012) UTC Ge p-in/Si**[69]	**Virot (2017) Lateral WG Ge p-i-n/Si**[70]	**Ahn (2007) WG Ge p-i-n/Si**[57]
J_d @ -1V ($\mu A/cm^2$)	780	$4.0*10^4$	$4.5*10^5$	$1.4*10^6$

Table 4-4. Comparison of dark current density to prior art in Ge-on-Si photodiodes.

4.5.3 Effects of low temperature annealing on device internal quantum efficiency

Internal quantum efficiency (IQE) is calculated from the photodiode's photoresponse via:

$$IQE = \frac{\frac{I_{light} - I_{dark}}{Q_{electron}}}{\frac{Power\ absorbed}{Energy_{1310nm\ photon}}} = \frac{\frac{I_{photo}}{1.602 * 10^{-19}\ C}}{\frac{Power * (1-R) * (1-\exp(-\alpha x))}{1.517 * 10^{-19}\ J}} \tag{4-12}$$

where I_{photo} is the photocurrent obtained by subtracting the dark current I_{dark} from the current while the diode is under illumination I_{light}, R is the reflectance calculated from Filmetrics Reflectance calculator[71] as 0.4 at 1310nm light, α is the absorption coefficient of Ge at 1310nm, 10^{-4} cm^{-2}, and x is the Ge thickness, 500nm. The incident power is calibrated via the beam profile correction detailed in Section 4.2.3. Figure 4-17a shows the impact of annealing time on IQE for LT Ge Gen2:B:5e17 and Gen3:B:1e17-1e19 devices with Si cap, for three different annealing temperatures, 450, 500, and 550°C. These annealing temperatures were chosen to investigate the point defect recovery temperature regime identified in Chapter 3. While temperatures at 450°C or lower are required for BEOL integration[5], the observed activation energy from Hall effect annealing experiments determined that over 8 hours would be necessary to completely remove the acceptor-like point defects (see Table 4-3). Thus, in order to investigate whether IQE to acceptor annealing ratio is 1-to-1 or more complex, we anneal at higher temperatures to investigate these point defect dynamics more effectively. IQE is plotted at a reverse bias of 8V for both Figure 4-17a and b because this was the voltage at which we saw saturation. We see an increase in IQE with annealing for Gen3 devices (Figure 4-17a), but the opposite effect for Gen2 devices – IQE appears to decrease with annealing except for the Gen2 devices annealed for at 550°C for 4 hours. All other annealing points which demonstrated a decrease in IQE for the uniformly doped devices (Gen2) could be due to not having a high enough doping density to compensate the n-type background that we revealed was present for LT Ge via Hall effect measurements in Chapter 3. This could result in growing n-type compensation comparable to the p-type doping density for the Gen2 devices which would then reduce quantum efficiency due to the energy band barrier present between n-type Ge and p-Si.

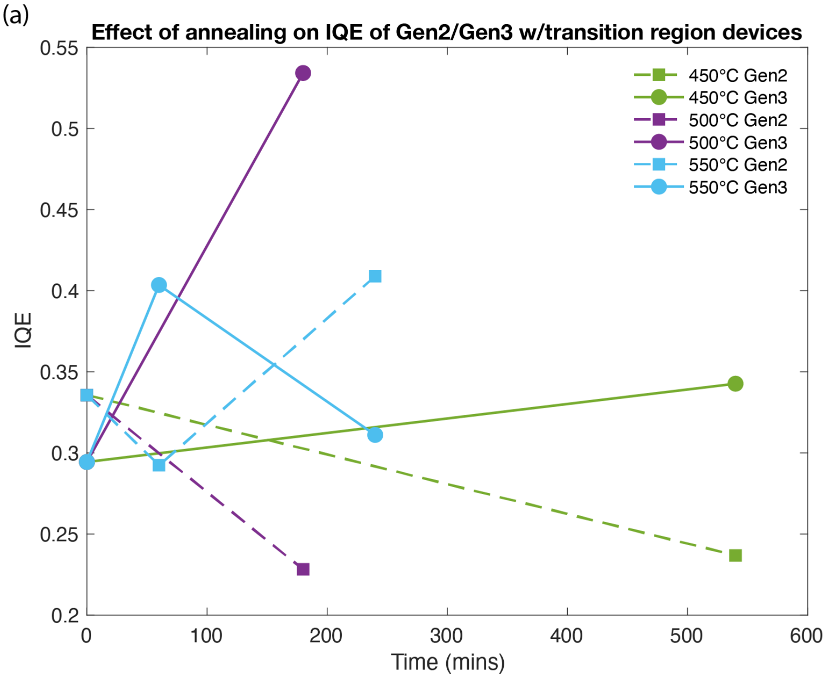
All Gen3 devices demonstrated an improvement in IQE upon annealing except for Gen3 annealed at 550°C for 240 minutes which demonstrated lower IQE than annealing at 550°C for 60 minutes. This is either due to experimental error for this point or deleterious effects of annealing at longer times and higher temperatures due to another mechanism beyond acceptor removal. The rate at which IQE improves with time is what we would expect in our Gen3 results – the 450°C trend line in Figure 4-17a has the lowest slope and the slowest improvement in IQE with time, with 550°C having the steepest and fastest improvement in IQE with time, and 500°C in between both. While more research is necessary to determine the cause of the different IQE annealing behavior in the different device structures, the IQE increase we observe upon annealing Gen3 device structures is very promising. This demonstrates that point defects which are being annealed out in these temperature regimes have an impact on LT Ge-on-Si photodetector IQE.

Figure 4-17b plots the impact of 60 minute isochronal anneals on Gen3 IQE. The blue circles are the measured data points for as-grown and 550°C annealed Gen3 devices, while the transparent red and purple circles are linearly extrapolated from the longer anneals at 450°C and 500°C since we are missing 60-minute isochronal data at these temperatures. We can fit the annealing IQE rate for this data set including the extrapolated data by assuming that annealing out the acceptor-like point defects we observed in Chapter 3 directly correlates to improvement in IQE performance – meaning one acceptor trap that is annihilated means we have one extra photocarrier that is collected at the junction. The annealing rate was calculated by:

$$\frac{dIQE}{dt * (IQE_{ag} + IQE_f)} = \frac{IQE - IQE_{ag}}{\left(IQE_f - IQE_{ag}\right) * \Delta t} \tag{4-13}$$

where IQE_f is 0.54, IQE_{ag} is 0.29 (both from experimental results detailed in Figure 4-17a), and Δt is the time in seconds (3600). This annealing rate is then plotted on an Arrehnius plot vs 1/T and fit to an exponential decay in Matlab with activation energy 0.3 eV, though the fit is not precise due to limited data. The IQE activation energy for Gen3 anneals is slower than that observed from Hall effect for acceptor annihilation, 1.7eV, either due to variation in the measurements, an incorrect assumption of a 1-to-1 ratio of acceptors annihilated to photocarriers collected/IQE increase, or due to another mechanism occurring in Gen3 device for longer/higher temperature anneals that drags down the overall activation energy of the acceptor annihilation process. More data points at isochronal and isothermal annealing times are necessary to determine the statistical error of the IQE activation energy fit.

(a)



(b)

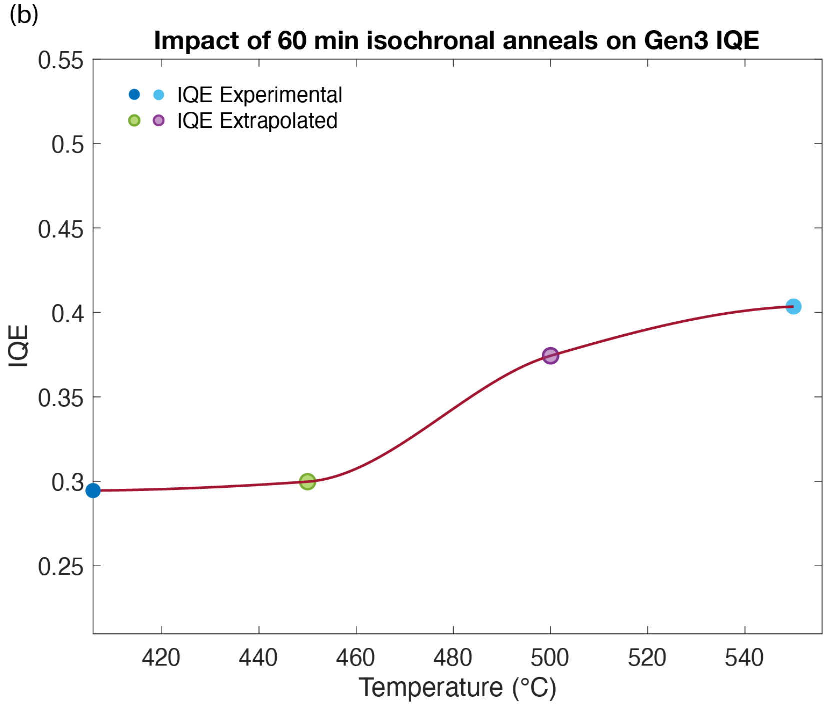


Figure 4-17. (a) Effect of low temperature annealing on IQE of Ge:B:5e17 Gen2 (dashed lines) and Ge:B:1e17-1e19 Gen3 (solid lines) devices at 450°C (green), 500°C (purple), and 550°C (blue). Gen2 IQE mainly decreases with annealing except for 500°C 240 mins annealing. Gen3 sees a definite benefit from annealing, though another mechanism is potentially causing IQE to

not be as high for 550°C 4h anneals. (b) Isochronal 60-minute anneals for Gen3 devices. Dark blue as-grown and light blue 550°C circles are experimental measured values, and the transparent green and purple circles are extrapolated values from experimental values in (a).

4.6 Summary and Conclusion

We developed a process to fabricate Ge-on-Si remote heterojunction photodetectors. We investigated processing effects on device performance – namely utilizing remote bond pads for wirebonding, sidewall passivation via Al_2O_3, and including a Si cap. We found that using insulation layer and metal thicknesses that were feasible given our processing capabilities resulted in very low yield for our wirebonded devices. Thus, we continued to probe all of our photodetectors in order to measure dark current and quantum efficiency. We found that including a Si cap on the Ge did not increase dark current in our devices compared to those whose Si cap was removed.

We also found that sidewall passivation via 20nm of Al_2O_3 resulted in areal leakage dependence and lowered the dark current density to 160 nA/cm^2 for the Ge:B:2e16 Gen2 devices that were annealed at 800°C. Our lowest observed dark current density was 50 nA/cm^2. These are the lowest ever dark current values reported in literature for Ge-on-Si photodiodes, which we attribute to the remote heterojunction structure, sidewall passivation, and the annealing scheme. Moving from a Gen1 to Gen2 structure lowers dark current density by 1-2 orders of magnitude. Sidewall passivation lowers dark current density by 2-3 orders of magnitude, and annealing at 800°C lowers dark current density by an additional 1-2 orders of magnitude. The annealing results are evidence that dislocations impact dark current density more than point defects as low temperature annealing did not improve dark current in our devices in the temperature and time scales we investigated. Sidewall passivation had the biggest impact on dark current density, thus peripheral leakage due to dangling bonds and etching-induced damage at the sidewalls has a bigger impact on dark current that bulk defects such as dislocations and point defects. However, future work annealing at lower temperatures for longer may further elucidate the role of point defects in Ge-on-Si dark current.

While we did not see an effect of point defects on device performance in terms of leakage, we were able to identify that point defects do play a role in IQE. IQE nearly doubles for our Gen3 devices with annealing in the 450-550°C point defect recovery regime identified via Hall effect carrier concentration measurements. The IQE annealing rate is fit to a 0.3 eV activation energy which is slower than that predicted by acceptor removal identified via Hall effect, 1.7 eV. This may be due to limited experimental data, the incorrect assumption that one annihilated acceptor results in one extra photocarrier, or another mechanism occurring beyond acceptor removal that drags down the activation energy. Future work will aid in solidifying the defect kinetics associated with IQE improvement.

Whereas prior work in Ge-on-Si photodetectors has focused on threading dislocations and their impact on device performance, we have very promising results showing that point defects have a significant impact on device performance. You may think all defects are bad and indeed, a perfect material would be ideal – but identifying point defects that can be reduced via annealing at BEOL-compatible temperatures much lower than those necessary to promote annihilation of dislocations is a huge step forward in determining optimal device performance under low temperature growth and processing requirements for BEOL integration.

Chapter 5: Room temperature evaporation of Ge on Si for amorphous Ge CMOS compatible waveguides

5.1 Introduction

The mid-infrared (mid-IR) to longwave infrared range (LWIR) encompass a variety of sensing applications, including toxic gas sensing, gases related to environmental effects, biomedical sensing, and sensing of essential molecules in the "fingerprint region". These molecules have unique absorption spectra in this wavelength range, allowing molecules like explosives, chemical weapons, pollutants, and biological markers to be readily identified.[72] Germanium's transparency in the mid-LWIR (2-14um)[73] makes it the ideal candidate for a variety of sensing platforms. Crystalline Ge waveguides on float-zone Si demonstrating low losses of less than 1db/cm in the for 10um wavelengths and greater and 5dB/cm or lower loss for the 7.5-11um wavelength range have recently been demonstrated by Gallacher et al.[72] However, crystalline Ge requires a crystalline template for growth. We explore room temperature physical deposition of Ge, resulting in amorphous Ge, for sensing applications in the mid-IR-LWIR. Depositing at room temperature not only lowers the thermal budget required for BEOL-compatibility, but also allows deposition on diverse substrates that augments vertical integration possibilities.

5.2 Operating Principles of Waveguides

Dielectric waveguides have two components: a high index core material which acts as the guiding material, and a cladding material that surrounds the core material and because of its lower index, confines the light inside of the core material due to the principle of total internal reflection. A part of the light is also guided evanescently in the cladding, which is referred to as the evanescent field. The simplest kind of waveguide used for modeling is a “slab” waveguide, where the layers are infinitely wide and deep. The kind of Ge waveguide we fabricated is a channel waveguide, where the waveguide is only presumed infinite in one direction, but has a defined height and width (see Figure 5-1). Unlike the slab waveguide, the light in this structure is confined in both in-plane and out-of-plane directions to facilitate in-plane optical routing.[74] Bending without incurring too much loss is essential in low cost fabrication of these waveguides as chip dimensions are constantly decreasing and bending allows more devices to be put on a chip. The channel waveguide also allows a particle landing on the surface or side of the waveguide structure to interact directly with the evanescent field, thus contributing to its sensing ability.

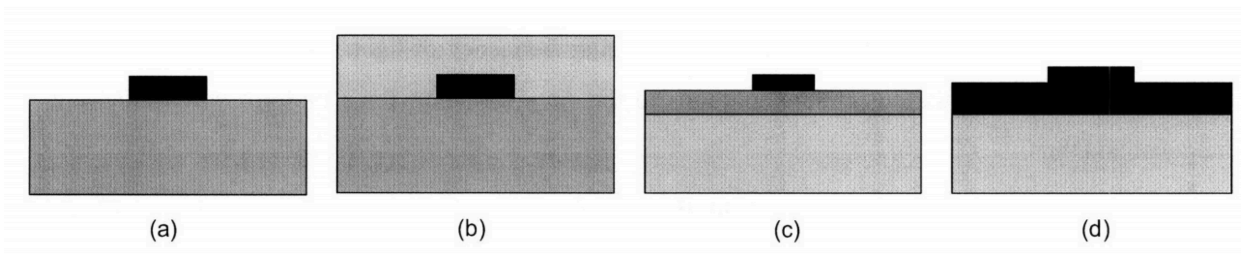


Figure 5-1. Types of waveguides (a) channel/strip, (b) buried channel or embedded strip, (c) strip load waveguide, and (d) ridge or rib waveguide. Schematic from Sparacin.[74]

Finite difference time domain (FDTD) simulations are conducted to solve the Helmholtz equations of the guided modes in channel waveguides. We will use FDTD simulations to determine the single mode cut-off dimensions for the waveguides. The advantages to using a single mode versus multi-mode waveguide are that have a smaller footprint and the lower confinement of multi-mode waveguides yields larger bending losses and increases substrate leakage which ultimately results in higher loss values for multi-mode compared to single-mode waveguides.[74]

Transmission loss is the most important operating metric of waveguides, and everything is designed to lower transmission loss. Loss in a waveguide can come from several different sources: intrinsic material absorption, bending loss, scattering from interface roughness or bulk inhomogeneities, and loss due to leakage to the substrate. The insertion loss is a sum of coupling loss and the transmission loss multiplied by the length of the optical path. The coupling loss arises from light lost when coupling in and out of the waveguide and is associated with losses incurred during measurement due to mode and size mismatch as well as reflection at the surface. Tapering the waveguide in from the edge is used to minimize mode and size mismatch issues whether the light is laser coupled or fiber coupled into the waveguide.

5.3 State-of-The-Art Amorphous Waveguides and Ge Waveguides

Evaporation at room temperature yields amorphous Ge due to the disordered assembly of atoms on the surface of the substrate. Amorphous Si (α-Si) acts as a source of comparison to amorphous Ge due to their similarity in structure. α-Si has had a lot of success as a waveguide material[75,76] due to its higher index of refraction than Si. Penadés et al report suspended Si waveguides with the underlying SiO_2 etched to use air as the bottom cladding with propagation loss of 3.1 dB/cm in the 7.5 micron range.[76] De Dood et al fabricated amorphous Si waveguides using low pressure O_2 reactive ion etching with no hydrogenation that exhibit 300 dB/cm loss at 1.5 microns, which is extremely large. They estimate the scattering loss from measured sidewall roughness of 10 nm, and the scattering loss appears negligible compared to the modal loss.[77] Takei et al show that hydrogenated amorphous Si, on the other hand, shows a transmission loss of 0.6 dB/cm at 1.55 μm wavelength most likely due to hydrogen passivation of defects.[78]

For reported results in Ge, Osman et al report 2.6 dB/cm loss for 7.57 μm wavelength in suspend germanium waveguides on Si. They estimate the scattering loss to be 0.3 dB/cm from a proposed 5 nm roughness in processing that has not been confirmed with Atomic Force Microscopy (AFM). They propose the remaining 2.3 dB/cm loss is due to threading dislocations from their epitaxially grown Ge.[79] Gallacher et al have reported effective losses of 1 dB/cm for 10 μm and greater wavelengths of TE polarized light for their Ge-on-Si waveguides.[72] The Ge was epitaxially grown by RPCVD on CZ-Si. CZ-Si contains O_2 bonds that absorb in the longwave infrared, so Gallacher et al subtracted these losses from their measured loss value to obtain the Ge waveguide loss. We explore using amorphous Ge on FZ-Si to avoid substrate loss and to implement a lower cost fabrication method that allows for flexible substrate integration and BEOL-compatible integration.

5.4 Experimental Methods

We explored various methods of amorphous Ge (a-Ge) growth including sputtering and electron beam (E-beam) evaporation, which are both physical vapor deposition methods conducted at effectively room temperature. We conducted Hall effect measurements to determine the carrier concentration and resistivity of our deposited a-Ge in order to gain insights into its free carrier concentration that may cause higher absorption and lower transparency. Since we deposited our a-Ge films in shared evaporation and sputtering chambers, we also evaluated the composition of the deposited films using Wavelength Dispersive Spectroscopy (WDS). Ellipsometry was also performed on the amorphous Ge films to understand their optical properties and gain initial insight into their absorption values at our wavelengths of interest.

In order to characterize the material's transmission loss, we fabricated a-Ge on FZ-Si waveguides. FZ-Si substrates were first RCA-cleaned and then loaded into the Ebeam evaporation chamber. Amorphous Ge films of 1 μm and 2 μm thickness were deposited at a base pressure of 10^{-6} torr. For the thicker films, two separate crucibles of Ge were used to obtain the required film thickness without piercing through the crucible. Each crucible was used to deposit 1 μm of Ge and crucibles were auto-exchanged in the chamber so there was no interruption in the vacuum during the amorphous Ge deposition. A high purity, silsesquioxane (HSQ)-based photoresist, FOx-16, was spun onto FZ-Si substrates that had been solvent-cleaned and ashed. Next, Ebeam-lithography was used to pattern the amorphous Ge waveguides based on Lumerical simulation dimensions (Figures 5-4 and 5-5). Waveguides were etched utilizing Reactive Ion Etching with Cl_2, BCl_4, and HBr gases. Finally, the waveguides were dipped in hydrofluoric acid (HF) after etching in order to remove the SiO_2 photoresist hard mask. Fabry-Perot measurements were conducted at 8 and 9.79 μm wavelengths to characterize transmission losses.

5.5 Results and Discussion

5.5.1 Optical and structural properties of amorphous Ge

We conducted Hall effect measurements to determine that sputtered a-Ge had very high p-type carrier concentration (1e18 – 1e19 cm^{-3}), which indicates high free carrier absorption. Ebeam-evaporated Ge had a much lower carrier concentration of 1e16 cm^{-3}, indicating lower losses from free carriers. Wavelength dispersive spectroscopy (WDS) was used to characterize composition of the thermally evaporated and Ebeam-evaporated films. We found that E-beam evaporation demonstrated the highest purity, with 98% Ge and the other 2% coming from carbon, silicon, tungsten, nickel, and copper contaminants most likely due to flaking from the Ebeam evaporator's chamber walls. This contamination could easily be reduced with denoting an evaporator for Ge use only in a CMOS production line. As can be seen in Figure 5-2, Ebeam evaporation demonstrates a refractive index of 4 – 4.5 and very low k values in the mid–LWIR range. Sputtered a-Ge has a lower refractive index in the range of 2.9 – 3.4 and higher k values, implying higher absorptive losses. The lower refractive index of sputtered a-Ge could be due to larger contamination in the film or to a structural difference in the sputtered and Ebeam-evaporated films, such as lower density for the sputtered Ge than for the Ebeam-evaporated films.

The aforementioned Hall effect results demonstrate there are large differences in the free carrier absorption between sputtered a-Ge and Ebeam-evaporated a-Ge, potentially due to different defect structures inherent in the two growth methods, or from contaminants that have a much higher free carrier concentration being present in the sputtering system. Sputtered material is usually more metallic in nature (less amorphous) due to higher surface mobility during deposition[80] which is consistent with a higher carrier density measured by Hall effect. Because of Ebeam-evaporated Ge's extremely low k values, we decided to move forward with this method of amorphous Ge growth. From the k value in the 10 µm range, $1.2*10^{-5}$, we can estimate the expected dB/cm loss from absorption. Using

$$\alpha = \frac{4\pi k}{\lambda} \tag{5-1}$$

to estimate the absorption coefficient, α (cm^{-1}), from the k value measured in ellipsometry, we can see that for the 2 – 15 µm wavelength range, we can expect an absorption coefficient of 0.36 cm^{-1} or lower. From these values, we can estimate dB/cm loss to be 1.56 dB/cm or lower for this range.

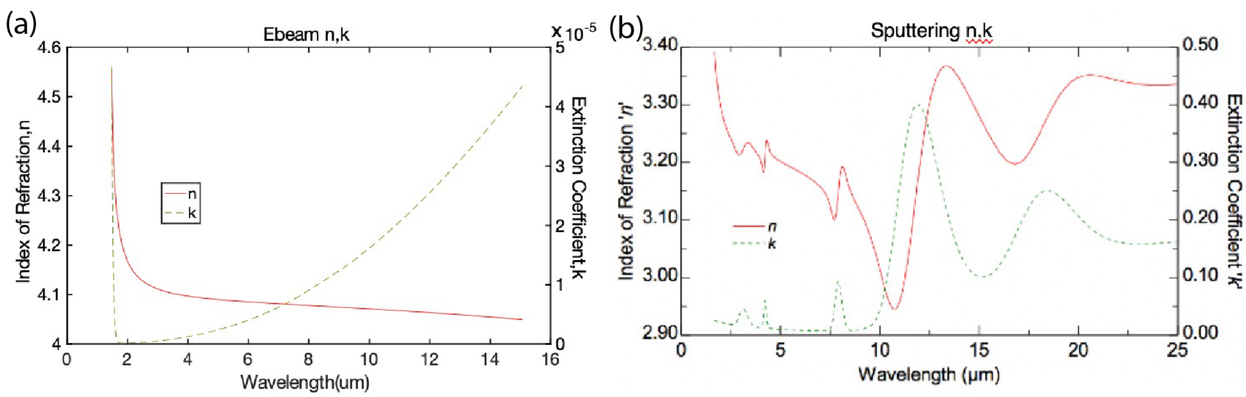


Figure 5-2. (a) Refrative index (n) and absorption coefficient (k) of Ebeam a-Ge and (b) Sputtered a-Ge, measurement courtesy of Ron Synowicki from J.A. Woollam and Vladimir Liberman from Lincoln Laboratory. Ebeam a-Ge demonstrates much lower extinction coefficient than sputtered Ge thus making it the preferred candidate for waveguide fabrication.

X-ray diffraction (XRD) was used to compare the structure of Ge deposited via Ebeam evaporation, a room temperature process, versus epitaxial Ge grown using UHVCVD at 730C, the industry standard for Ge growth. As can be seen from Figure 5-3, amorphous Ge exhibits much broader and irregular peaks than the sharp peaks exhibited by epitaxial Ge. Germanium grown epitaxially is crystalline in nature and thus highly ordered, rendering these sharp peaks in XRD. This confirms that Ge grown via Ebeam evaporation is amorphous and has a highly irregular structure. It's also important to note that the amorphous Ge shows a much lower count value on the left axis than the epitaxial Ge (right axis). This indicates that the light is scattered in many more directions off of the Ebeam evaporated Ge, lowering the measured count value throughout the angular spectrum and further confirming the amorphous nature of Ebeam evaporated Ge.

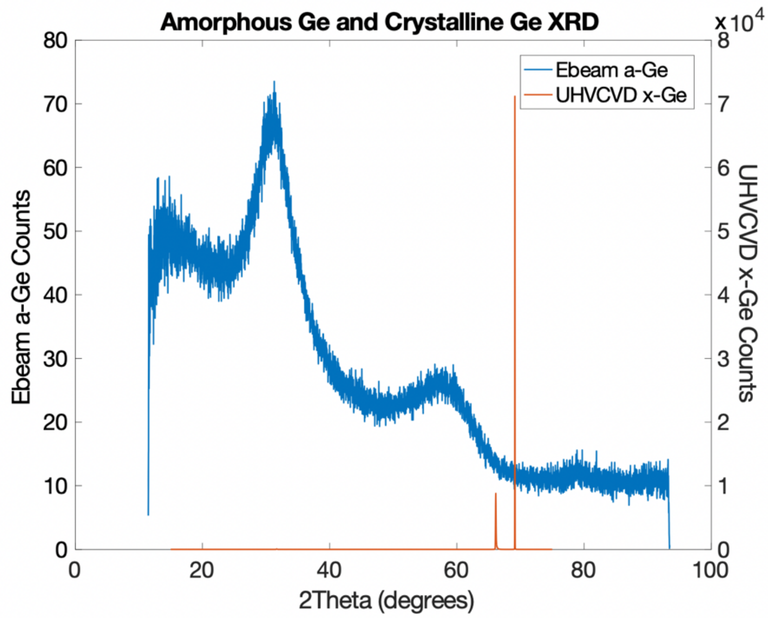


Figure 5-3. XRD plot of a-Ge (blue) and x-Ge (orange). The broad, low-amplitude peaks of a-Ge compared to the sharp, high-amplitude peaks of x-Ge demonstrate that the structure of Ebeam-evaporated Ge is amorphous in nature. Crystalline Ge XRD courtesy of Dr. Ruitao Wen.

5.5.2 Waveguide Design

Germanium on silicon waveguides were simulated using Lumerical to find the guided mode cut-off for 1μm (Figure 5-4) and 2μm-thick (Figure 5-5) a-Ge on FZ-Si. For 1 um-thick Ge, 6.5 um – 15 um is the width range that allows for a guided mode and single mode operation. The only mode supported at this thickness is the TE mode, which is not optimal for sensing but is used to characterize waveguide loss for this initial investigation. We chose to fabricate waveguides in the 6.5 – 10.5 um width range in order to optimize confinement factor in air and waveguide footprint on chip. CZ-Si is most commonly used as a substrate for CMOS applications but has very high losses in our wavelength range of interest, 8-10 um. Due to the large confinement factor in the cladding, we chose to deposit amorphous germanium on float zone silicon to mitigate losses from cladding absorption. Bending loss simulations were also conducted and we found that 50 um was the bending radius that resulted in less than 1dB/cm bending loss. Based on these simulations, we fabricated waveguides in the 8 – 10.5 um width range in both straight and paperclip configurations (see Figure 5-6).

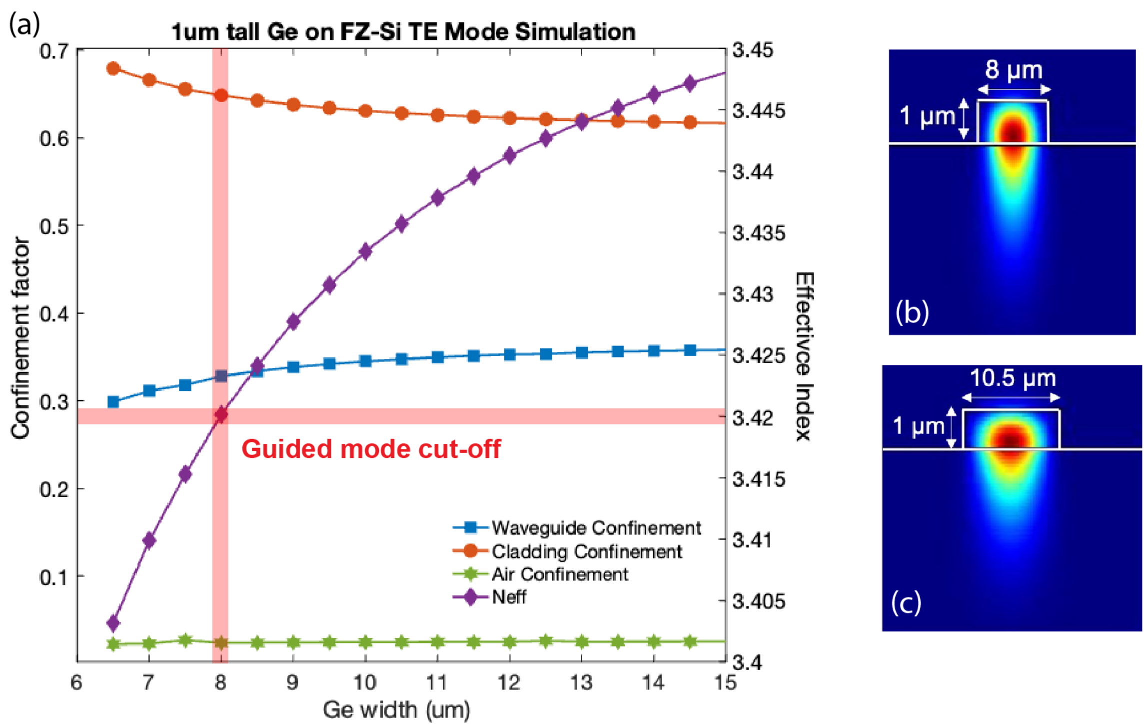


Figure 5-4. Ge on Si channel waveguides simulated in Lumerical demonstrate in (a) that for 1um tall Ge, the guided mode cut-off width is 8um. Only the TE mode is supported for 1um thick Ge. Confinement factor in the air is around 1-2%, and most of the mode is confined in the cladding (~65%). (b) 8um wide waveguide profile (c) 10.5 um wide waveguide profile.

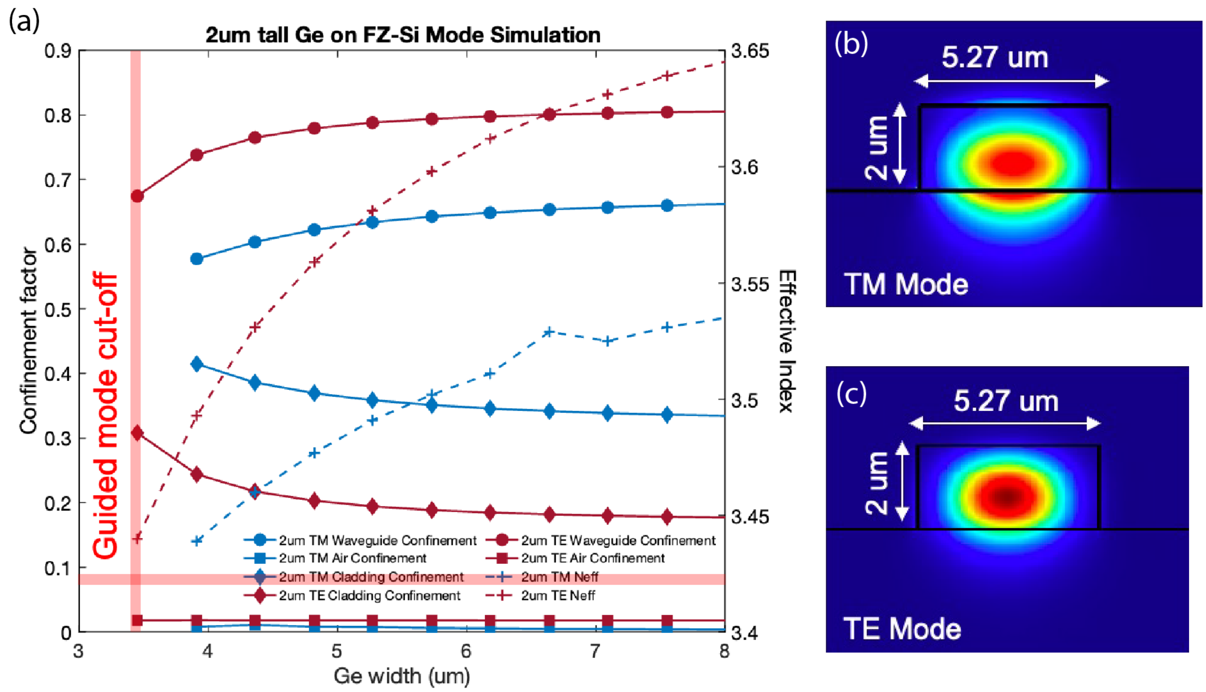


Figure 5-5. Ge on Si channel waveguides simulated in Lumerical demonstrate in (a) that for2um tall Ge, the guided mode cut-off width is 3.5um. (b) 5.27um wide waveguide TM mode profile (c) 5.27 um wide waveguide TE mode profile.

5.5.3 Transmission Loss Measurements via Fabry-Perot

Germanium was deposited on float-zone silicon using E-beam evaporation and waveguides were fabricated using Ebeam-lithography followed by reactive ion etching to form air-clad channel waveguides (see Figure 5-6). Transmission loss measurements were performed in the 8-10 µm wavelength utilizing the Fabry-Perot (FP) method (see Figure 5-7). The loss was determined using:

$$\alpha\left(\frac{dB}{cm}\right) = -\frac{4.34}{Length} * Ln\left(\frac{\gamma}{2R}\right) \tag{5-2}$$

where the fringe contrast, γ, is given from the ratio of peaks and valleys:

$$\gamma = \frac{Peak - Valley}{Peak + Valley} \tag{5-3}$$

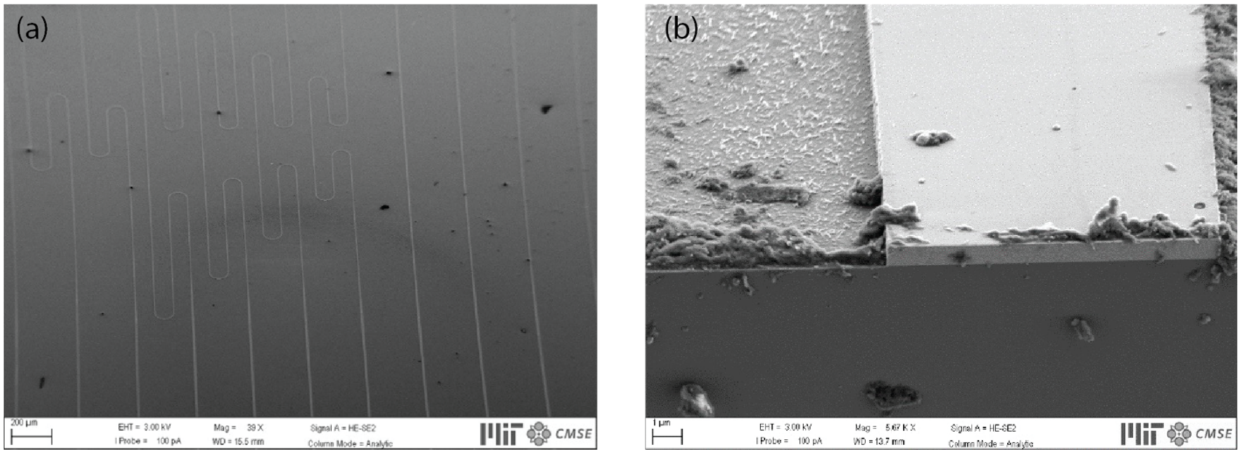


Figure 5-6. (a) Top view SEM image of fabricated paperclip co-linear and asymmetric paperclip structures as well as straight waveguides for FP measurements. (b) SEM image of 1um thick Ge profile courtesy of Samarth Aggarwal.

The reflectivity, R, is given by:

$$R = \left(\frac{1 - n_g}{1 + n_g}\right)^2 \tag{5-4}$$

where n_g is determined from the Free Spectral Range (FSR), shown in Figure 5-7b, for amorphous Ge on FZ-Si waveguides. The FSR is given by:

$$FSR = \frac{\lambda^2}{2 * Length * n_g} \tag{5-5}$$

We chose the FP method to measure straight waveguides as well as bent waveguides rather than traditional cut-back measurements due to the difficulty in coupling at such high wavelengths. Alignment is very difficult in the LWIR because the mode is difficult to see on the LWIR camera, so FP was used due to our ability to confirm a guided mode by seeing FP fringes. Waveguide bends do cause small internal reflections that influence the resonant conditions and thus produce uncertainty in the FP measurements.[74] Feuchter et al demonstrate by numerical simulation that for intracavity reflections lower than -20 dB, the average value of the FP fringe contrast is unaffected by internal reflections around bends. They demonstrate 0.02 dB variation in loss data due to intracavity reflections, which is less than standard deviation from their average measured loss using FP.[81] Our paperclip FP (Fabry-Perot) measurements also exhibit small deviations from the straight waveguide measurement. However, the average value of the paperclip transmission losses (11 dB/cm) measured at a wavelength of 9.79 µm is within a standard deviation (3.1 dB/cm) of the straight waveguide measured at a wavelength of 8 µm (11.5 dB/cm). The ellipsometry results show that the transmission loss calculated from the measured extinction coefficients for 8 and 9.79 µm are 0.61 dB/cm and 0.86 dB/cm. Given that the losses are so close in range, the standard deviation of the 9.79 µm measurements is acceptable in comparison to the 8 µm straight waveguide measurement. Thus, the spread from intracavity reflections is within an acceptable error range for our FP loss calculations.

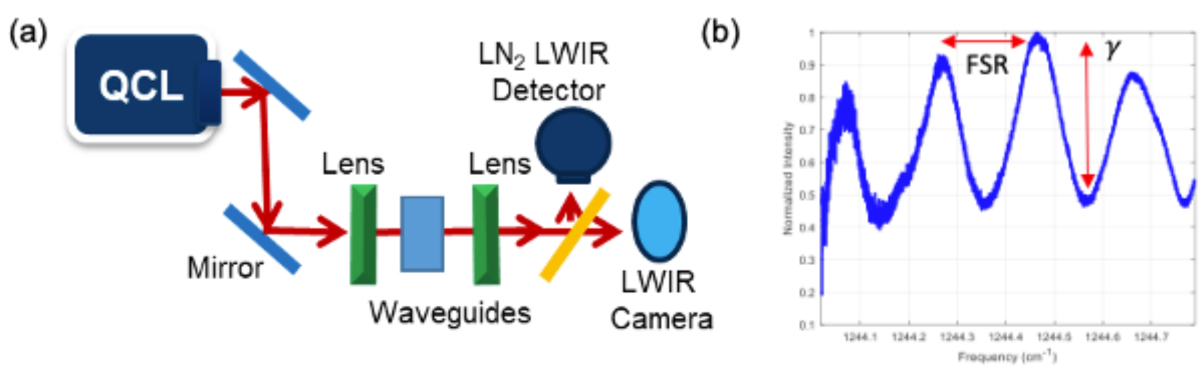
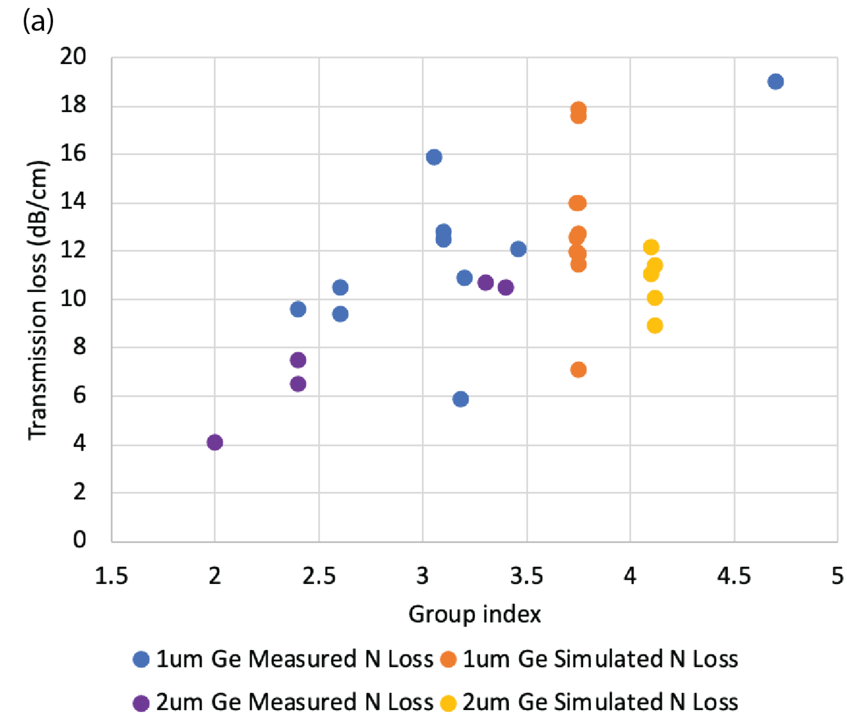


Figure 5-7. (a) Transmission loss set-up for Fabry-Perot measurements. Light is directed from the 9.79µm QCl to the input objective lens which then focuses the light on the waveguide input facet. The second objective lens is focused on the output facet and collects the light transmitted through the waveguides. The light is then passed through a beam splitter and travels to the LWIR camera as well as the liquid nitrogen-cooled LWIR detector. The detector outputs the signal to an

oscilloscope which measures the transmission fringes while the QCL scans. (b) FP transmission loss measurement of 1µm-thick a-Ge on FZ-Si 10µm-wide straight waveguide. By calculating the average peak-to-peak fringe constrast, γ, and free spectral range (FSR), we obtain the transmission loss. Measurement set-up and schematic courtesy of Michelle Clark.

The large spread in loss measurements (Figure 5-8) could be due to non-uniformity in the structure of a-Ge among various batch depositions, fluctuations in etching conditions resulting from shared tool use, as well as aging since the a-Ge waveguides are air-clad and thus susceptible to damage from exposure to humidity. The measurements also demonstrate a wide spread of measured group index (2-4.6), potentially due to coupling differences between each measurement. Due to the difficulty of coupling at such long wavelengths, the sample position was modified each time to achieve the best fringe contrast. Thus, we may have coupled further into the cladding or in the air thus resulting in much lower group indices than anticipated (we anticipate group index between that of FZ-Si and Ge, 3.4-4, or higher). It is also possible that the damage from dry etching may have revealed porosity in the a-Ge structure, thus resulting in lower group indices.



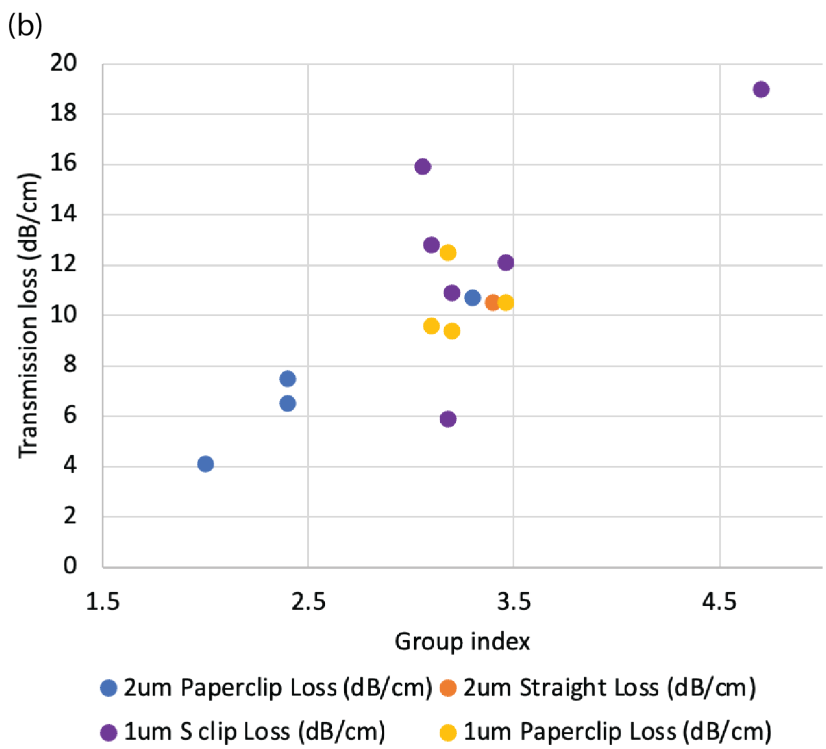


Figure 5-8. (a) Transmission loss measured via Fabry-Perot for 1µm and 2µm tall Ge on FZ-Si waveguides utilizing group index calculated from the FSR of measurements as well utilizing group index simulated via Lumerical. (b) Transmission loss measured via Fabry-Perot organized by

height and type of waveguide structure measured. We see that we have lower losses generally for 2μm tall structures and about the same spread in measurements for S clips vs co-linear paperclip structures.

A number of sources could be responsible for the resultant loss value – radiative loss due to absorption, coupling loss, or scattering loss due to sidewall roughness to name a few. Based on our ellipsometry results, the loss due to absorption in the core should be very low. To understand the dependence of loss on sidewall roughness, we used the Payne-Lacey model[82] to estimate an upper bound for scattering loss:

$$\alpha \leq \frac{\sigma^2}{k_0 d^4 n_1} * 0.48 \tag{5-6}$$

where σ is the root man square roughness of the sidewalls, k_0 is the free-space wavenumber, d is ½ of the waveguide width, and n_1 is the index of the core. Based on the parameters of our waveguide, the sidewall roughness corresponding to 1dB/cm is around 100 nm root mean square value. To explain the average 15 dB/cm loss measured for the group index range that align with expected values (3.4 and above), we can see from Figure 5-9 that a sidewall roughness of around 300-350 nm would produce scattering loss in the 15 dB/cm range. Future work should include utilizing AFM measurements to confirm loss dependence on sidewall roughness.

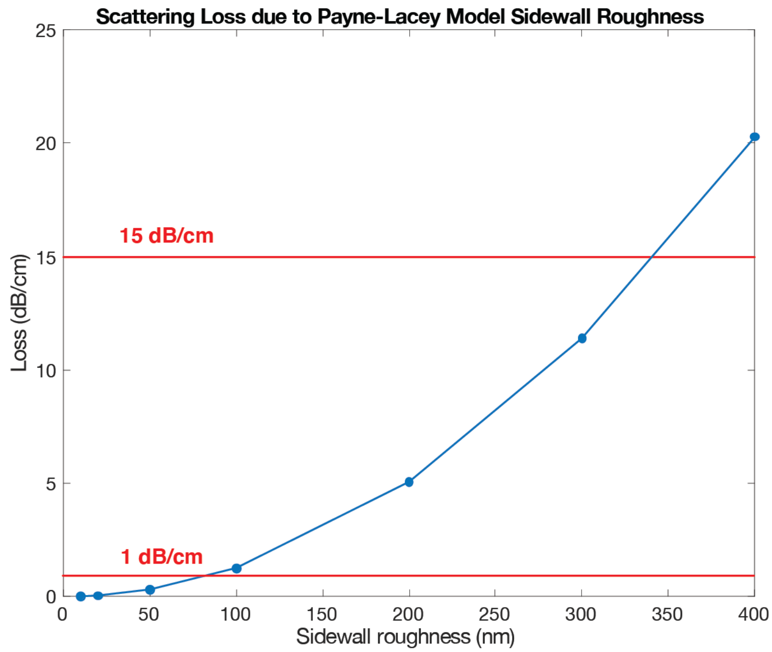


Figure 5-9. Scattering loss due to sidewall roughness calculated using Payne-Lacey model. To achieve a sidewall roughness of 1 dB/cm would require sidewall roughness lower than 100nm rms. Amorphous Ge on FZ-Si waveguides exhibited an average loss of 15 dB/cm which corresponds to a sidewall roughness of 300-350nm rms.

5.6 Summary and Conclusion

Room temperature deposited amorphous Ge has been evaluated for evanescent sensing applications in the MWIR and LWIR ranges. Thin films of amorphous Ge were obtained using various deposition approaches: sputtering and E-beam evaporation. These growth approaches were characterized using WDS, Hall effect, and ellipsometry. Ellipsometry performed on the amorphous Ge samples at a wavelength range of 1.5 to 15 μm showed that the index of refraction, n, of sputtered Ge had a lower range (2.9 – 3.4) than that of E-beam Ge (4 – 4.5). The extinction coefficients for both sputtered Ge and E-beam Ge are in a comparable range, 0 to 0.4, for the same measured wavelength range (1.5 to 15 µm), indicating low absorption losses within the material. Due to the higher levels of contamination identified via WDS and the higher free carrier concentration via Hall effect, E-beam evaporated Ge showed more promise than sputtered Ge for low-loss waveguides in the mid-LWIR range.

We fabricated amorphous germanium waveguides using E-beam evaporation on a CMOS compatible LWIR-transparent substrate (float zone silicon). We developed various etching recipes and tested a variety of structures aimed at reducing loss in amorphous Ge. Fabry-Perot measurements were not ideal for characterizing waveguide loss due to the difficulty in measuring reflectivity of facets and resulting group index values that did not fit the expected range of group index from Lumerical simulations. Though data points are limited for group indices that match simulated values – losses obtained are between 12-19 dB/cm at 9.79 um with 15 dB/cm average transmission loss for the 3.4-4.6 group index range. The Payne-Lacey model identifies sidewall roughness on the order of 300 nm would result in transmission losses in the observed range. Reducing sidewall roughness to 100 nm rms roughness would reduce transmission loss to the 1dB/cm range predicted via ellipsometry. While we had limited data due to difficulties with Fabry-Perot measurements, the average transmission loss of 15 dB/cm demonstrates that amorphous Ge is a good candidate for sensing in the LWIR. Future improvements to sidewall roughness and scattering centers via optimized processing conditions will lower observed transmission loss.

Chapter 6: Summary and Outlook

We have evaluated the material properties of germanium grown at low temperatures both epitaxially via UHVCVD and amorphously via Ebeam-evaporation. Throughout our exploration of optimal growth temperature for BEOL-integration of near-IR Ge-on-Si PDs, we identified three different strain regimes: I) the low temperature growth regime, II) the point defect recovery regime, and III) the dislocation recovery regime. We identified a residual compressive strain of 0.086% in Ge-on-Si that exists at a wider temperature range than previously investigated, which is evidence for its origin as an intrinsic growth phenomenon rather than a post-processing effect of strain relaxation. TEM was used to correlate threading dislocations to misfit dislocations and showed that we have lower misfit dislocation density (10^{11} cm^{-2}) in LT Ge than required for complete relaxation (10^{12} cm^{-2}). A model is developed for homogeneous nucleation of misfit dislocations which includes dislocation repulsion interaction energy that increases the nucleation activation energy barrier significantly. Strain fluctuations of 4.77% are required to overcome this barrier to misfit dislocation nucleation. LT Ge-on-Si pitting surface morphology is prevalent in the low temperature growth regime despite the use of a 350°C buffer layer. The incomplete relaxation phenomenon observed via the residual compressive strain that remains in the material, the lower MDD density, and the impact of dislocation repulsions on driving up the misfit dislocation nucleation barrier is evidence that this pitting surface morphology is necessary to allows higher strain concentrations at the edges of the pits which lower the barrier for misfit dislocation nucleation to be surmountable by available thermal energy.

Aside from growth temperature optimization, we also explore optimizing a post-growth low temperature anneal to improve Ge-on-Si photodetector performance for the near-IR. We found that as-grown LT Ge is initially p-type, but type converts upon annealing at 500°C and higher. Using TEM, we observe that 90° dislocation density is reduced upon annealing in the 500-600°C range and these 90° dislocations are identified as pure edge of mixed character via g•b analysis. Annihilation of sessile edge dislocations at unprecedented low temperatures is evidence for acceptors facilitating 90° dislocation climb as the mechanism behind this type conversion. Based on the defect compensation level present in the material, we determined we had to intentionally dope Ge to carrier concentrations of $5*10^{17}$ or higher to overcome the n-type background revealed upon annealing. We found that low (BEOL-compatible) temperature annealing does not affect dark current, but does improve device IQE by nearly twofold compared to as-grown IQE. Thus, point defects impact photocarrier collection efficiency but do not dominate recombination/generation pathways that affect leakage current as do dislocations and sidewall contributions. Future work should seek to further decouple the effects of point defects, dislocations, and sidewall dangling bonds on device performance to optimize the low temperature post-processing step further.

Switching gears, we looked at passive devices that were BEOL-compatible in amorphous Ge-on-Si waveguides. Because passive Ge devices operate in the mid-LWIR regimes, the defects we explored in our active BEOL Ge-on-Si device such as point defects or dislocations are not as concerning due low scattering loss at these large wavelengths. Instead, we care about surface roughness affecting scattering loss, voids, and non-uniformities/contaminants in the amorphous material that could lead to higher absorption losses. Ellipsometry yielded promisingly low absorption coefficients that estimated 1.56 dB/cm or lower transmission loss. We characterized transmission losses to be between 12-19 dB/cm for amorphous Ge on FZ-Si waveguides at a wavelength of 9.79 μm. However, we discovered that measuring transmission loss via Fabry-Perot led to inaccuracies due to the different facet reflectivity for each sample. Future work should use cut-back measurements to measure transmission loss more accurately. Initial loss measurements of 12-19 dB/cm are very promising for amorphous Ge, and future optimization of waveguide design and sidewall roughness, as well as accurate coupling, could lower these loss values dramatically.

Ultimately, we provide a thorough characterization of BEOL-compatible Ge-on-Si material properties and avenues to optimize these material defect characteristics and device designs in tandem. Future directions should seek to optimize further on low temperature post-processing and determine the extent to which dislocations prevalent in low temperature growth can participate in point defect annihilation reactions. Effects of point defects vs dislocations on dark current should be examined further and decoupled from effects of sidewall passivation, potentially via selective growth of Ge-on-Si. The bounds of the elastic strain acommodation regime and how it relates to point defect-dislocation interactions should also be explored further in order to find the ultimate recipe for BEOL-compatible Ge-on-Si. Low-temperature grown Ge has demonstrated its device performance is well within range of standard high-temperature Ge-on-Si photodiodes, and now we must determine how much further we can push both the material and device optimization avenues.

References

1. Moore, G. E. Cramming more components onto integrated circuits With unit cost. *Electronics* **38**, 114 (1965).
2. Pearson, B. Germanium Photodetectors on Amorphous Substrates for Electronic-Photonic Integration. (Massachusetts Institute of Technology, 2016).
3. Magen, N., Kolodny, A., Weiser, U. & Shamir, N. Interconnect-power dissipation in a microprocessor. in *System Level Interconnect Prediction* (2004).
4. Wada, K. *et al.* Silicon and silica platform for on-chip optical interconnection. in *Proceedings of LEOS Annual Meeting* (2002).
5. Beals, M. *et al.* Process flow innovations for photonic device integration in CMOS. *Silicon Photonics III* **6898**, 689804 (2008).
6. Michel, J., Liu, J. & Kimerling, L. C. High-performance Ge-on-Si photodetectors. *Nat. Photonics* **4**, 527–534 (2010).
7. Liu, J. GeSi Photodetectors and Electro-absorption Modulators for Si Electronic-photonic Integrated Circuits. (Massachusetts Institute of Technology, 2007).
8. Fitzgerald, E. A. Dislocations in strained-layer epitaxy: theory, experiment, and applications. *Mater. Sci. Reports* **7**, 87–142 (1991).
9. Albert, B. Germanium on Silicon Heteroepitaxy for High Efficiency Photovoltaic Devices. (MIT, 2016).
10. Luryi, S., Kastalsky, A. & Bean, J. C. New infrared Detector on a Silicon Chip. *IEEE Trans. Electron Devices* **31**, 1135–1139 (1984).
11. Fitzgerald, E. A. *et al.* Dislocations in relaxed SiGe/Si heterostructures. *Phys. Status Solidi Appl. Res.* **171**, 227–238 (1999).

12. Luan, H. C. *et al.* High-quality Ge epilayers on Si with low threading-dislocation densities. *Appl. Phys. Lett.* **75**, (1999).

13. Hirth, J. P. & Lothe, J. *Theory of Dislocations*.

14. Kimerling, L. C., Patel, J. R., Benton, J. L. & Freekabd, P. E. Dislocations in Silicon. in *Defects and Radiation Effects in Semiconductors* 401–406 (1980).

15. Samavedam, S. B. & Fitzgerald, E. A. Novel dislocation structure and surface morphology effects in relaxed Ge/Si-Ge(graded)/Si structures. *J. Appl. Phys.* **81**, 3108–3116 (1997).

16. Patel, J. R. & Kimerling, L. C. Dislocation energy levels in deformed silicon. *Cryst. Res. Technol.* **16**, 187–195 (1981).

17. Cunningham, B., Chu, J. O. & Akbar, S. Heteroepitaxial growth of Ge on (100) Si by ultrahigh vacuum, chemical vapor deposition. *Appl. Phys. Lett.* **59**, 3574–3576 (1991).

18. Ma, D. Ge and GeSi electroabsorption modulator arrays via strain and composition engineering. (MIT, 2020).

19. Burr, T. Electrical properties of silicon surfaces and interfaces. (MIT, 1998).

20. Colace, L. *et al.* Metal – semiconductor – metal near-infrared light detector based on epitaxial Ge / Si Metal – semiconductor – metal near-infrared light detector based on epitaxial Ge / Si. **3175**, 1–4 (2004).

21. Srolovitz, D. J. On the stability of surfaces of stressed solids. *Acta Metall.* **37**, 621–625 (1989).

22. Jaccodine, R. J. Surface Energy of Germanium and Silicon. *J. Electrochem. Soc.* **110**, (1963).

23. Giovane, L. M. Strain-balanced silicon-germanium materials for near IR photodetection in silicon-based optical interconnects. (MIT, 1998).

24. Dolbak, A. E. & Olshanetsky, B. Z. Ge diffusion on Si surfaces. *Cent. Eur. J. Phys.* **4**, 310–317 (2006).

25. Eaglesham, D. J. & Cerullo, M. Dislocation-free Stranski-Krastanow growth of Ge on Si(100). *Physical Review Letters* vol. 64 1943–1946 (1990).

26. Ye, H. & Yu, J. Germanium epitaxy on silicon. *Sci. Technol. Adv. Mater.* **15**, (2014).

27. Colace, L. *et al.* Metal–semiconductor–metal near-infrared light detector based on epitaxial Ge/Si. *Appl. Phys. Lett.* **72**, 3175–3177 (1998).

28. Hull, R. & Bean, J. C. Misfit dislocations in lattice-mismatched epitaxial films. *Crit. Rev. Solid State Mater. Sci.* **17**, 507–546 (1992).

29. Halbwax, M. *et al.* Kinetics of Ge growth at low temperature on Si(001) by ultrahigh vacuum chemical vapor deposition. *J. Appl. Phys.* **97**, (2005).

30. Marzegalli, A. *et al.* Unexpected dominance of vertical dislocations in high-misfit Ge/Si(001) films and their elimination by deep substrate patterning. *Adv. Mater.* **25**, 4408–4412 (2013).

31. Kimerling, L. C. & Patel, J. R. Defect states associated with dislocations in silicon. *Appl. Phys. Lett.* **34**, 73–75 (1979).

32. Ono, H. & Sumino, K. Defect states in plastically deformed p-type silicon crystals investigated by Hall effect. *J. Appl. Phys.* **54**, 4426–4432 (1983).

33. Grazhulis, V. A., Kveder, V. V. & Mukhina, V. Y. Investigation of the energy spectrum and kinetic phenomena in dislocated Si crystlas (I). *Phys. Status Solidi* **43**, 407–415 (1977).

34. Tweet, A. G. Electrical properties of plastically deformed Ge. *Phys. Rev.* **99**, 1245–1248 (1955).

35. Grillot, P. N., Ringel, S. A., Michel, J. & Fitzgerald, E. A. Structural, electronic, and

luminescence investigation of strain-relaxatrion induced electrical conductivity type conversion in GeSi/Si heterostructures. *J. Appl. Phys.* **80**, 2823–2832 (1998).

36. Levinshtein, M. E., Rumyantsev, S. L. & Shur, M. S. *Properties of Advanced Semiconductor Materials: GaN, AlN, InN, BN, SiC, SiGe*. (Wiley, 2001).

37. Cannon, D. D. *et al.* Tensile strained epitaxial Ge films on Si(100) substrates with potential application in L-band telecommunications. *Appl. Phys. Lett.* **84**, 906–908 (2004).

38. Hull, R. Properties of Strained and Relaxed Silicon Germanium. in (ed. Kasper, E.) 28–45 (INSPEC, 1995).

39. Bharathan, J., Narayan, J., Rozgonyi, G. & Bulman, G. E. Poisson ratio of epitaxial germanium films grown on silicon. *J. Electron. Mater.* **42**, 40–46 (2013).

40. Wortman, J. J. & Evans, R. A. Young's Modulus, Shear Modulus, and Poisson's Ratio in Silicon and Germanium. *J. Appl. Phys.* **36**, 153–156 (1964).

41. Liu, Q. *et al.* Strain Field Mapping of Dislocations in a Ge/Si Heterostructure. *PLoS One* **8**, 4–9 (2013).

42. Hull, R. & Bean, J. C. Nucleation of misfit dislocations in strained-layer epitaxy in the Ge x Si 1− x /Si system . *J. Vac. Sci. Technol. A Vacuum, Surfaces, Film.* **7**, 2580–2585 (1989).

43. Dong, L., Schnitker, J., Smith, R. W. & Srolovitz, D. J. Stress relaxation and misfit dislocation nucleation in the growth of misfitting films: A molecular dynamics simulation study. *J. Appl. Phys.* **83**, 217–227 (1998).

44. Tersoff, J. & Legoues, F. K. Competing relaxation mechanisms in strained layers. *Phys. Rev. Lett.* **72**, 3570–3573 (1994).

45. Luan, H. C. Ge photodetectors for Si microphotonics. (MIT, 2001).

46. Patel, N. & Monmeyran, C. Point defect states in Sb-doped germanium. *J. Appl. Phys.* **118**,

(2015).

47. Kor, S. K., Mishra, P. K. & Pandey, S. K. Study of effective Debye temperature for germanium. *J. Appl. Phys.* **45**, 1482–1483 (1974).

48. Marzegalli, A. *et al.* Onset of plastic relaxation in the growth of Ge on Si(001) at low temperatures: Atomic-scale microscopy and dislocation modeling. *Phys. Rev. B - Condens. Matter Mater. Phys.* **88**, 1–6 (2013).

49. Chuang, S. L. Chapter 14: Photodetectors. in *Physics of Optoelectronic Devices* (John Wiley & Sons, 1995).

50. Chapter 4: p-n Junctions. https://ecee.colorado.edu/~bart/book/book/chapter4/ch4_2.htm.

51. Saleh, B. E. A. & Teich, M. C. 18.4 Avalanche Photodiodes. in *Fundamentals of Photonics* 766–771 (Wiley, 2007).

52. Masini, G., Colace, L., Assanto, G., Luan, H. C. & Kimerling, L. C. High-performance p-i-n Ge on Si photodetectors for the near infrared: From model to demonstration. *IEEE Trans. Electron Devices* **48**, 1092–1096 (2001).

53. Cannon, D. D. *et al.* Near-infrared Ge Photodetectors Fabricated on Si Substrates with CMOS Technology. in *MRS Proceedings* 770 (2003).

54. Giovane, L., Luan, H. C., Agarwal, A. M. & Kimerling, L. C. Correlation between leakage current density and threading dislocation density in SiGe p-i-n diodes grown on relaxed graded buffer layers. *Appl. Phys. Lett.* **78**, 541–543 (2001).

55. Yi, X. *et al.* Simulation of the effects of defects in low temperature Ge buffer layer on dark current of Si-based Ge photodiodes. *J. Semicond.* **38**, 042001 (2017).

56. Li, C. *et al.* Effects of Interface States on Ge-On-SOI Photodiodes. *IEEE J. Electron Devices Soc.* **7**, 7–12 (2019).

57. Ahn, D. *et al.* High performance, waveguide integrated Ge photodetectors. *Opt. Express* **15**, 3916 (2007).

58. Feng, N. *et al.* Vertical p-i-n germanium photodetector with high external responsivity integrated with large core Si waveguides. *Opt. Express* **18**, 96 (2009).

59. Bandaru, P. R. *et al.* Fabrication and characterization of low temperature (<450 °C) grown p-Ge/n-Si photodetectors for silicon based photonics. *Mater. Sci. Eng. B* **113**, 79–84 (2004).

60. Masini, G., Sahni, S., Capellini, G., Witzens, J. & Gunn, C. High-Speed Near Infrared Optical Receivers Based on Ge Waveguide Photodetectors Integrated in a CMOS Process. *Adv. Opt. Technol.* **2008**, (2008).

61. Littlejohns, C. G. *et al.* Ge-on-Si Plasma-Enhanced Chemical Vapor Deposition for Low-Cost Photodetectors. *IEEE Photonics J.* **7**, (2015).

62. Ge Band structure and carrier concentration. *Ioffe*.

63. Vincent, B. *et al.* Si passivation for Ge pMOSFETs: Impact of Si cap growth conditions. *Solid. State. Electron.* **60**, 116–121 (2011).

64. Hunter, S. *Recent Development Work in IC Bond Pad Structure and Circuit Under Pad*. (2011).

65. Bowers, J. E., Dai, D., Kang, Y. & Morse, M. High-gain high-sensitivity resonant Ge/Si APD photodetectors. *Infrared Technol. Appl. XXXVI* **7660**, 76603H (2010).

66. Osmond, J. *et al.* Ultralow dark current Ge/Si(100) photodiodes with low thermal budget. *Appl. Phys. Lett.* **94**, (2009).

67. Samavedam, S. B., Currie, M. T., Langdo, T. A. & Fitzgerald, E. A. High-quality germanium photodiodes integrated on silicon substrates using optimized relaxed graded buffers. *Appl. Phys. Lett.* **73**, (1998).

68. Lin, Y., Son, B., Lee, K. H., Michel, J. & Tan, C. S. Sub-ma/cm2 dark current density, buffer-less germanium (Ge) photodiodes on a 200-mm Ge-on-insulator substrate. *IEEE Trans. Electron Devices* **68**, 1730–1737 (2021).

69. Piels, M. & Bowers, J. E. Si/Ge uni-traveling carrier photodetector. *Opt. Express* **20**, 7488 (2012).

70. Virot, L. *et al.* Integrated wavegguide pin photodiodes eploiting laterial Si/Ge/Si heterojunction. *Opt. Express* **25**, (2017).

71. Filmetrics. Spectral Thin Film Reflectance Calculator for Thin-Film Stacks. https://www.filmetrics.com/reflectance-calculator?wmin=1290&wmax=1320&wstep=1&angle=0&pol=s&units=nm&mat[]=Air&d[]=0&mat[]=Al2O3&d[]=20&mat[]=Si&d[]=2&mat[]=Ge&d[]=500&mat[]=Si&d[]=0&sptype=r.

72. Gallacher, K. *et al.* Low loss Ge-on-Si waveguides operating in the 8–14 μm atmospheric transmission window. *Opt. Express* **26**, 25667–25675 (2018).

73. Soref, R. Mid-infrared photonics in silicon and germanium. *Nat. Photonics* **4**, 495–497 (2010).

74. Sparacin, D. K. Process and design techniques for low loss integrated silicon photonics. (MIT, 2006).

75. Gat, N., Subramanian, S., Sheffield, M., Erives, H. & Barhen, J. Chemical detection using the airborne thermal infrared imaging spectrometer (TIRIS). in *SPIE International Conference* (1997).

76. Penadés, J. S. *et al.* Suspended silicon waveguides for long-wave infrared wavelengths. *Opt. Lett.* **43**, 795 (2018).

77. De Dood, M. J. A. & Polman, A. Amorphous silicon waveguides for microphotonics. *J. Appl. Phys.* **92**, 649 (2002).

78. Takei, R. *et al.* Sub-1 dB/cm submicrometer-scale amorphous silicon waveguide for backend on-chip optical interconnect. *Opt. Express* **22**, 4779–4788 (2014).

79. Osman, A. *et al.* Suspended low-loss germanium waveguides for the longwave infrared. *Opt. Lett.* **43**, 5997–6000 (2018).

80. Doolittle, A. Physical Vapor Deposition: Evaporation and Sputtering.

81. Feuchter, T. & Thirstrup, C. High Precision Planar Waveguide Propagation Loss Measurement Technique Using a Fabry-Perot Cavity. *IEEE Photonics Technol. Lett.* **6**, 1244–1247 (1994).

82. Payne, F. P. & Lacey, J. P. R. A theoretical analysis of scattering loss from planar optical waveguides. *Opt. Quantum Electron.* **26**, 977–986 (1994).